

Enterprise Active Directory Infrastructure Lab

Windows Server 2025 • Windows 11 Enterprise • VMware Workstation Pro

Technical Deployment Guide

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Executive Summary

This document describes the design, deployment and validation of a Windows Server 2025 Active Directory environment. The lab was built in VMware Workstation Pro to simulate a small business infrastructure. The completed environment includes a domain controller, DNS, organisational units, users, groups, Group Policy, shared folders, NTFS permissions and a Windows 11 domain-joined client. It also demonstrates common IT support procedures including password resets, account lockouts and access control.

1. Introduction

1.1 Project Purpose

The purpose of this project is to demonstrate practical Windows Server administration skills by deploying and documenting a functional Active Directory environment from installation through validation.

1.2 Objectives

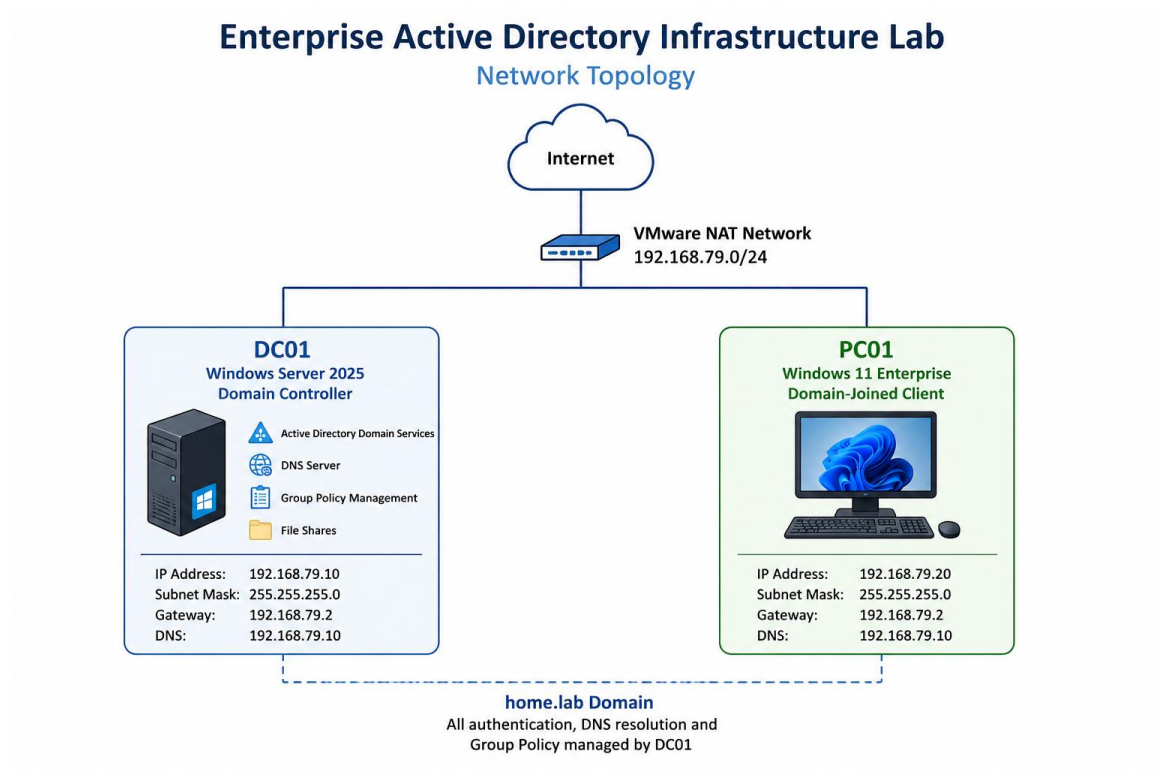
- Deploy Windows Server 2025 in VMware Workstation Pro.
- Configure Active Directory Domain Services (AD DS).
- Configure DNS for the home.lab domain.
- Create organisational units, users and security groups.
- Join a Windows 11 Enterprise client to the domain.
- Configure Group Policy Objects (GPOs).
- Configure departmental file shares and NTFS permissions.
- Validate the environment using realistic IT support scenarios.

2. Lab Environment

Component	Configuration
Host OS	Windows 11
Hypervisor	VMware Workstation Pro
Server	DC01 - Windows Server 2025
Client	PC01 - Windows 11 Enterprise
Domain	home.lab
Primary Service	Active Directory Domain Services

3. Solution Architecture

3.1 Network Topology



4. Implementation

Step 1

Purpose

Create a Windows Server 2025 virtual machine that will become the domain controller.

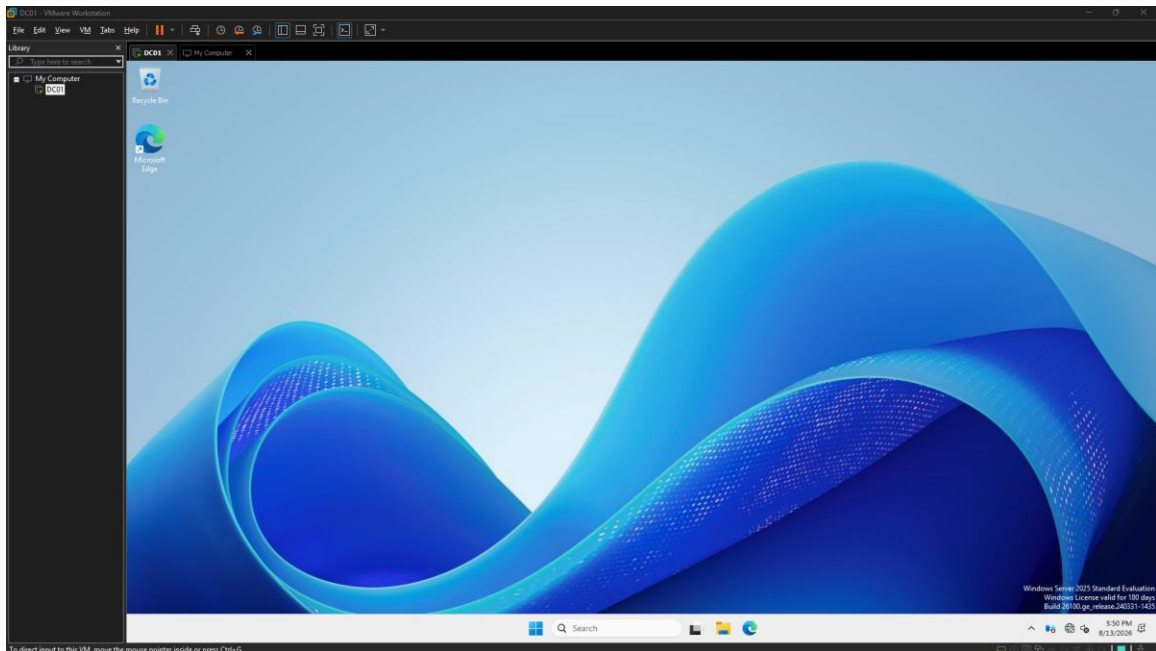
Procedure

1. Create a new virtual machine in VMware Workstation Pro.
2. Mount the Windows Server 2025 ISO.
3. Assign hardware resources.
4. Configure NAT networking.
5. Boot the virtual machine.

Verification

The virtual machine boots successfully into the Windows Server installation.

Evidence



Step 2

Purpose

Install Windows Server 2025 and complete the initial operating system configuration.

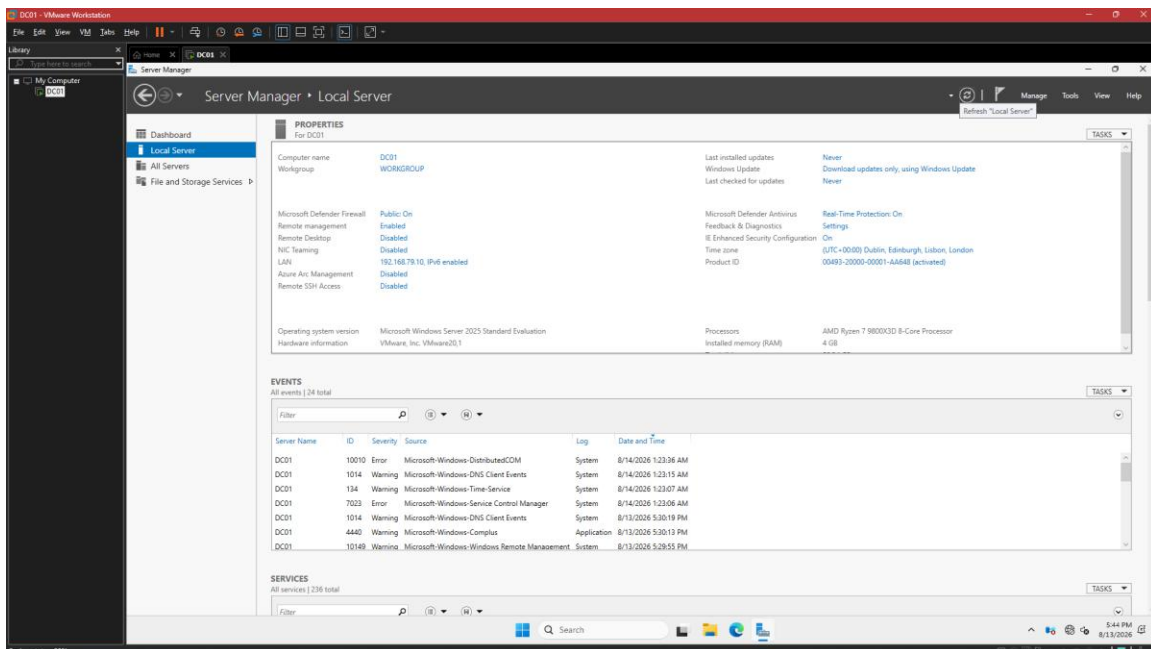
Procedure

6. Complete the Windows installation wizard.
7. Configure the local Administrator account.
8. Sign in and verify Server Manager opens.

Verification

Windows Server is installed successfully.

Evidence



Step 3

Purpose

Prepare the server for Active Directory deployment.

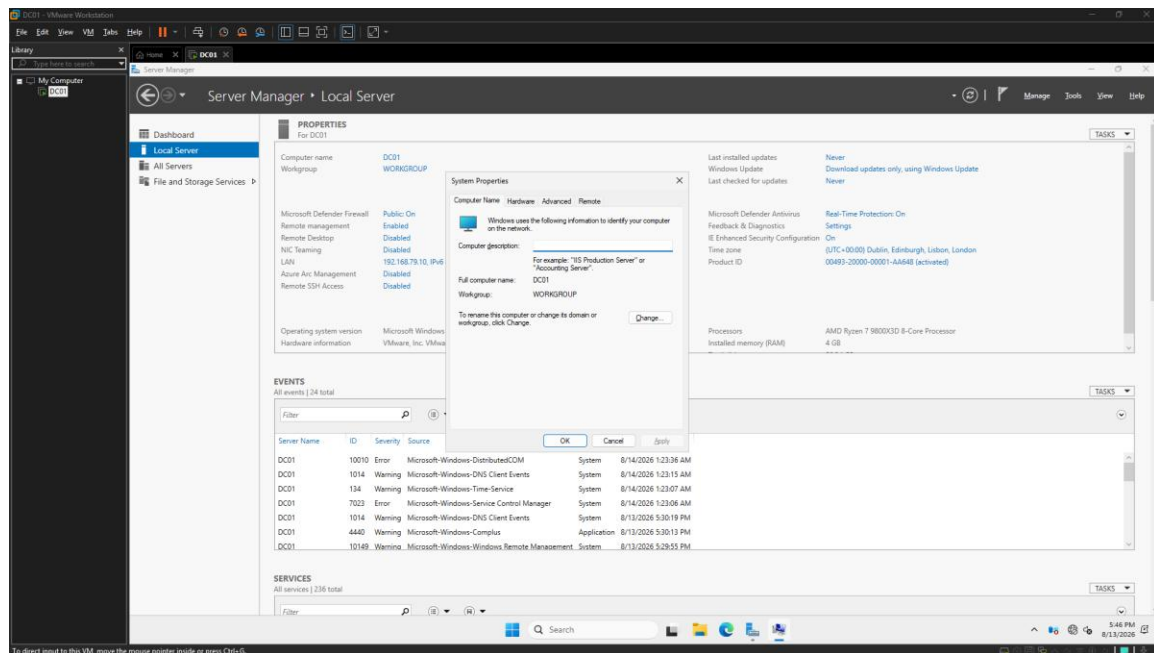
Procedure

9. Rename the server to DC01.
10. Configure a static IPv4 address.
11. Restart the server.

Verification

DC01 is configured with a static address and ready for AD DS.

Evidence



Implementation

Step 4 – Rename the Network Adapter

Purpose

Rename the default network adapter to improve administration and readability.

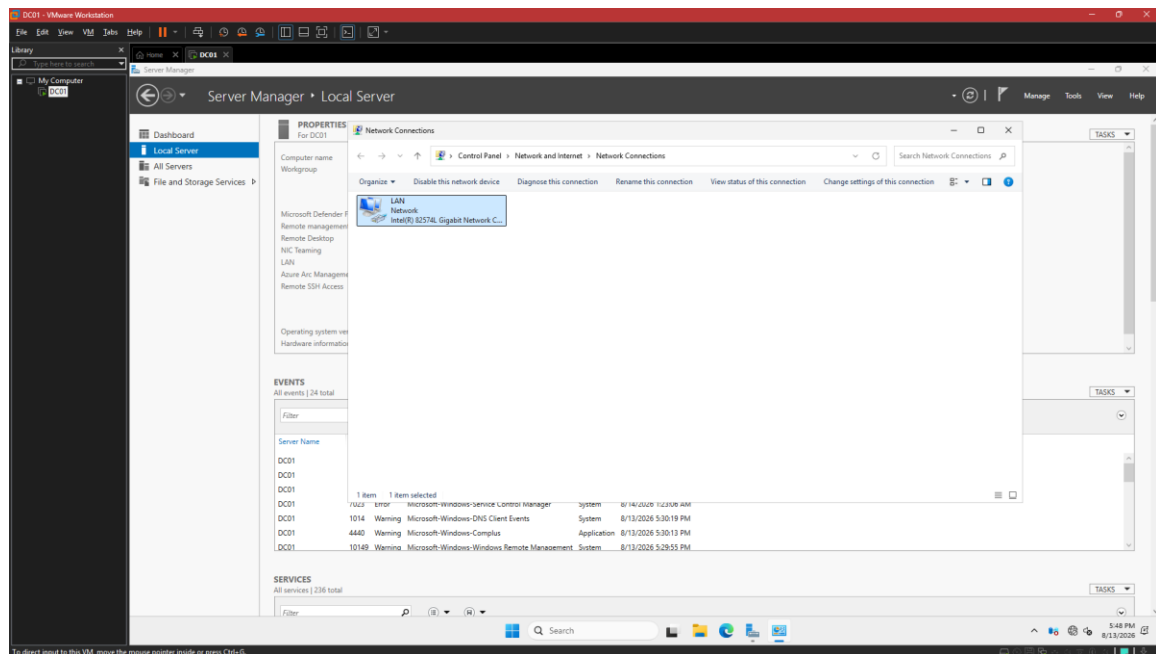
Procedure

12. Open Network Connections.
13. Rename the default Ethernet adapter to 'LAN'.
14. Verify the new adapter name is displayed.

Verification

The server network adapter is clearly identified as LAN.

Evidence



Step 5 – Configure a Static IPv4 Address

Purpose

Assign a static IP address to the domain controller before installing Active Directory.

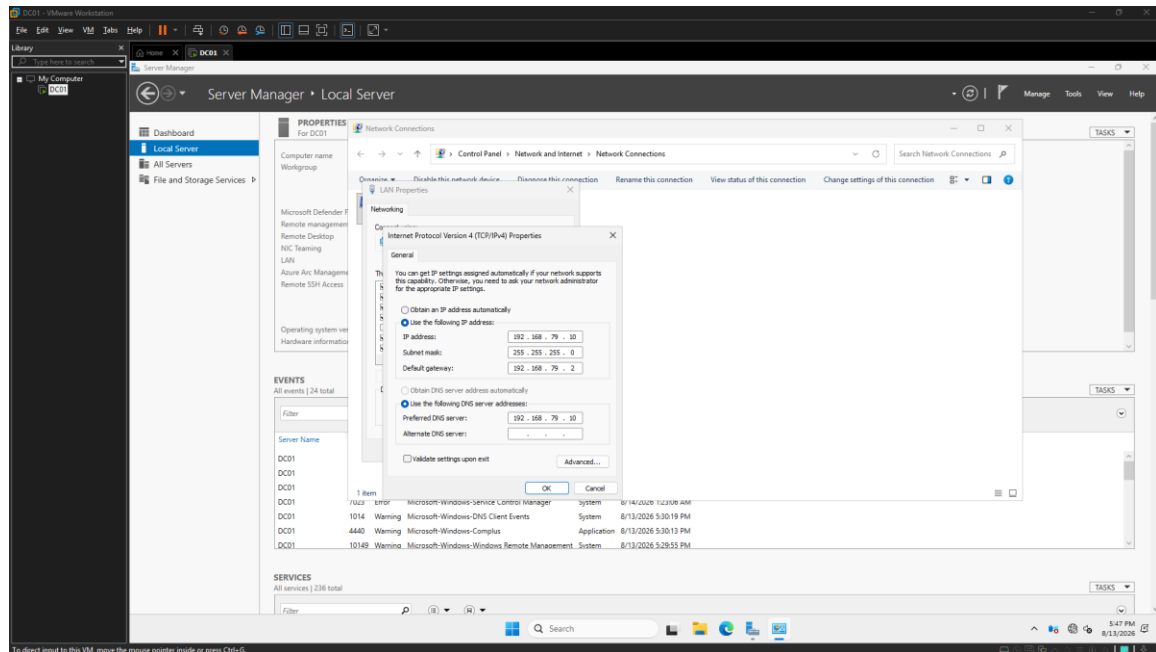
Procedure

15. Open IPv4 properties.
16. Assign 192.168.79.10/24.
17. Configure gateway 192.168.79.2.
18. Set Preferred DNS Server to 192.168.79.10.
19. Verify connectivity.

Verification

DC01 has a permanent IPv4 configuration suitable for Active Directory.

Evidence



Step 6 – Create a VMware Snapshot

Purpose

Create a recovery point before making major configuration changes.

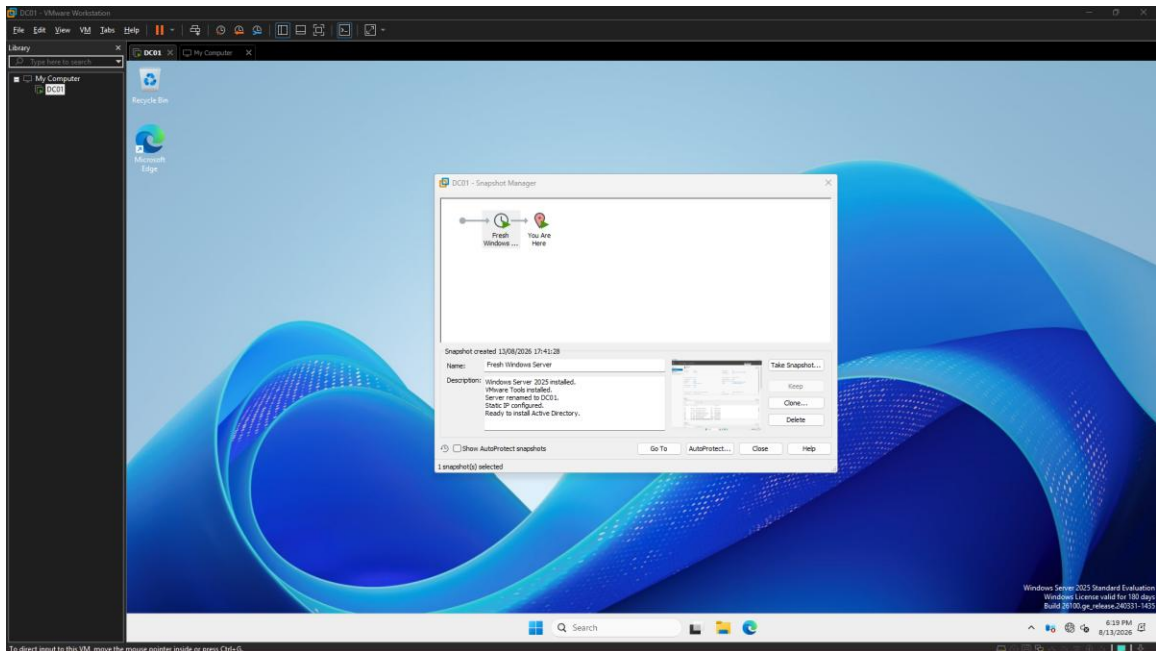
Procedure

20. Open Snapshot Manager.
21. Create a snapshot named 'Fresh Windows Server Installation'.
22. Record an appropriate description.

Verification

A restore point exists for rapid recovery if later configuration fails.

Evidence



Step 7 – Install Active Directory Domain Services

Purpose

Launch the Add Roles and Features Wizard and begin installing AD DS.

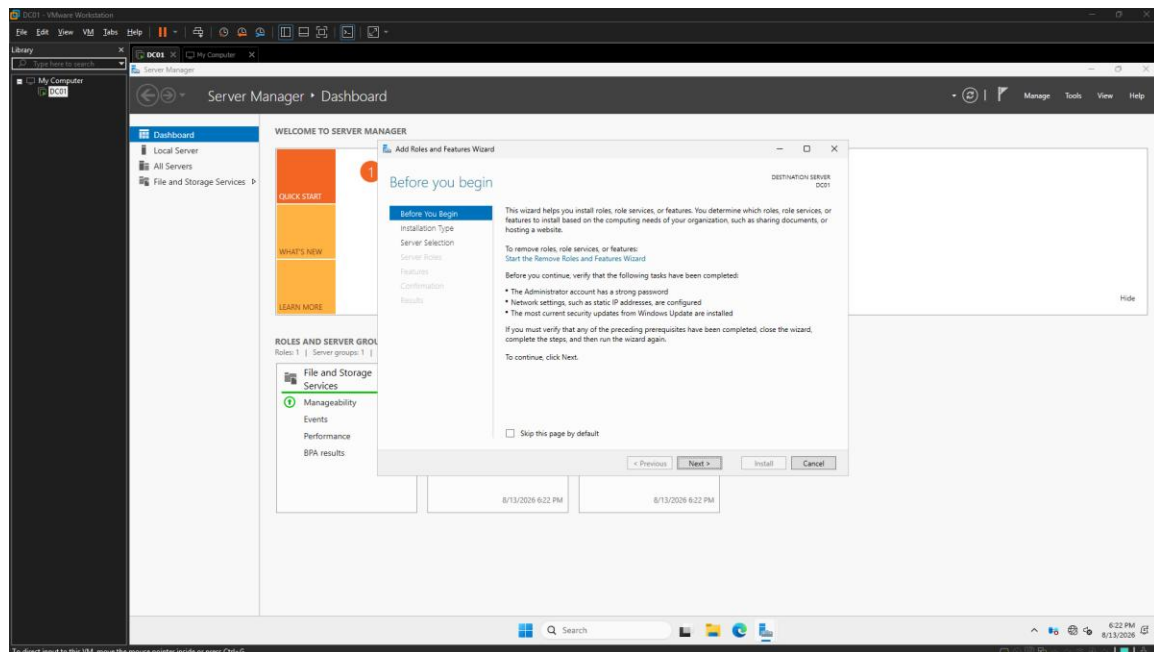
Procedure

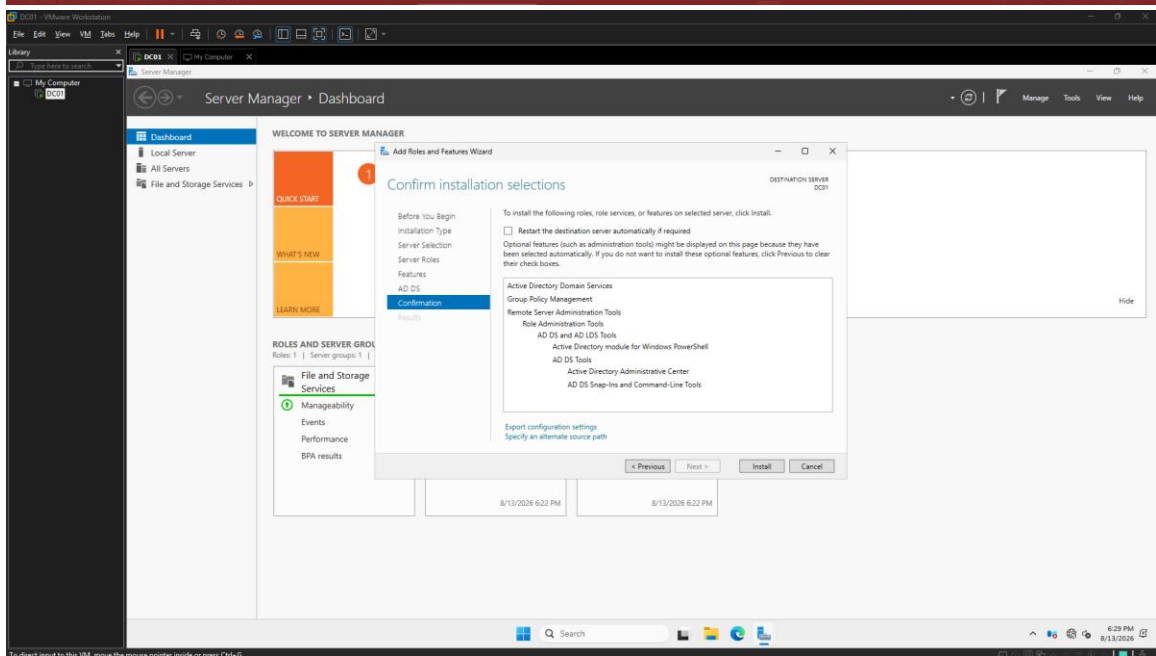
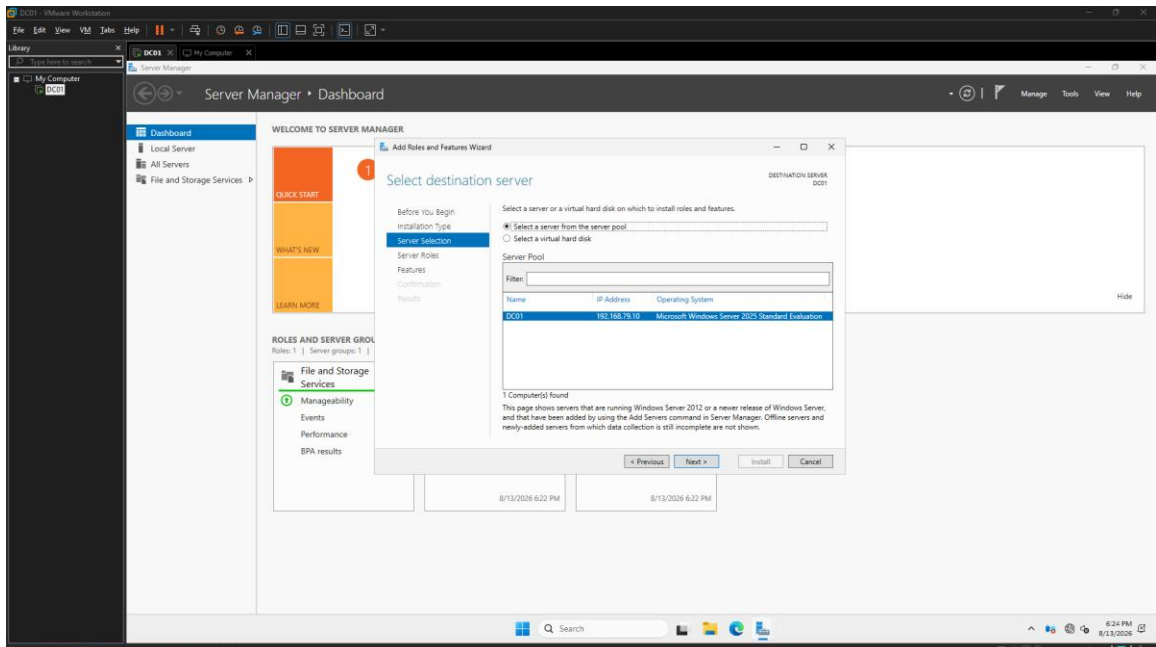
23. Open Server Manager.
24. Select Manage > Add Roles and Features.
25. Choose Role-based installation.
26. Select Active Directory Domain Services.
27. Accept the required management tools and continue.

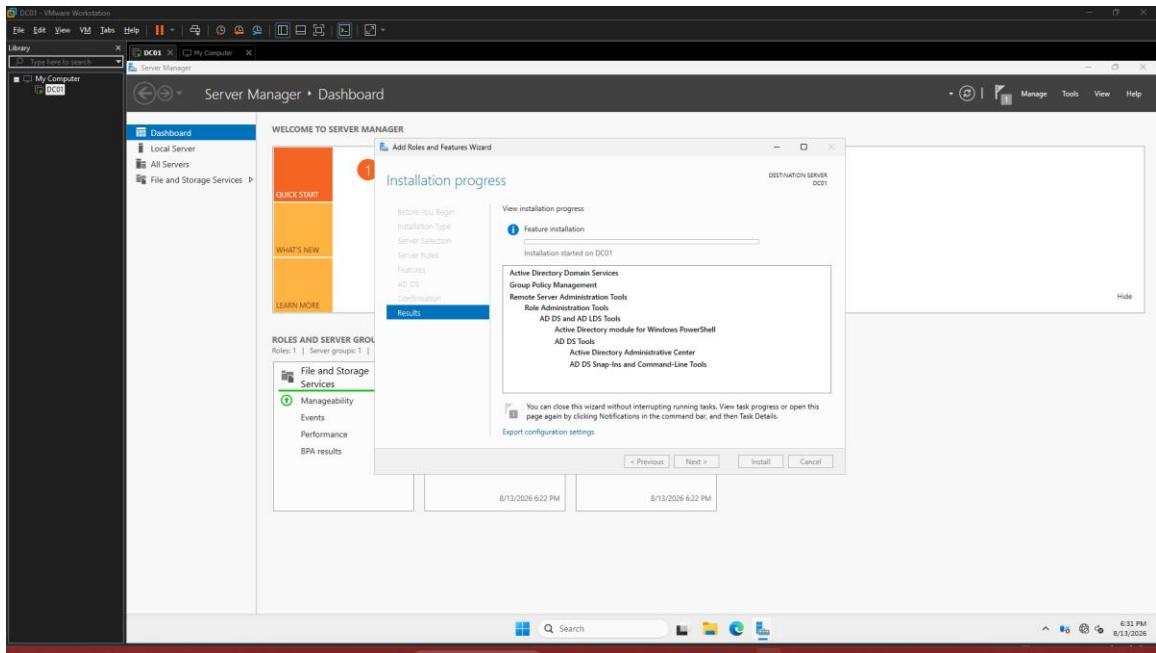
Verification

AD DS installation completes successfully and the server is ready for promotion.

Evidence







Step 8 – Promote the Server to a Domain Controller

Purpose

Create a new Active Directory forest and promote DC01 to the first domain controller.

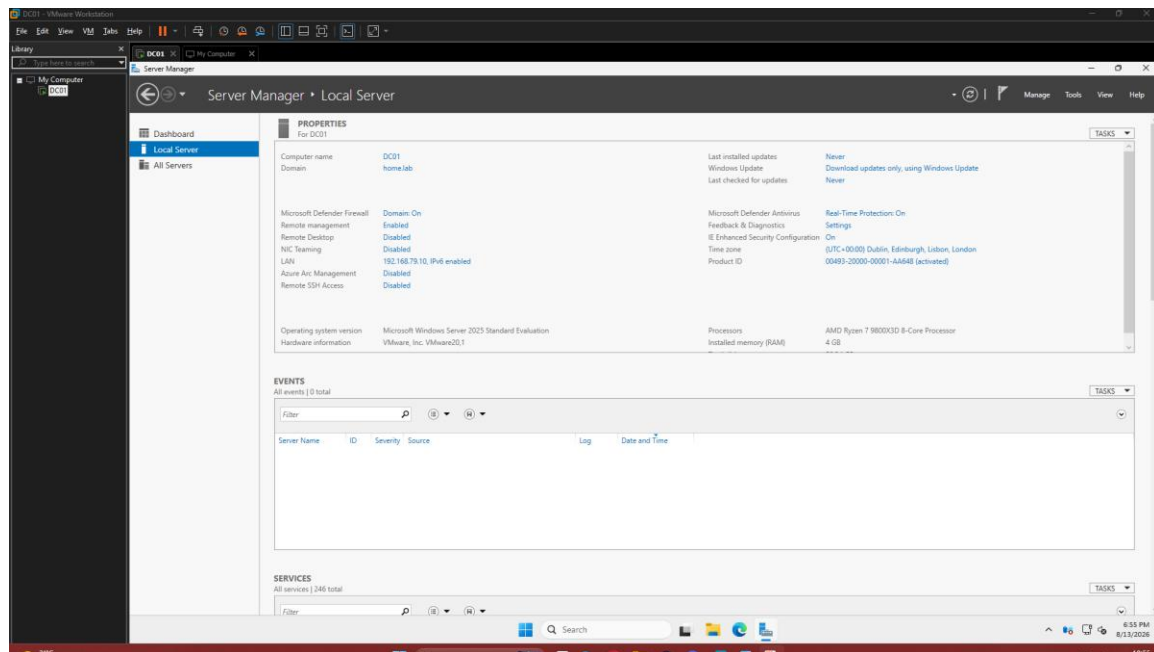
Procedure

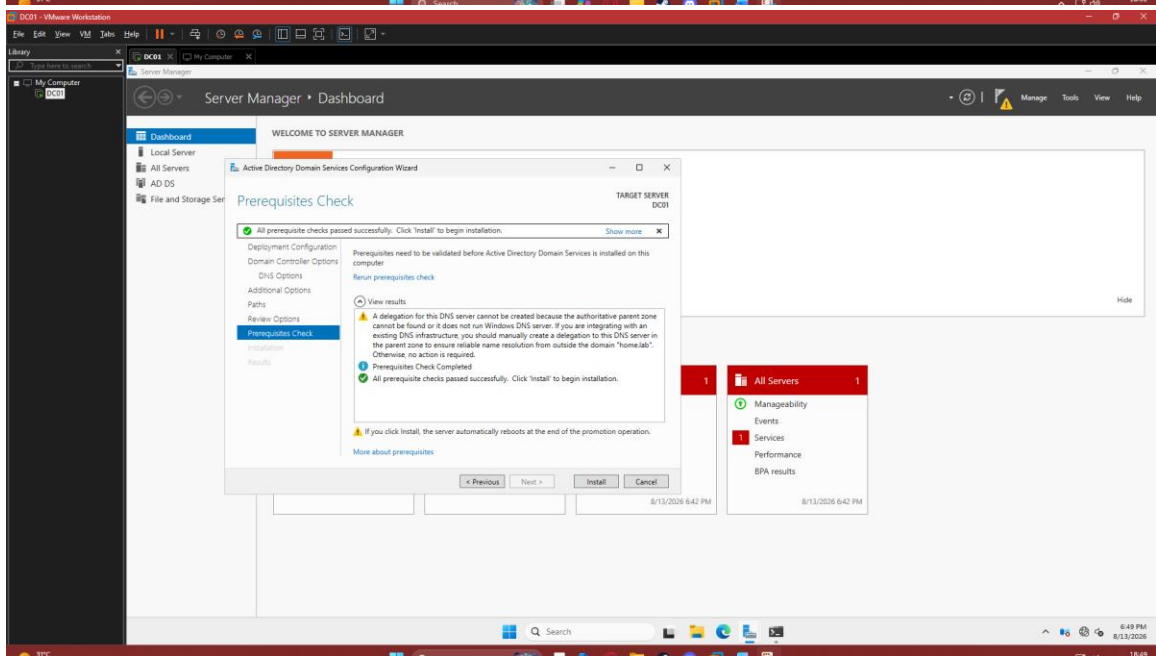
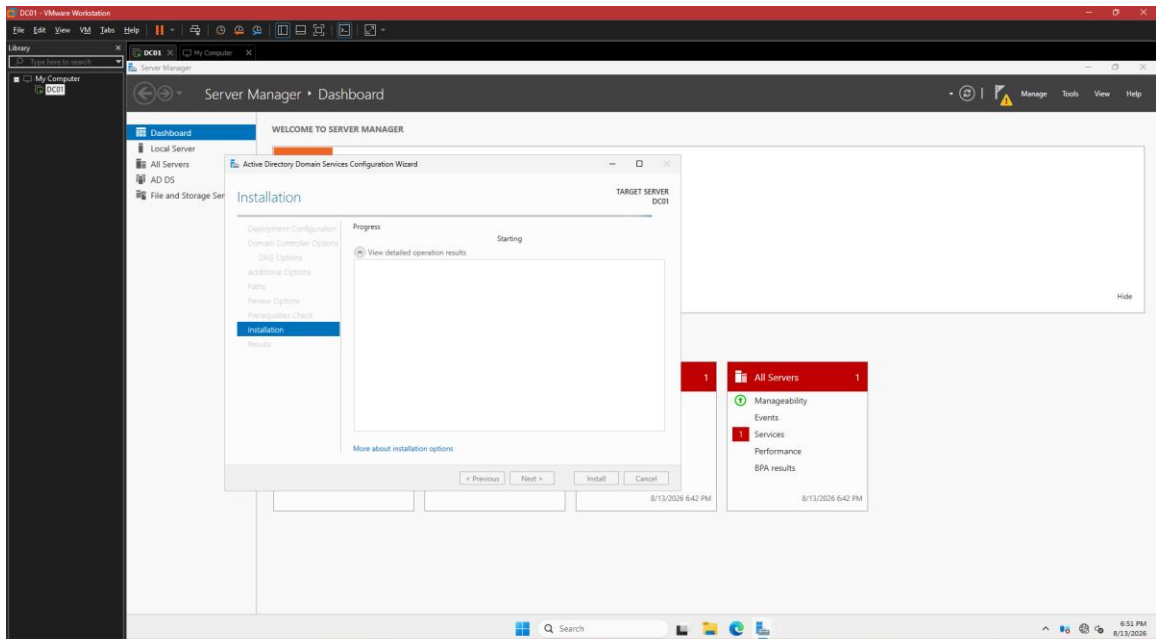
28. Launch the AD DS Configuration Wizard.
29. Create a new forest named home.lab.
30. Configure Domain Controller options and DSRM password.
31. Accept the default database and SYSVOL paths.
32. Run prerequisite checks.
33. Complete the installation and allow the server to restart.

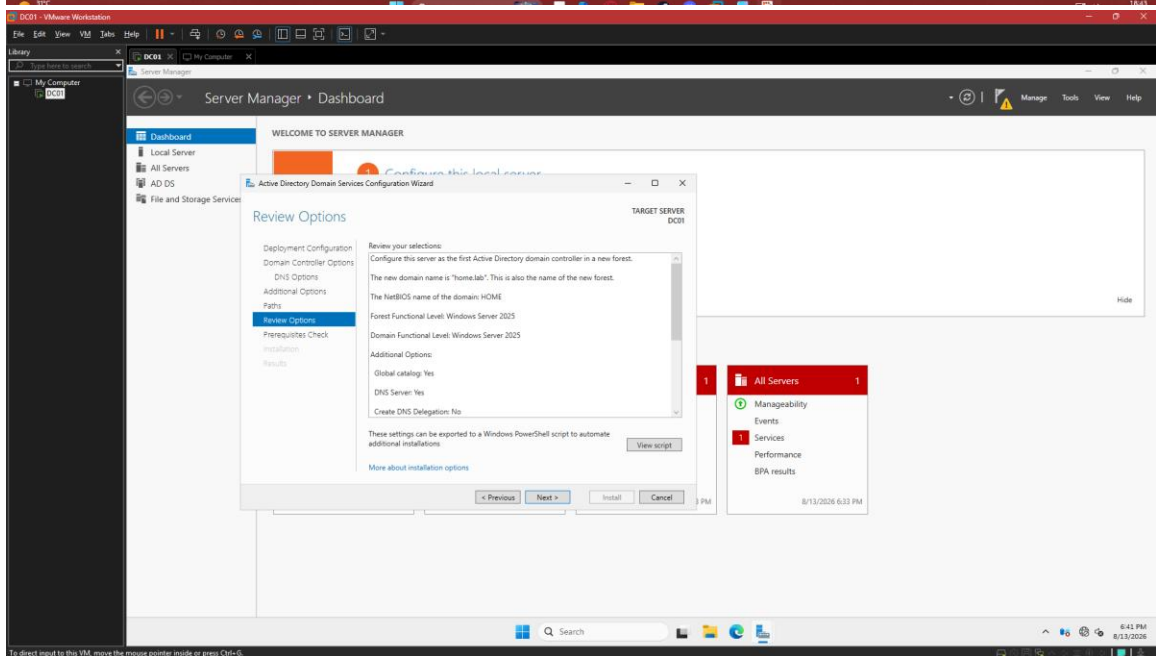
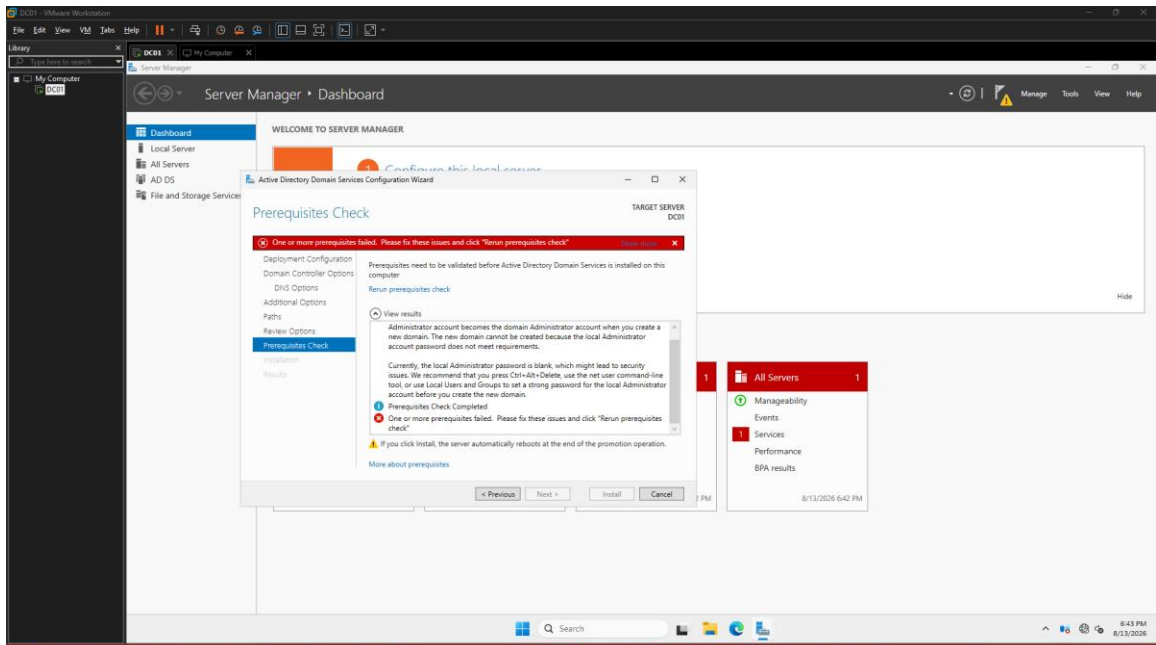
Verification

DC01 is promoted successfully and becomes the first Domain Controller for the home.lab forest.

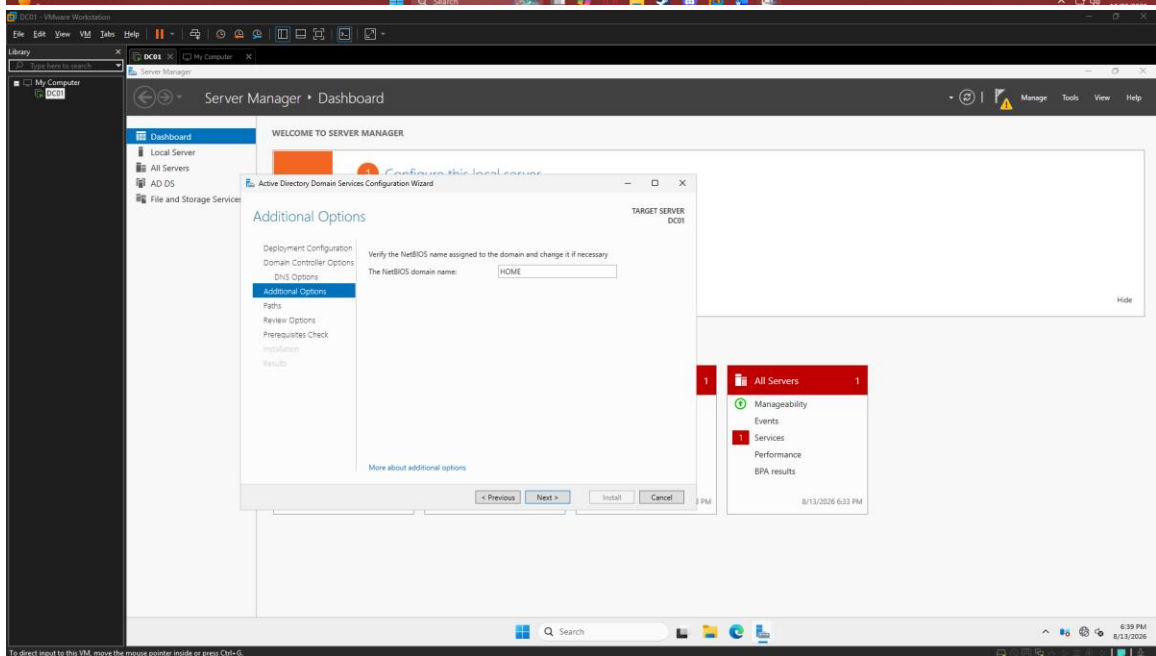
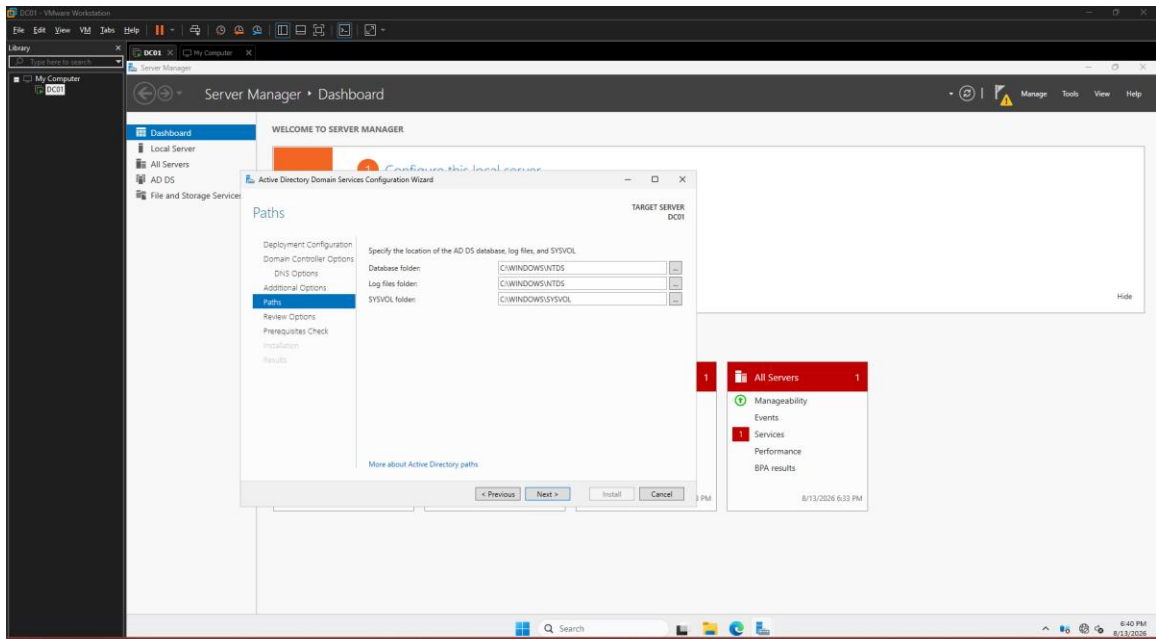
Evidence



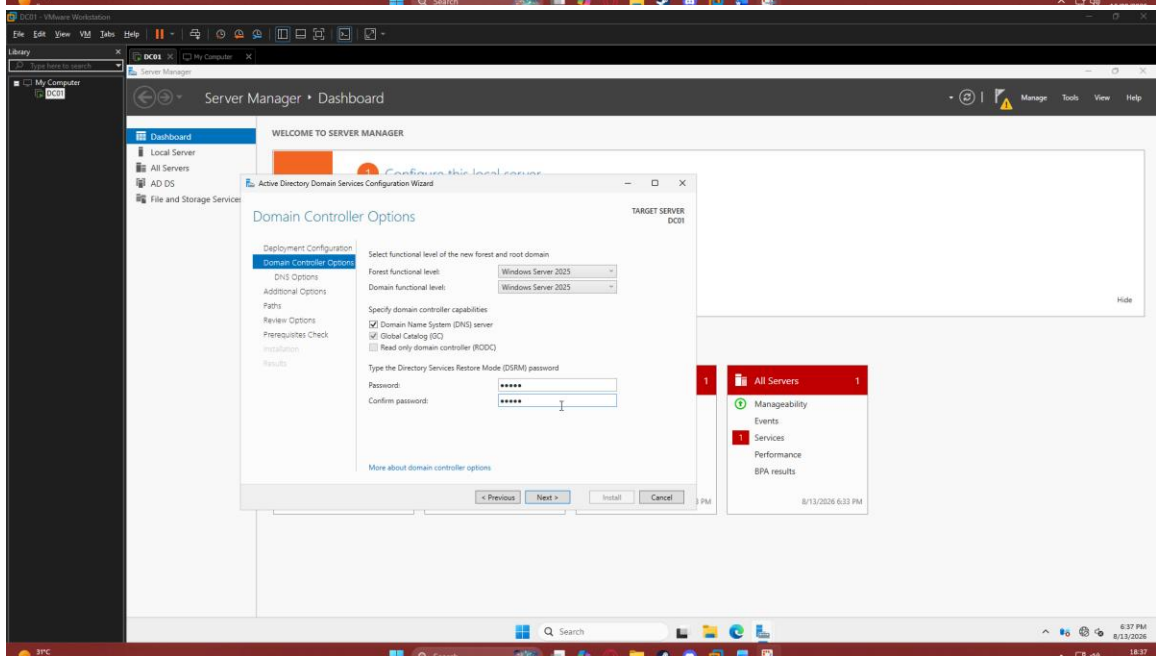
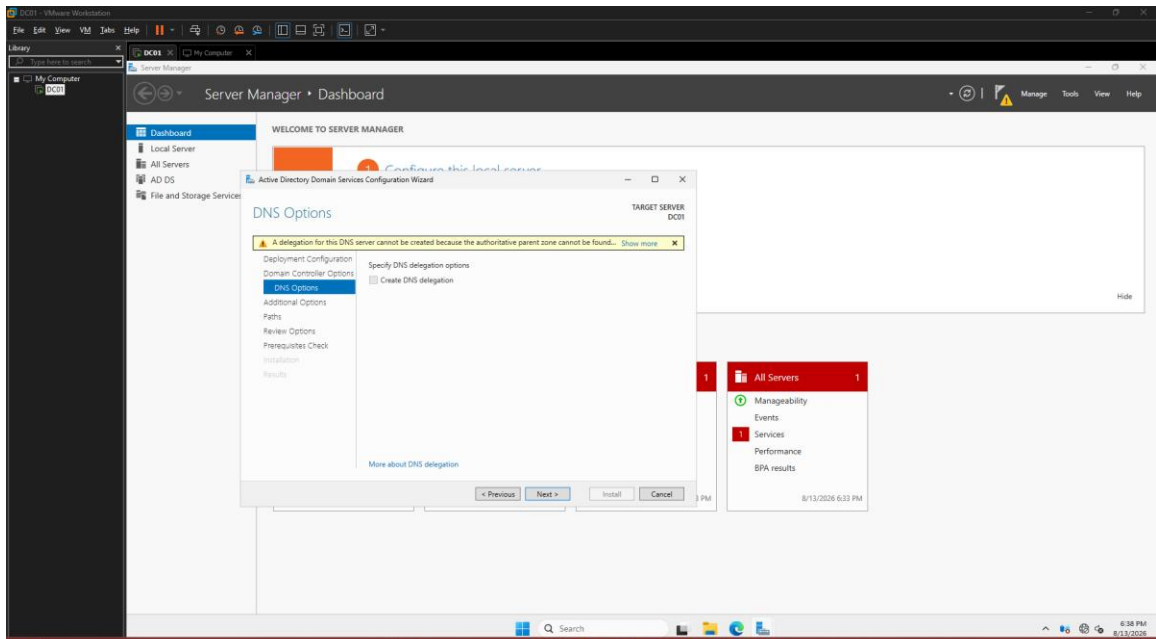


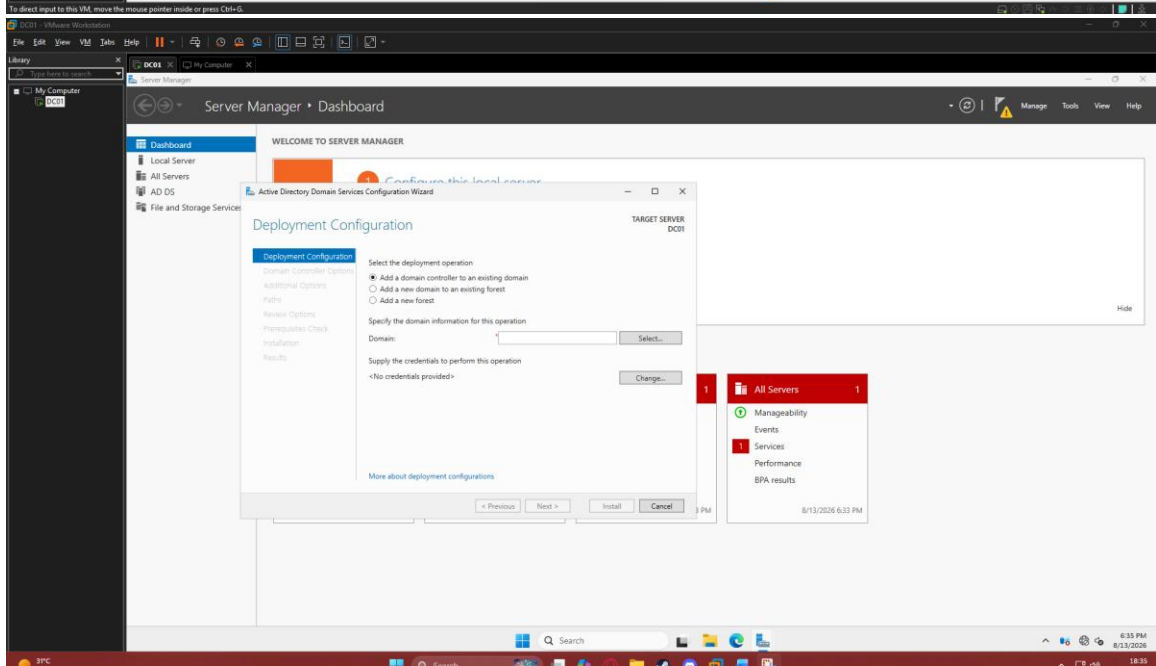
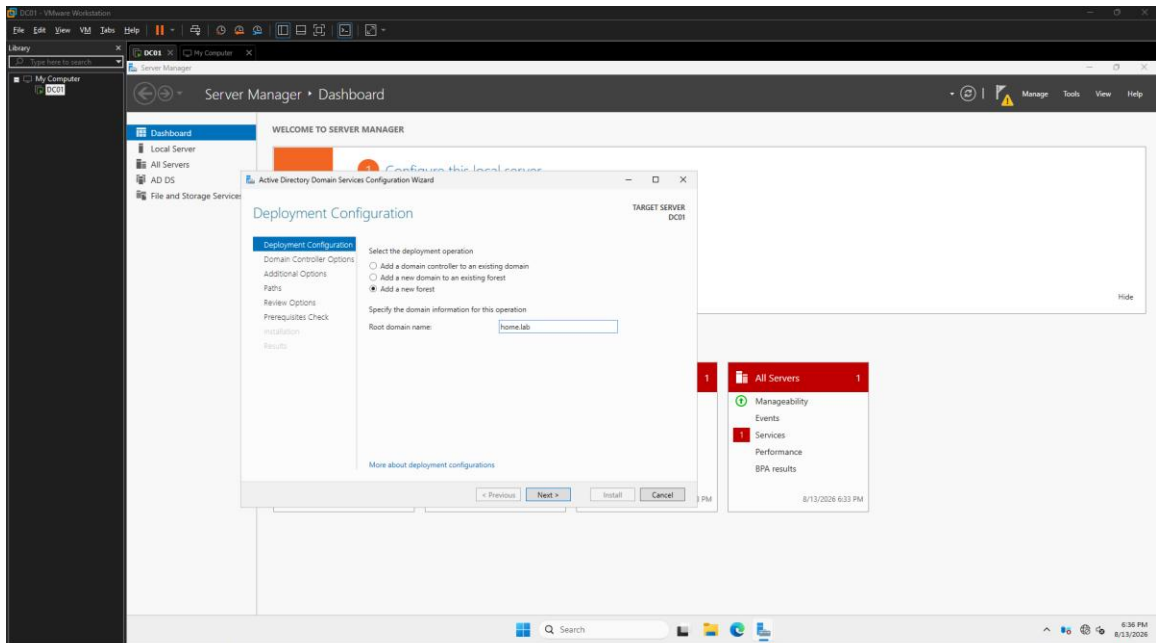


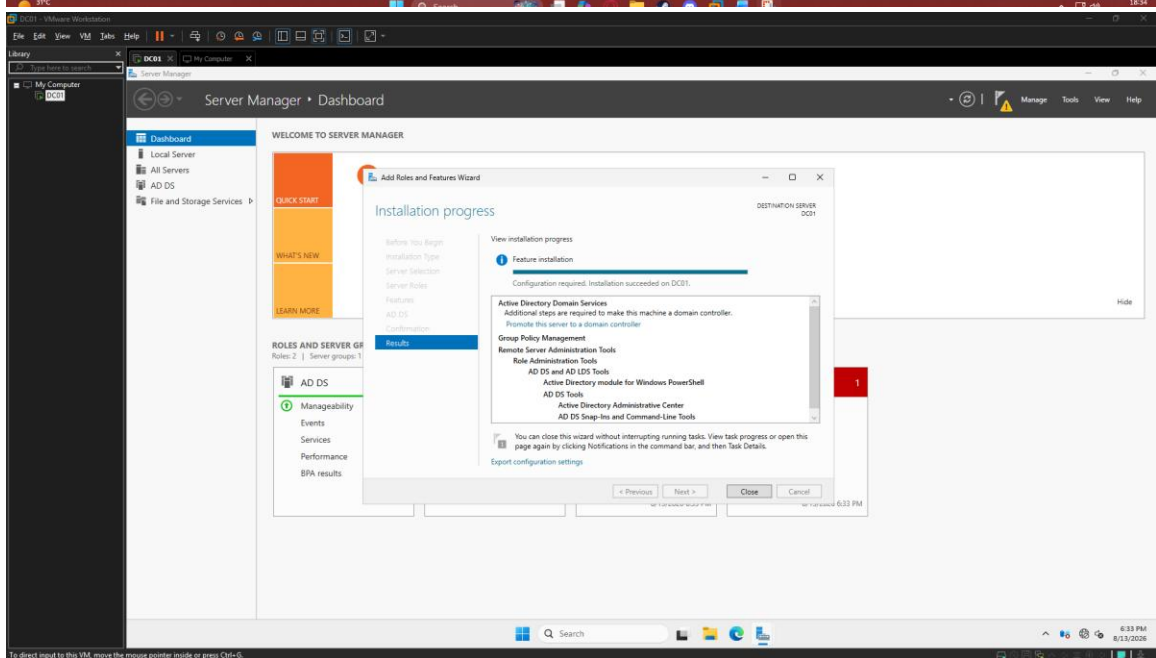
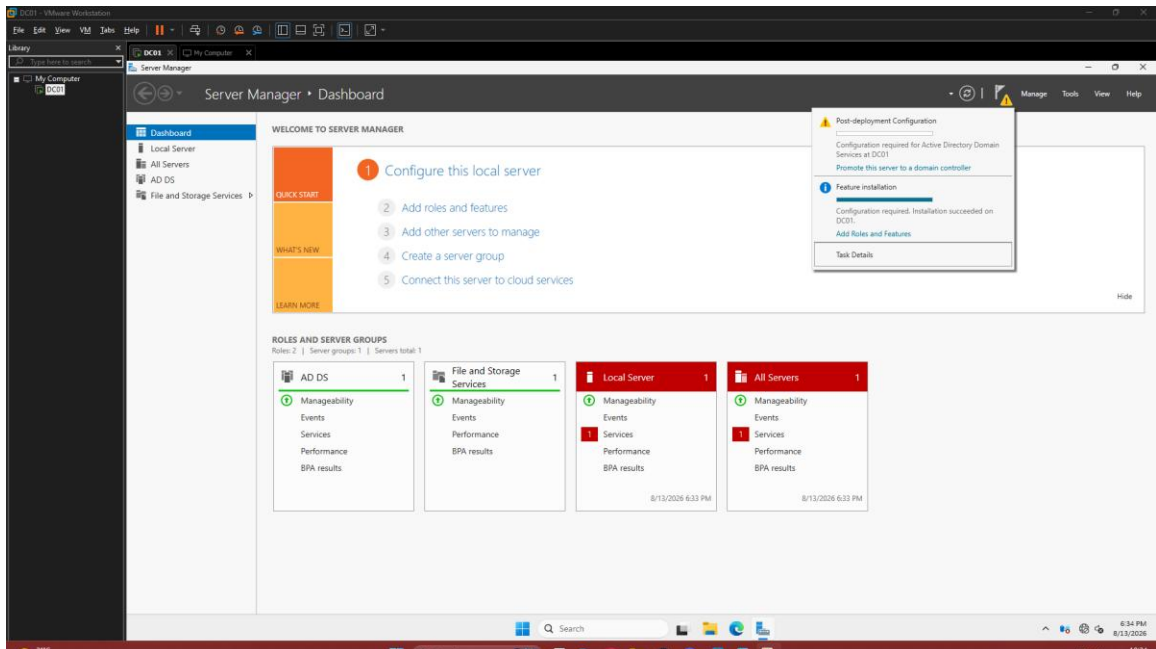
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



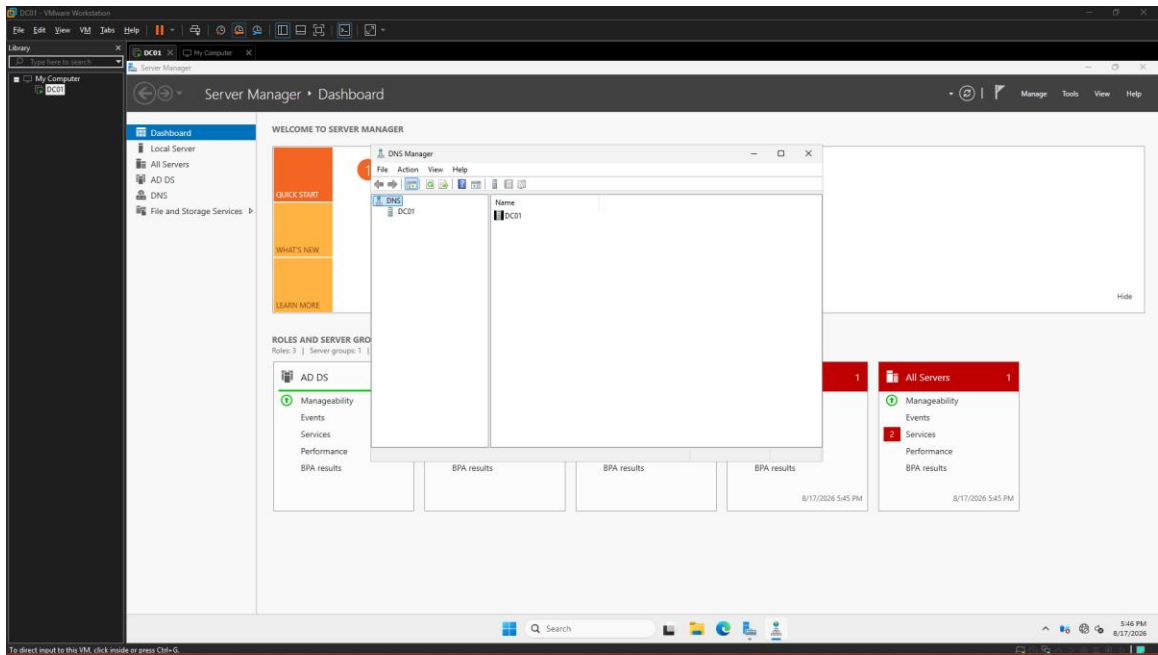
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.







To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



Deployment Notes

This deployment intentionally follows enterprise administration practices by assigning a static IP address before installing Active Directory, using a dedicated DNS server, creating restore points before major changes, and validating each stage before progressing to the next. These practices reduce troubleshooting time and reflect common procedures used in production Windows Server environments.

Implementation

Step 9 – Verify AD DS Installation

Purpose

Confirm that AD DS and supporting management tools were installed successfully.

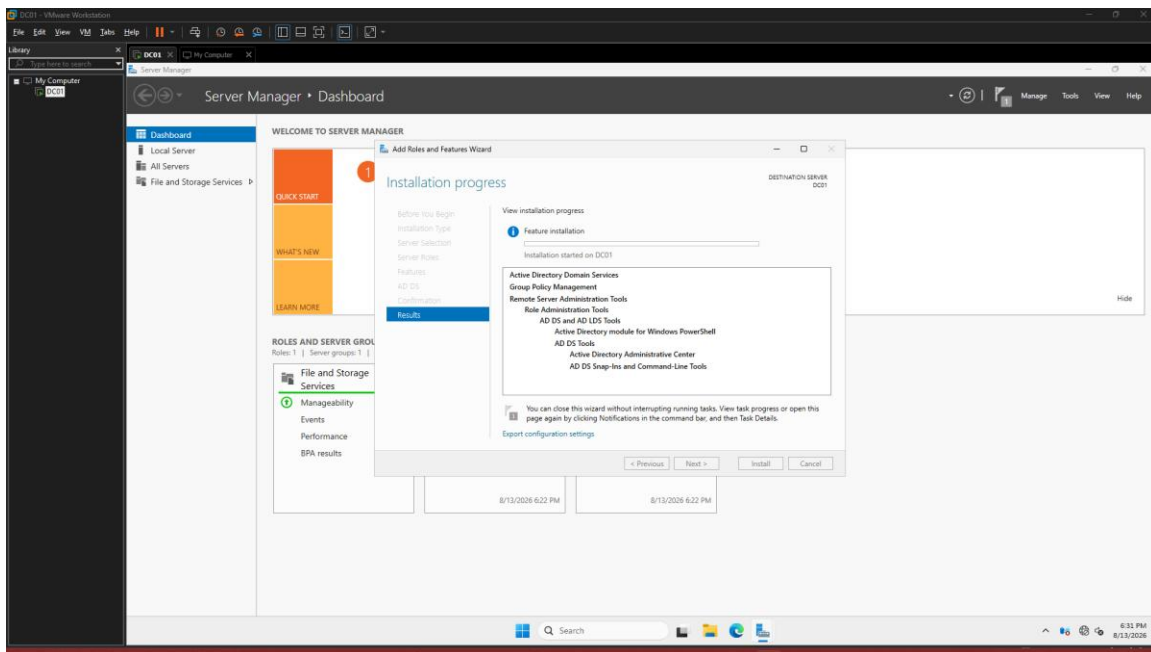
Procedure

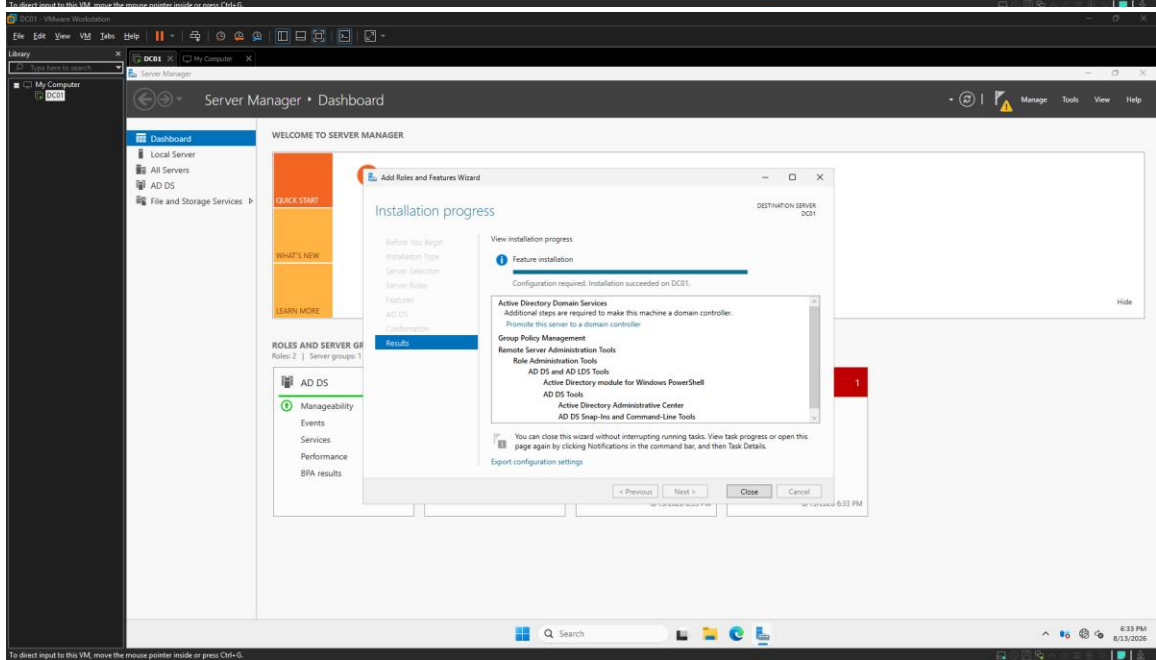
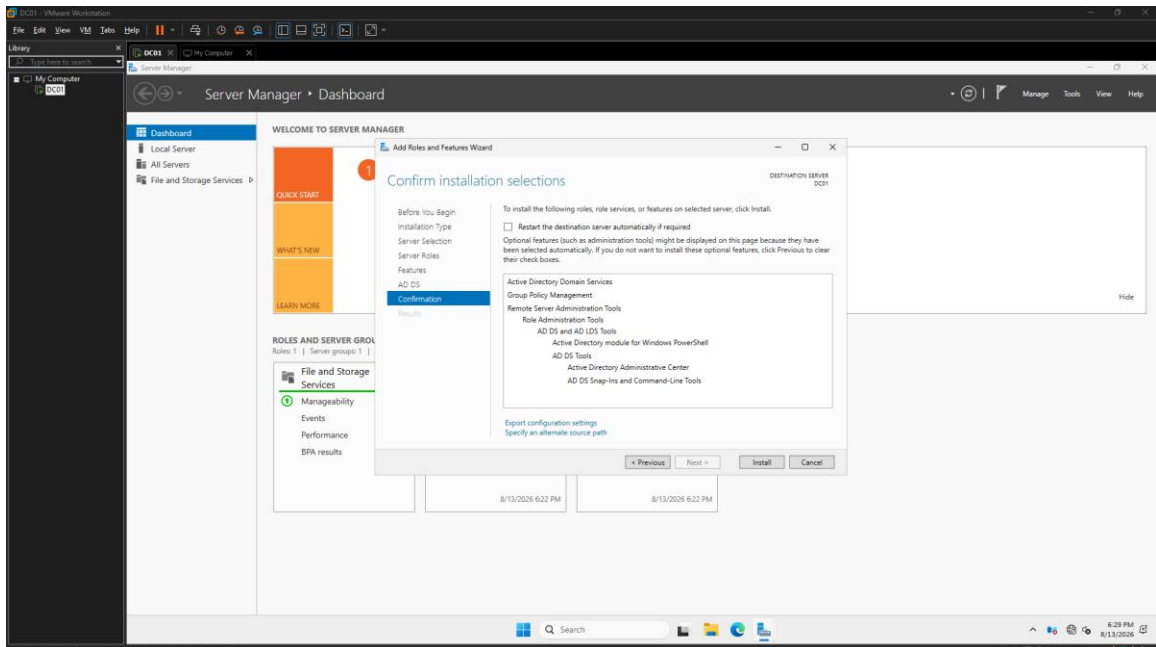
34. Review the installation summary.
35. Confirm AD DS installation completed successfully.
36. Verify the 'Promote this server to a domain controller' notification appears.

Verification

The server is ready to be promoted to the first Domain Controller.

Evidence





Step 10 – Create the Active Directory Forest

Purpose

Deploy a new Active Directory forest named home.lab.

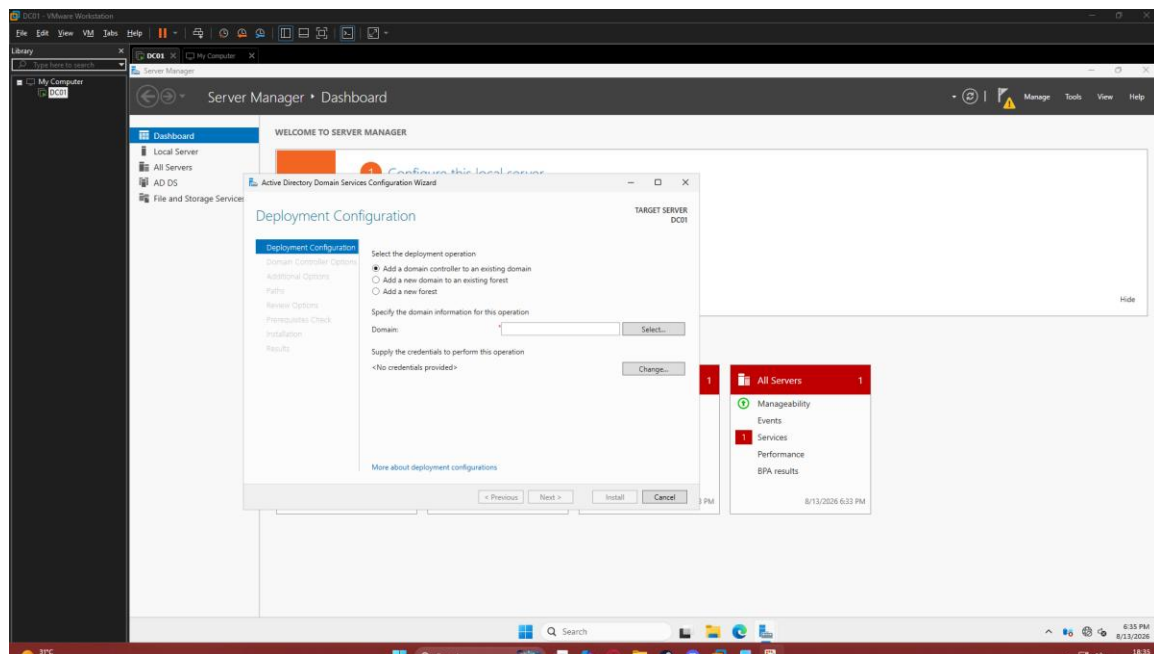
Procedure

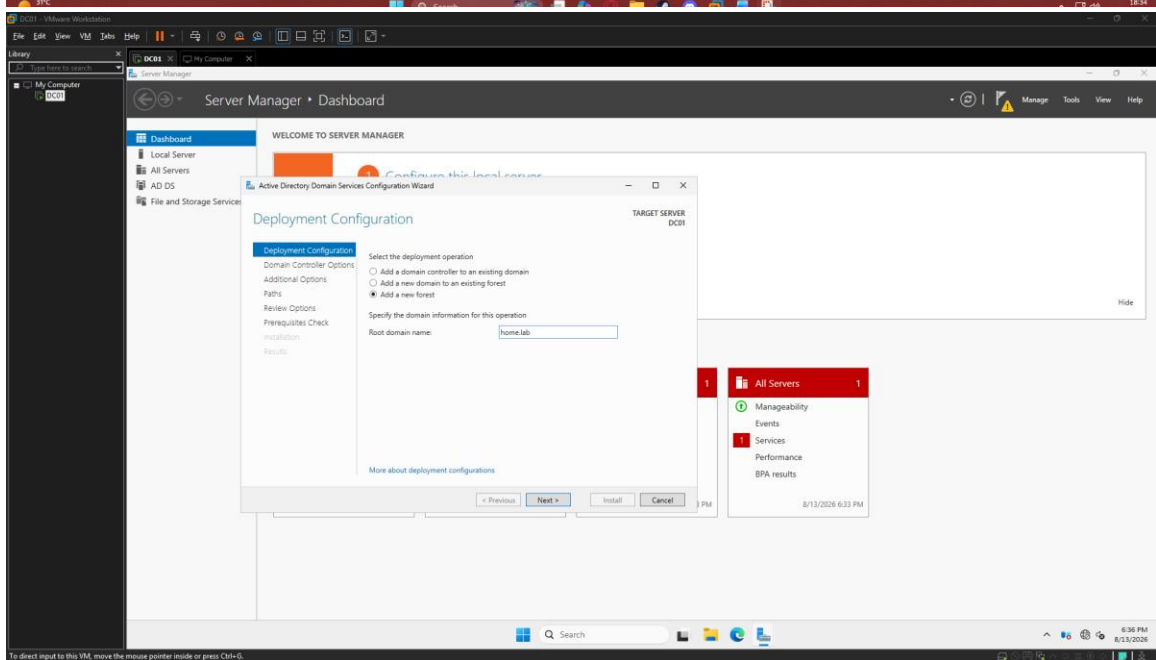
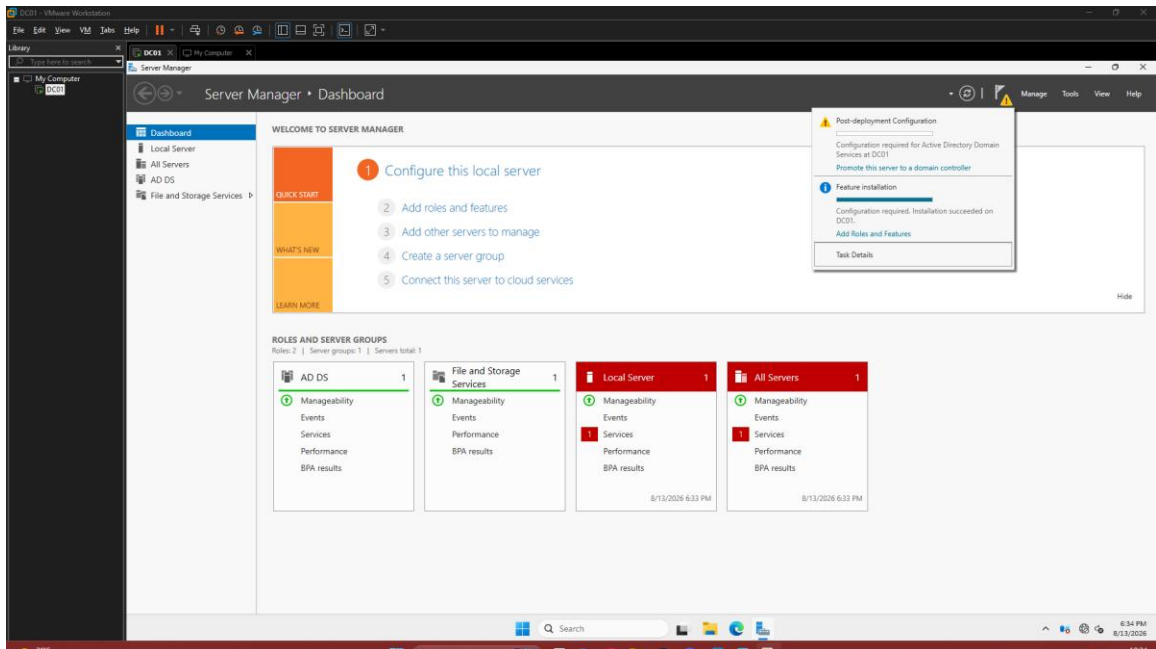
37. Launch the AD DS Configuration Wizard.
38. Select 'Add a new forest'.
39. Enter 'home.lab' as the root domain name.
40. Review deployment settings.

Verification

The new Active Directory forest is configured and ready for deployment.

Evidence





To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Step 11 – Configure Domain Controller Options

Purpose

Configure functional levels, DNS and recovery settings.

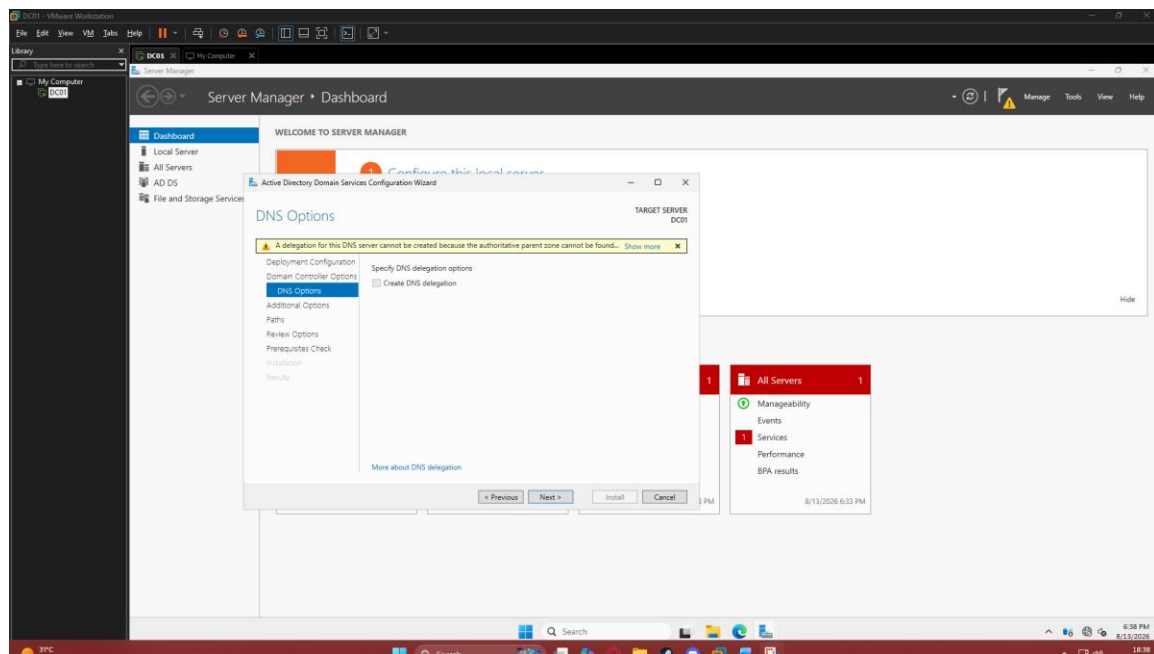
Procedure

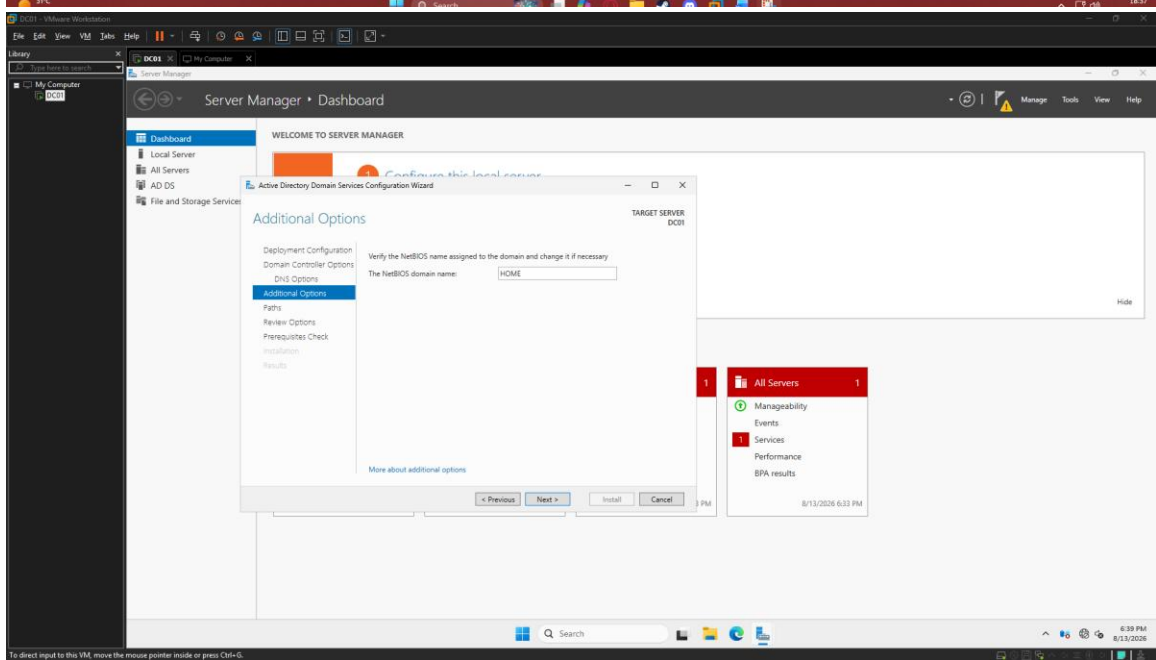
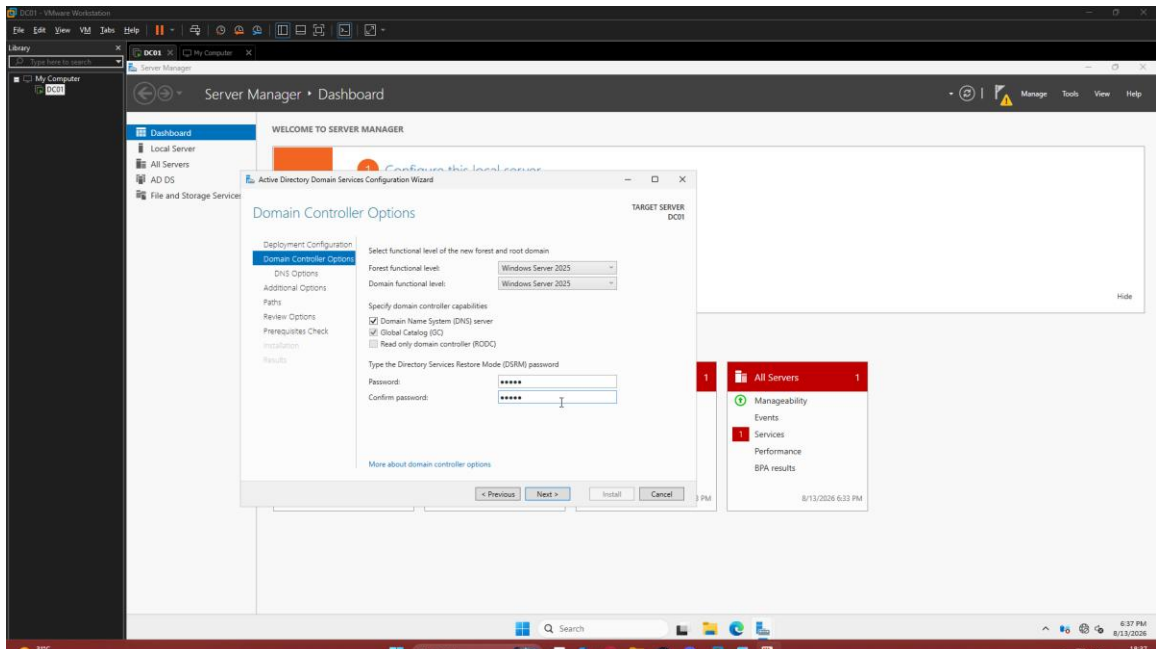
41. Select the Forest and Domain Functional Levels.
42. Enable the DNS Server role.
43. Configure the Directory Services Restore Mode (DSRM) password.
44. Review additional domain controller settings.

Verification

The domain controller configuration satisfies the deployment requirements.

Evidence





To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Step 12 – Review Database Paths

Purpose

Review storage locations for the AD database, log files and SYSVOL.

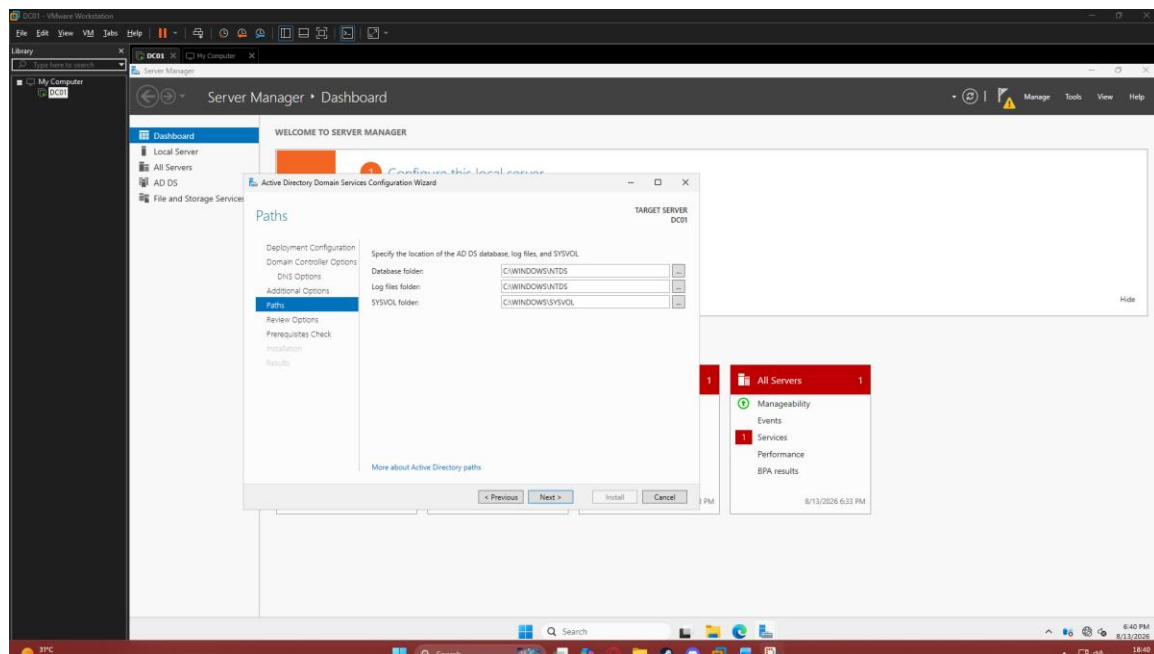
Procedure

45. Review the default NTDS database location.
46. Review transaction log location.
47. Review SYSVOL location.
48. Accept the default paths.

Verification

The default Microsoft-recommended storage locations are retained.

Evidence



Step 13 – Validate the Deployment

Purpose

Review the deployment summary and perform prerequisite checks.

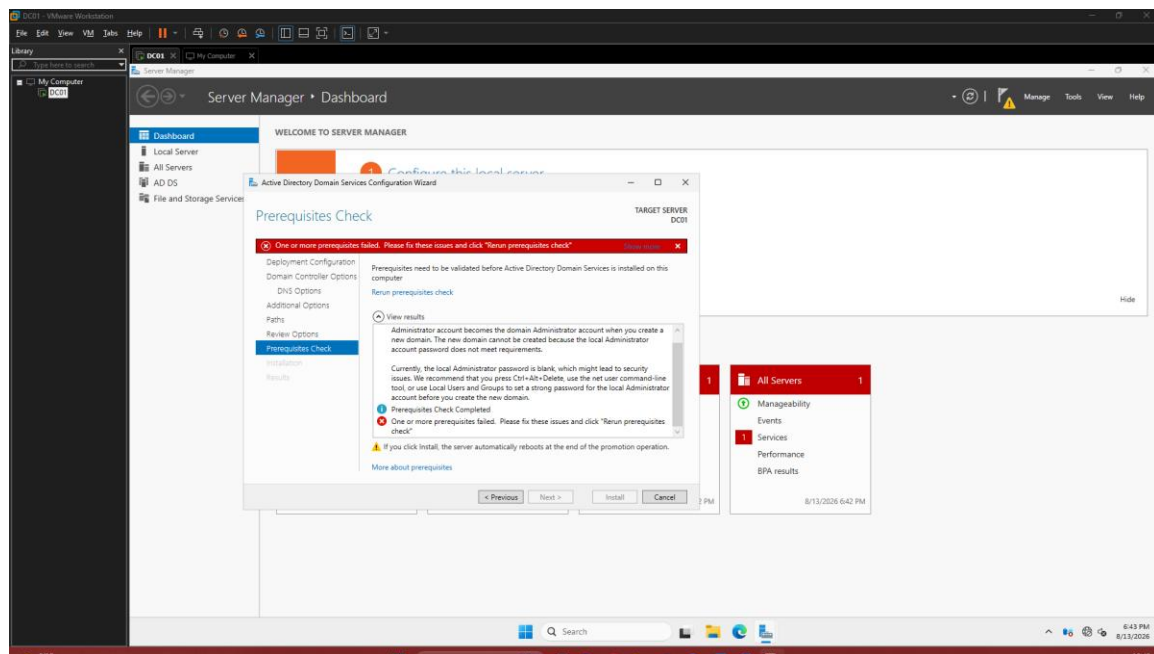
Procedure

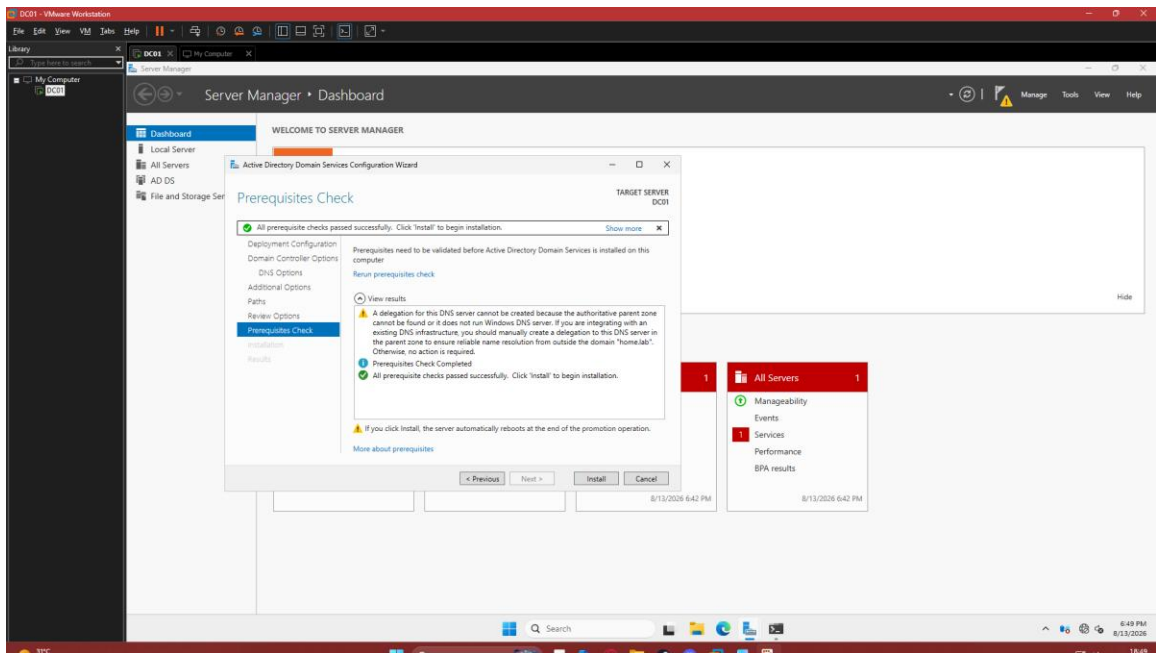
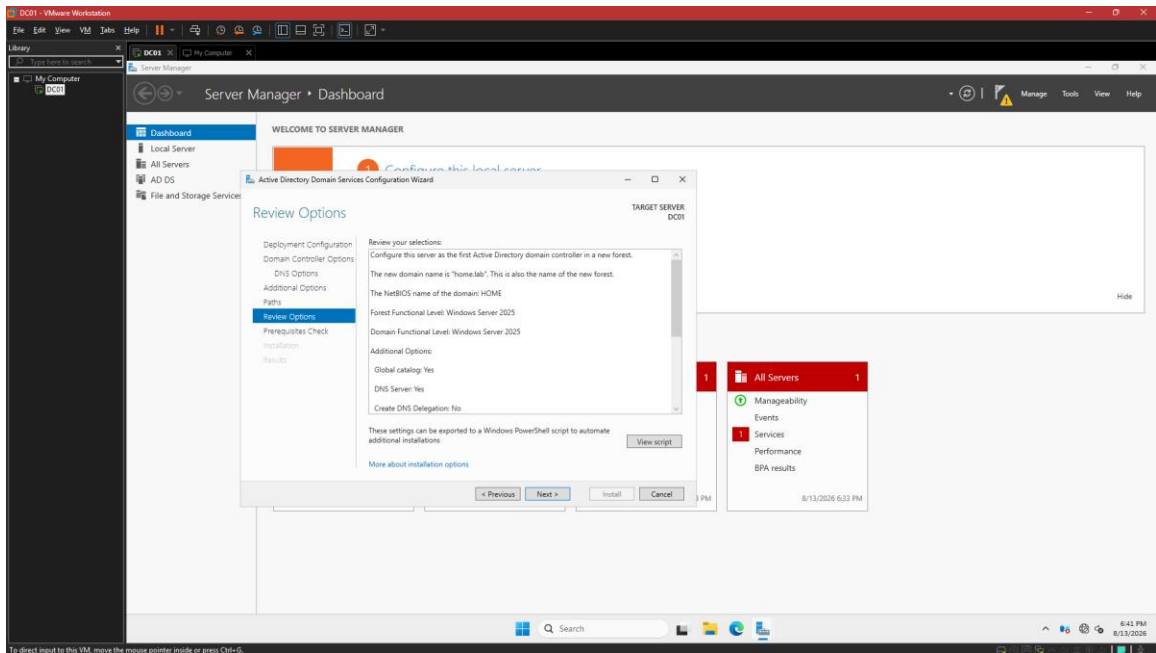
49. Review the configuration summary.
50. Run prerequisite validation.
51. Resolve any reported issues.
52. Confirm the deployment is ready.

Verification

All prerequisite checks pass successfully before installation.

Evidence





Step 14 – Complete Domain Controller Promotion

Purpose

Install Active Directory and restart the server.

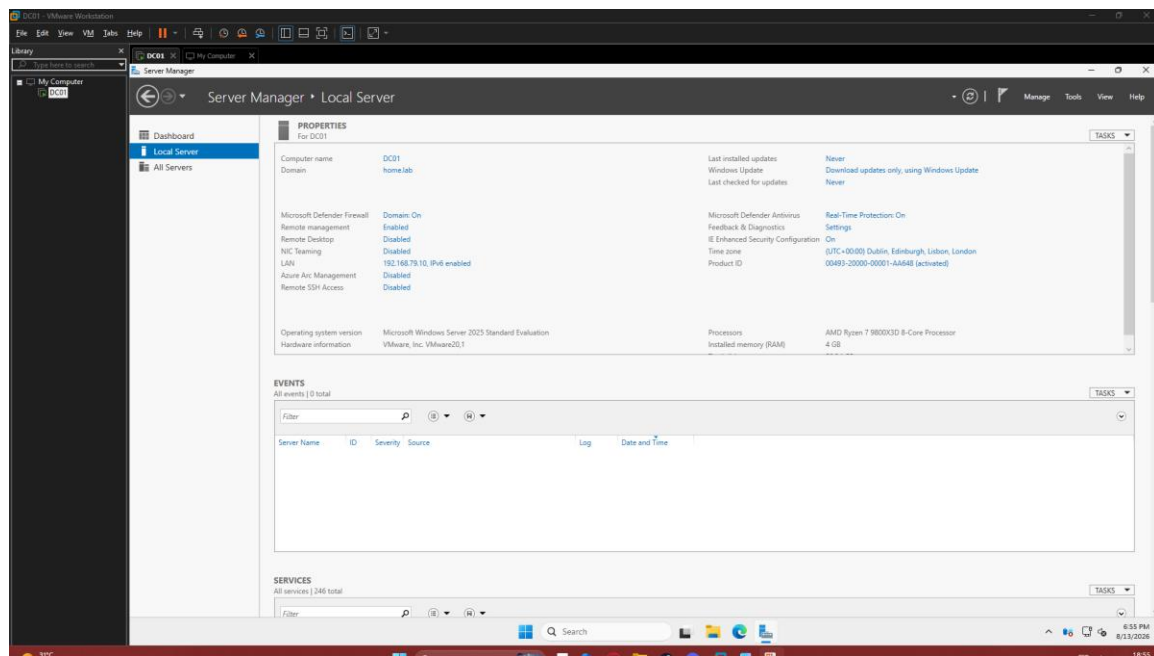
Procedure

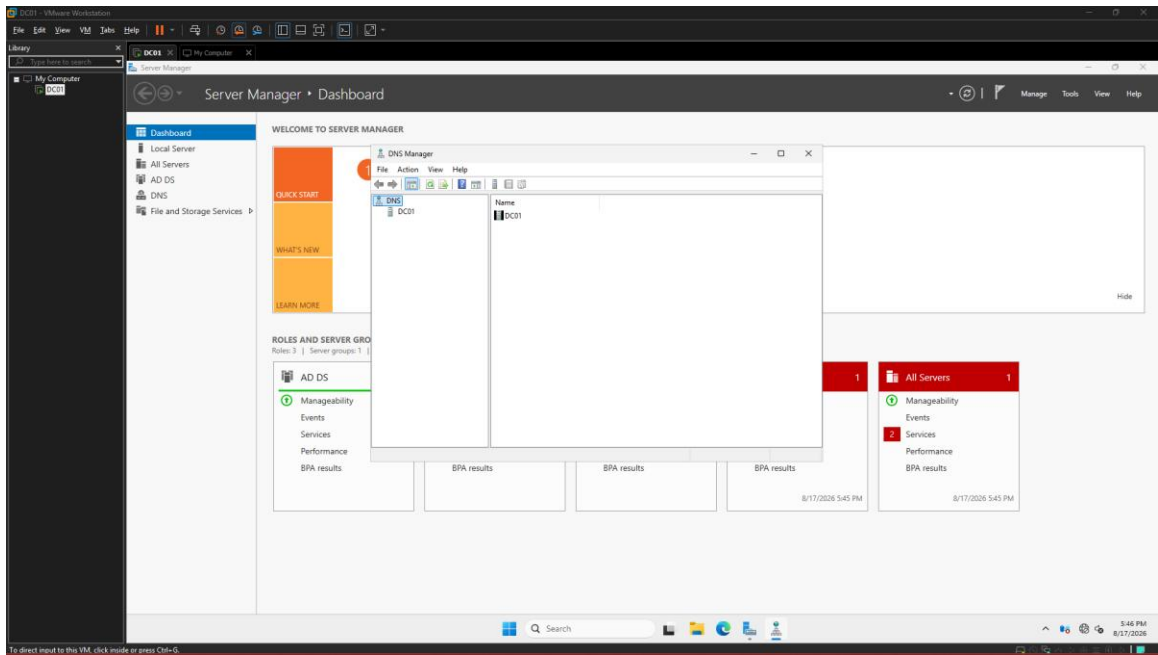
53. Start the installation.
54. Allow the server to restart automatically.
55. Sign in using the domain administrator account.
56. Open Server Manager and verify the domain.

Verification

DC01 is successfully promoted as the first Domain Controller for the home.lab forest.

Evidence





Step 15 – Verify DNS Configuration

Purpose

Confirm that DNS zones and records were created automatically.

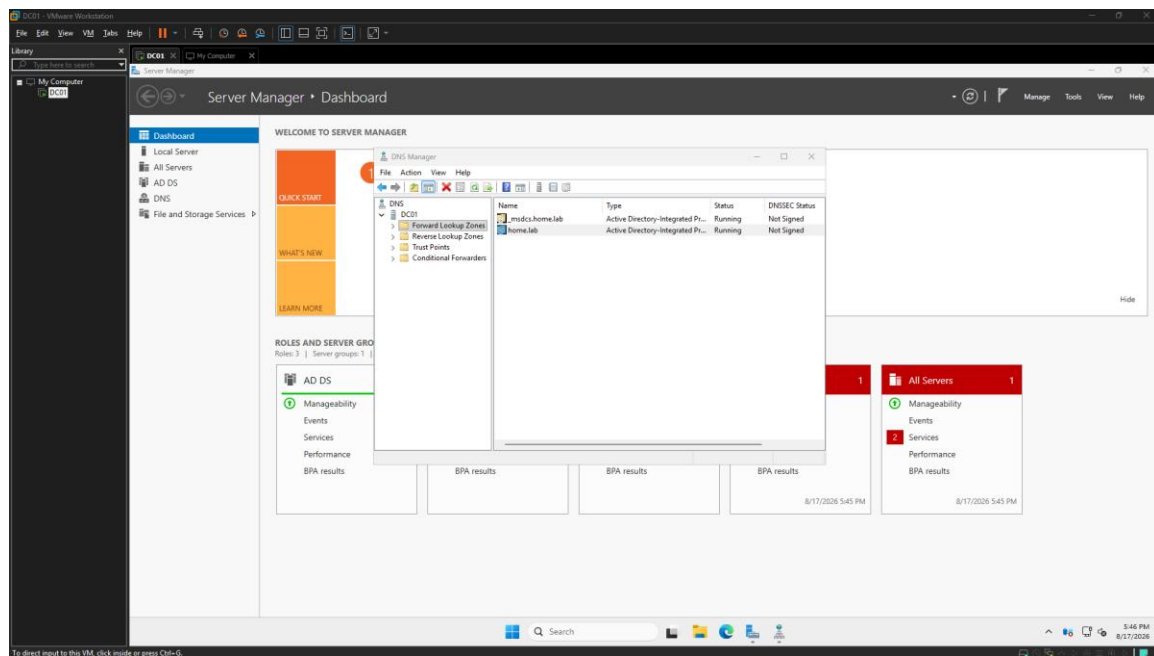
Procedure

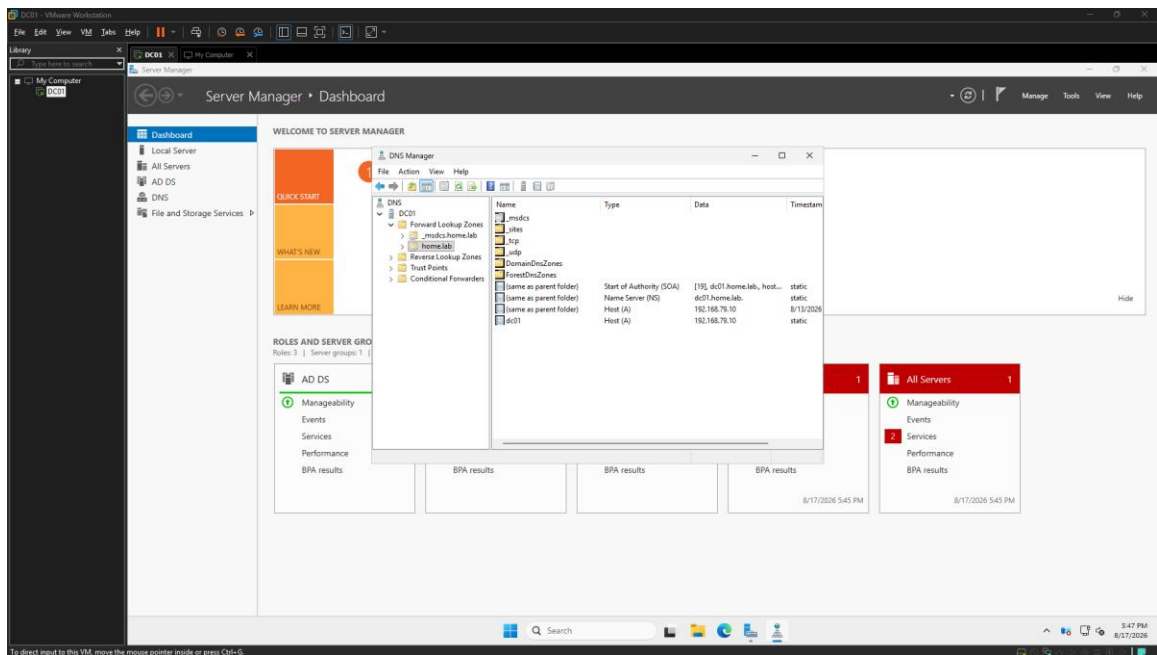
57. Open DNS Manager.
58. Expand the Forward Lookup Zone.
59. Review host and SRV records.
60. Confirm Active Directory integration.

Verification

DNS is operational and supports Active Directory name resolution.

Evidence





Implementation

Step 16 – Create the Organisational Unit Structure

Purpose

Create a logical Active Directory hierarchy to organise users, computers, servers and groups.

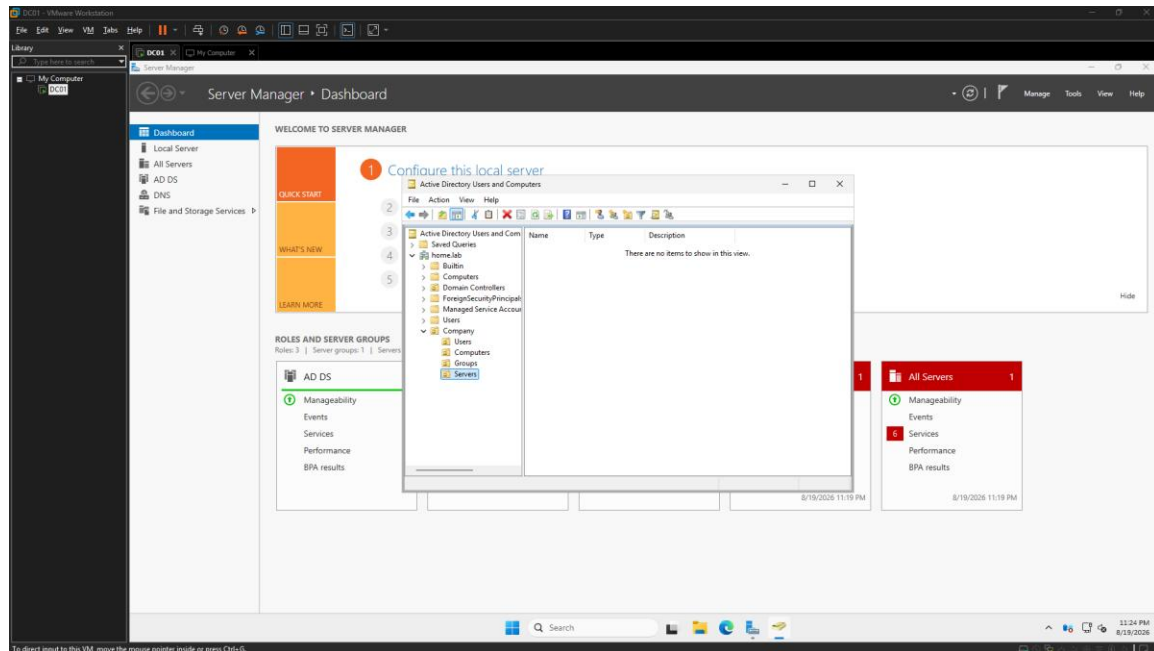
Procedure

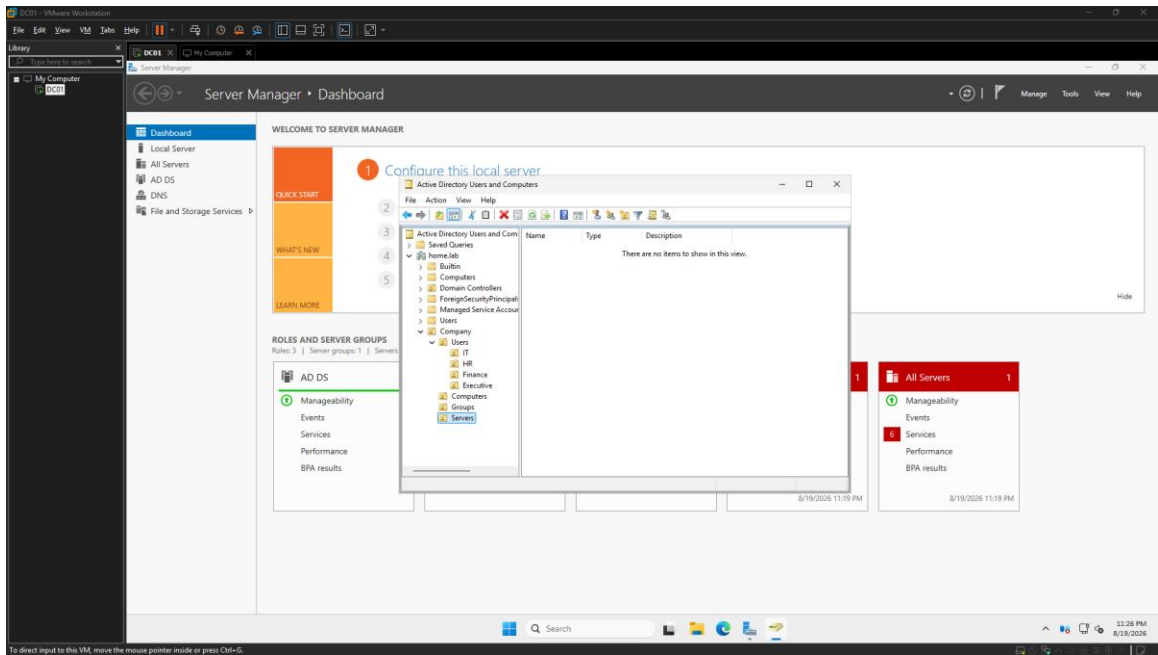
61. Open Active Directory Users and Computers.
62. Create a top-level OU named Company.
63. Create Users, Computers, Groups and Servers OUs.
64. Create departmental OUs: IT, HR, Finance and Executive.
65. Verify the hierarchy.

Verification

The OU structure provides a scalable foundation for administration and Group Policy.

Evidence





Step 17 – Create Security Groups

Purpose

Create departmental Global Security Groups to simplify permission management.

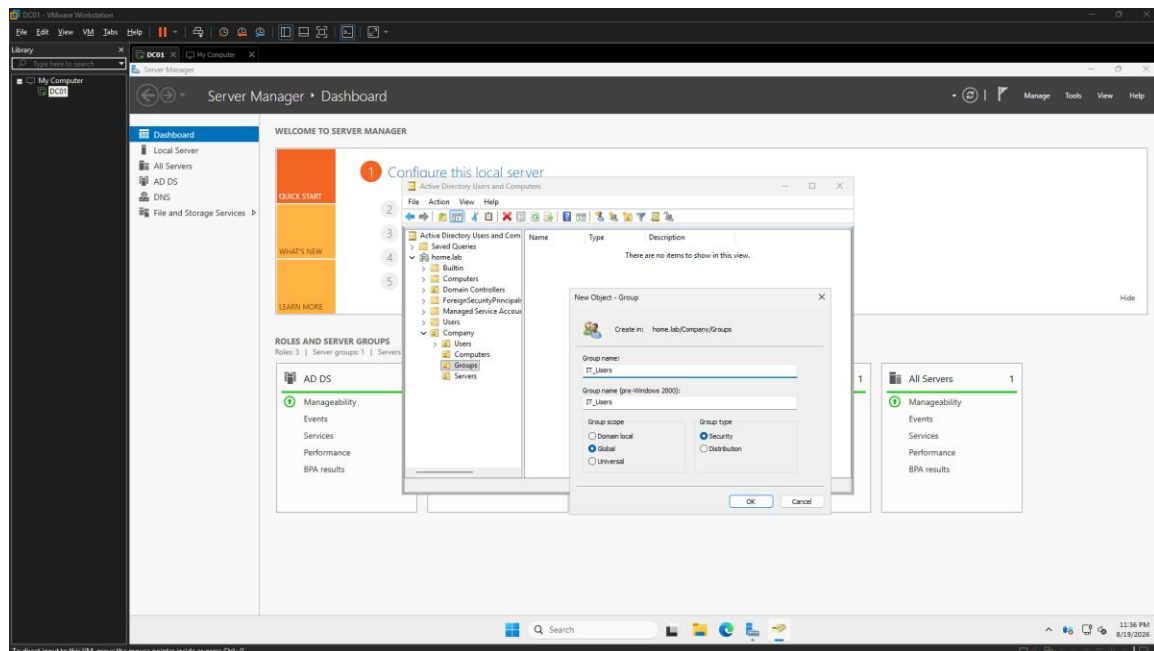
Procedure

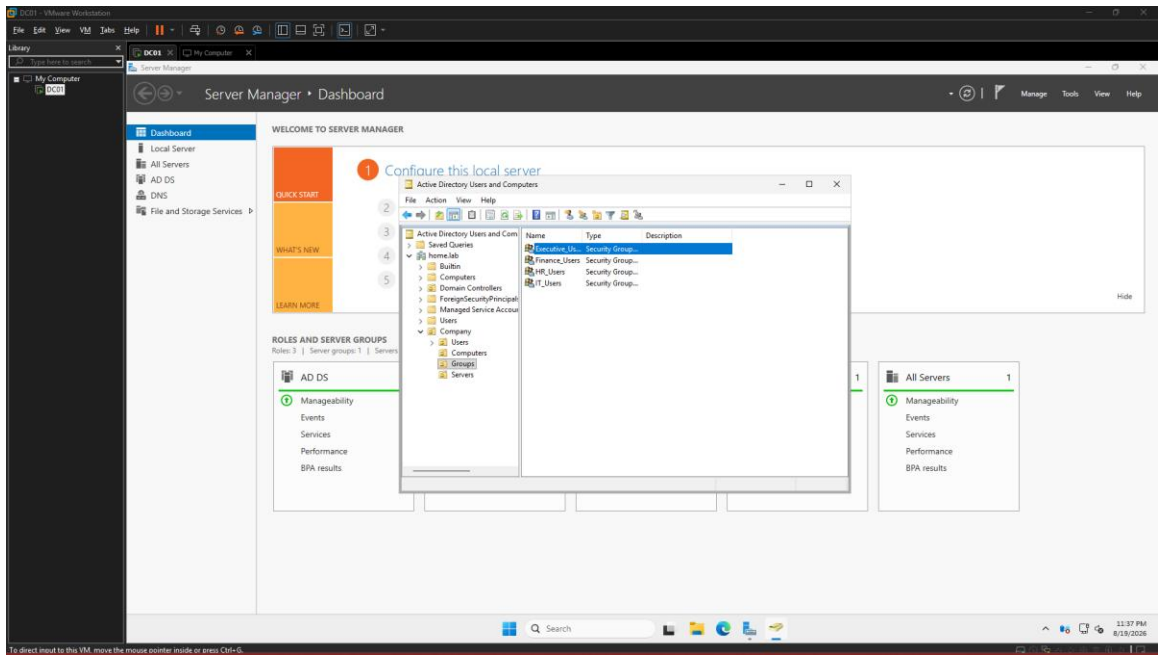
66. Open the Groups OU.
67. Create IT_Users, HR_Users, Finance_Users and Executive_Users.
68. Verify group scope and type are configured correctly.

Verification

Departmental security groups are available for permissions and access control.

Evidence





Step 18 – Create User Accounts

Purpose

Create domain user accounts and assign them to the appropriate departmental groups.

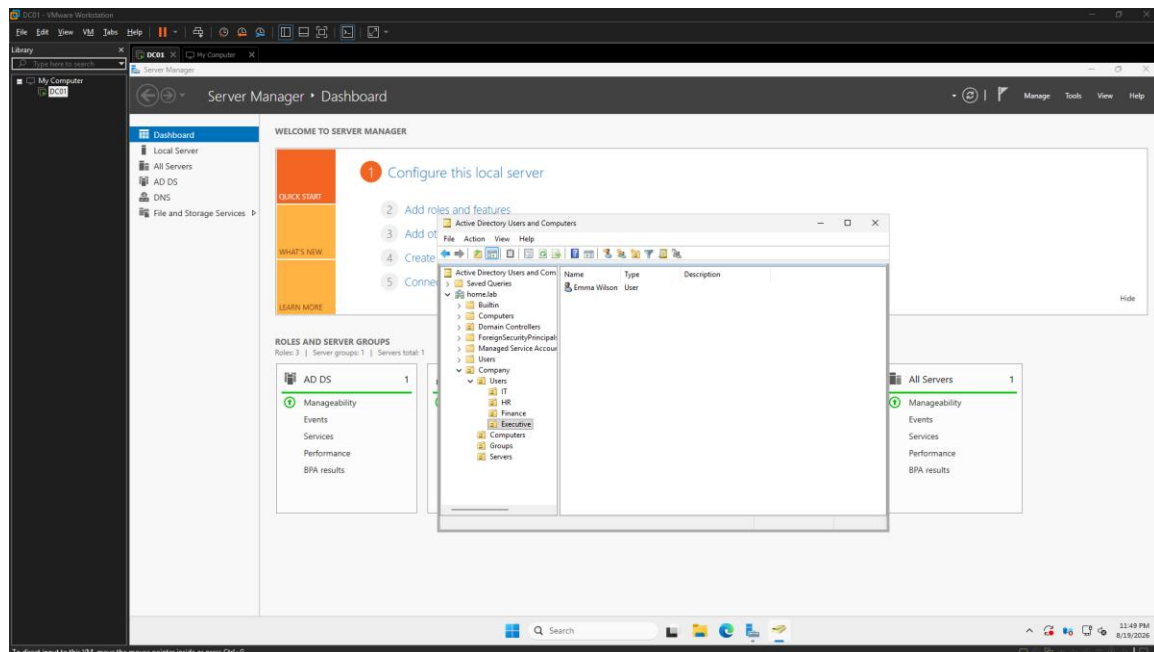
Procedure

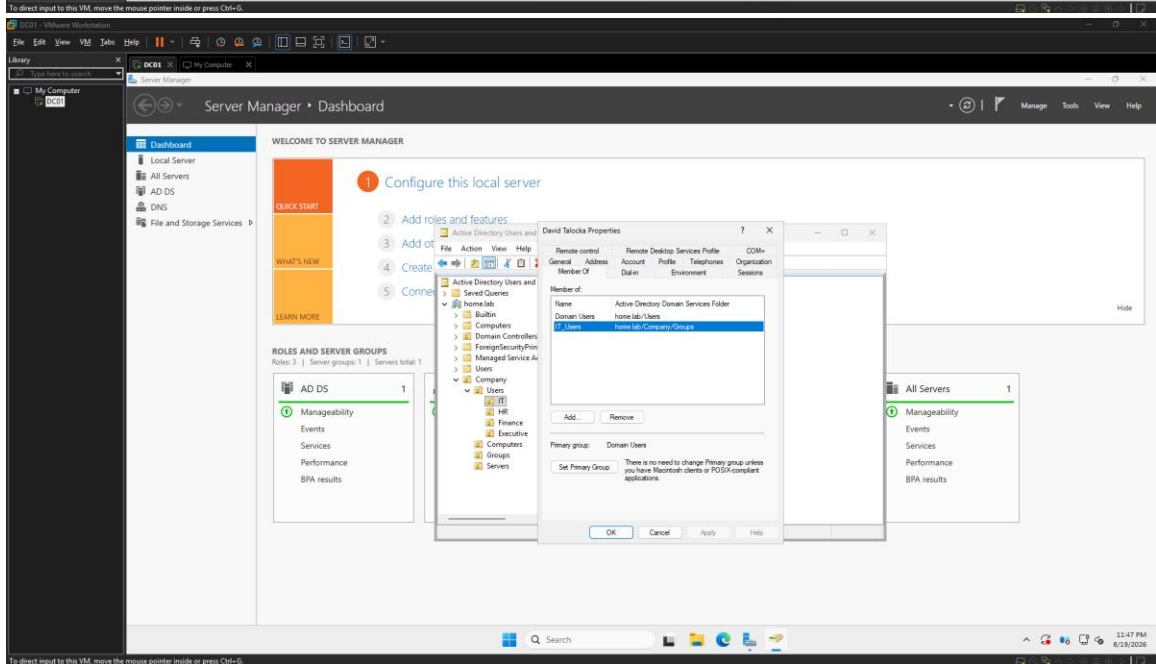
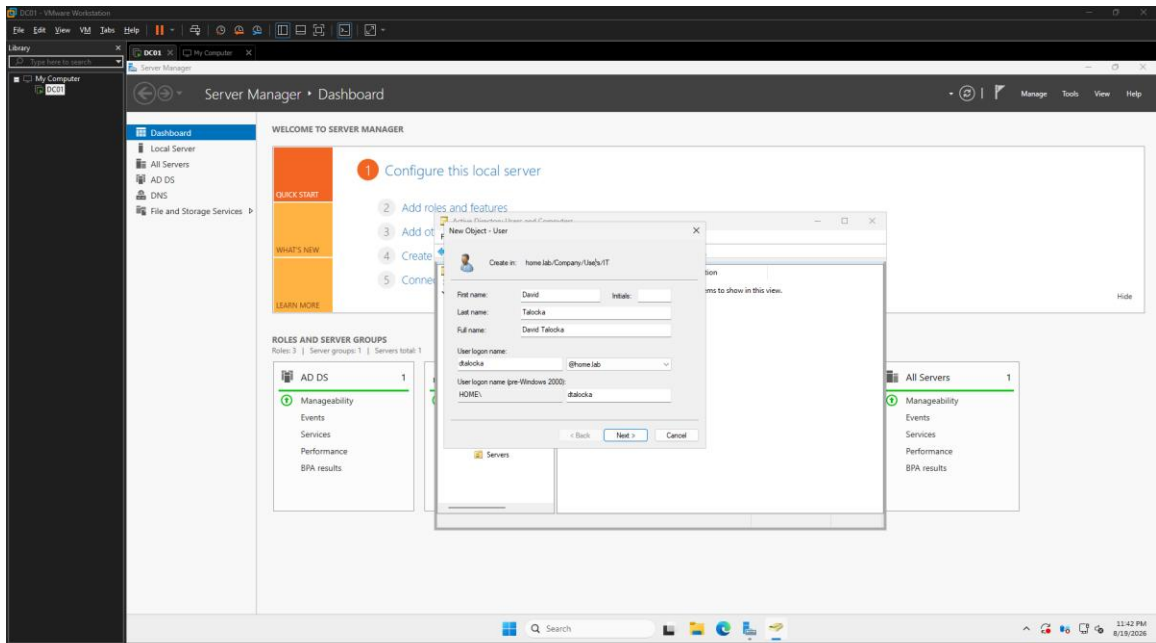
69. Create user accounts within each departmental OU.
70. Configure usernames and passwords.
71. Add each user to the correct security group using the Member Of tab.
72. Verify group memberships.

Verification

Users authenticate through Active Directory and inherit permissions through group membership.

Evidence





Step 19 – Deploy the Windows 11 Client

Purpose

Create the Windows 11 Enterprise virtual machine that will be joined to the domain.

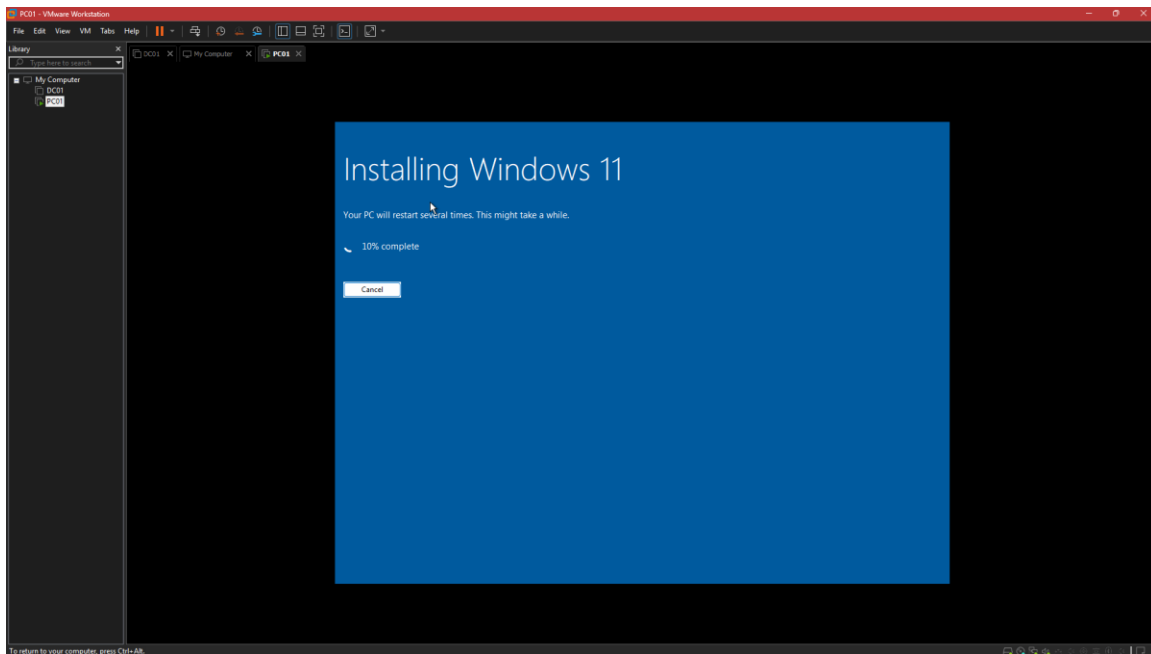
Procedure

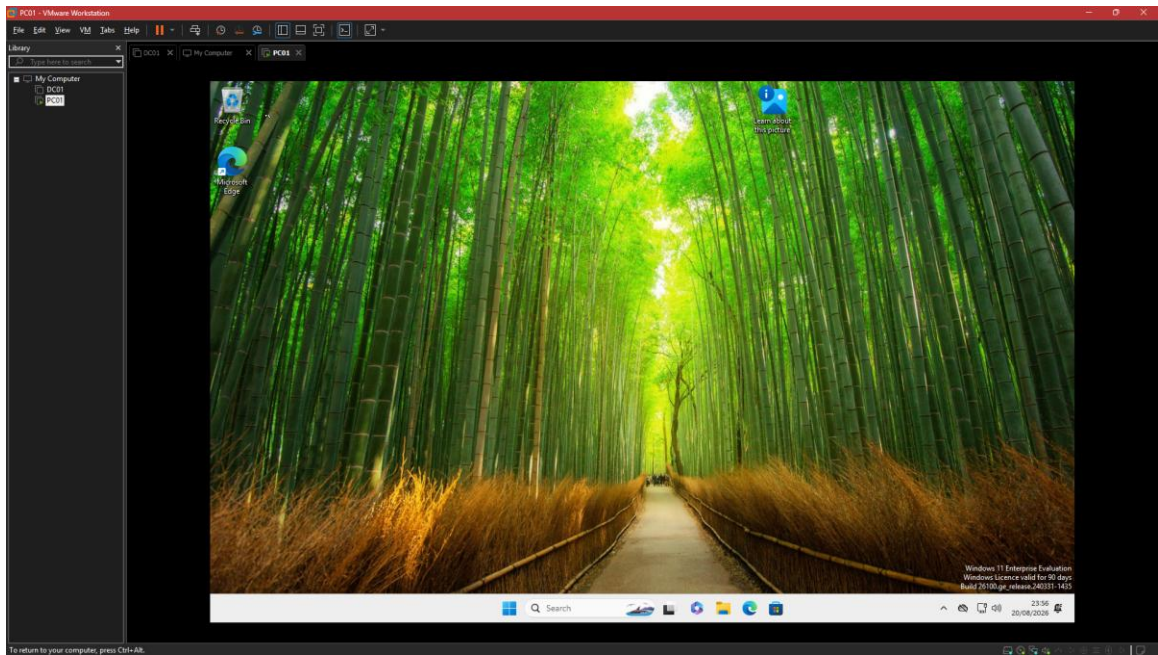
73. Create a new virtual machine in VMware Workstation Pro.
74. Install Windows 11 Enterprise.
75. Complete the Out-of-Box Experience using a local account.
76. Install VMware Tools if required.

Verification

PC01 is ready for network configuration and domain integration.

Evidence





Step 20 – Configure Client Networking

Purpose

Configure PC01 so it can communicate with the domain controller.

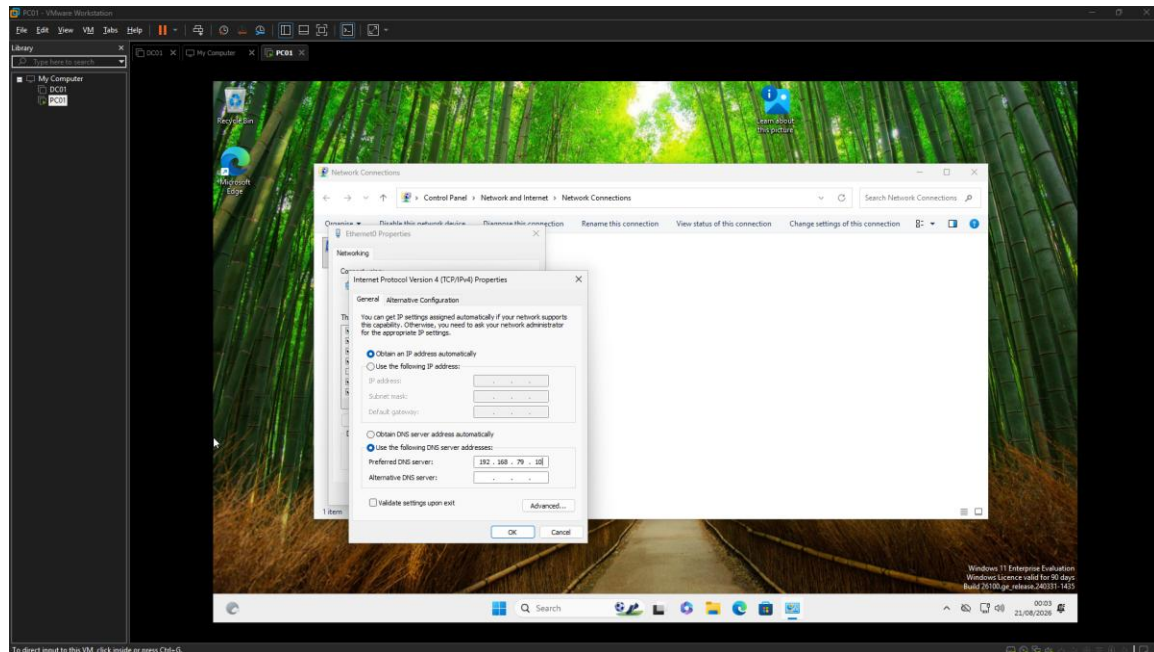
Procedure

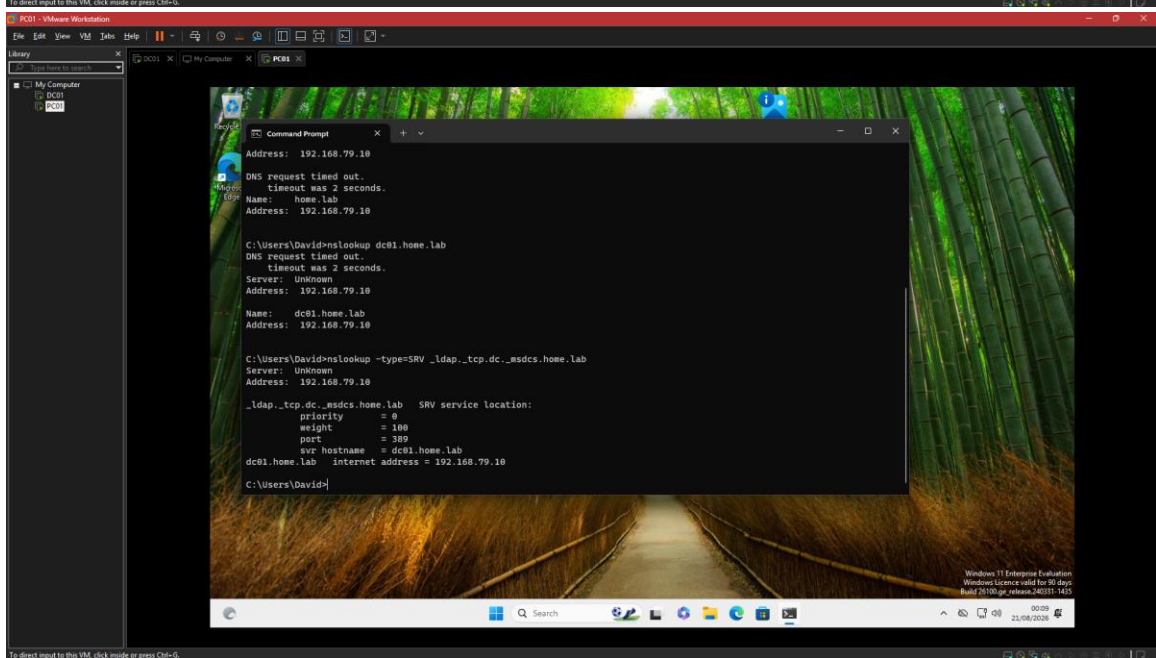
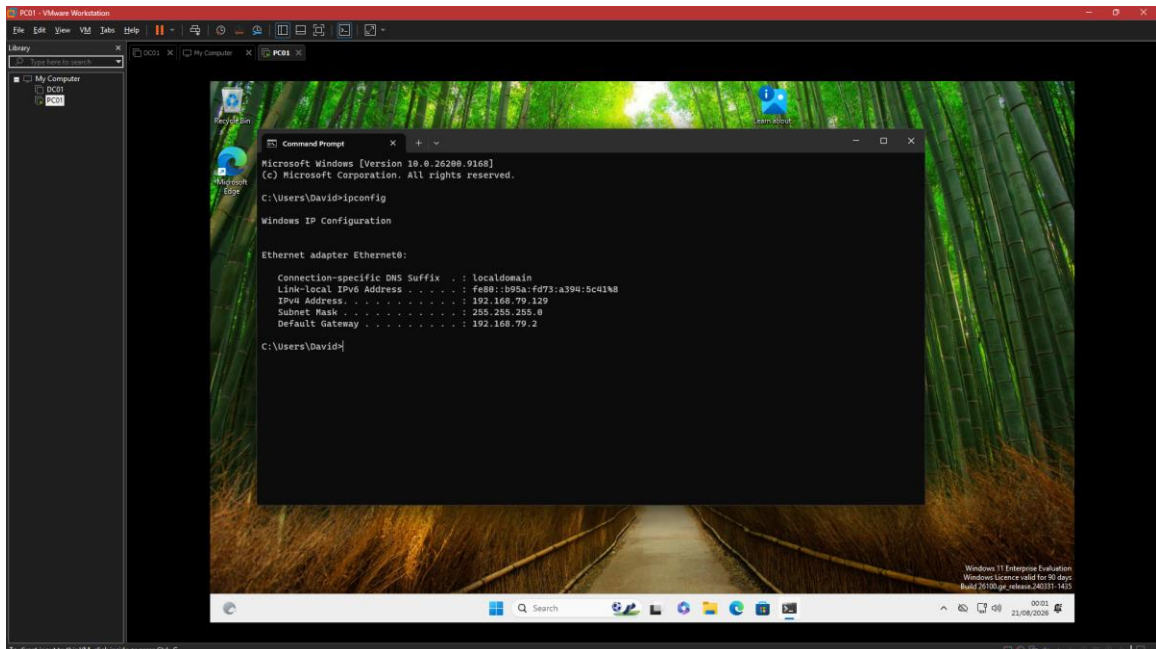
77. Review the current IP configuration using ipconfig.
78. Configure the preferred DNS server as 192.168.79.10.
79. Verify connectivity using ping.
80. Verify DNS resolution using nslookup.
81. Confirm LDAP SRV records can be resolved.

Verification

PC01 successfully communicates with DC01 and can locate Active Directory services.

Evidence





Step 21 – Join PC01 to the Domain

Purpose

Join the Windows 11 workstation to the home.lab Active Directory domain.

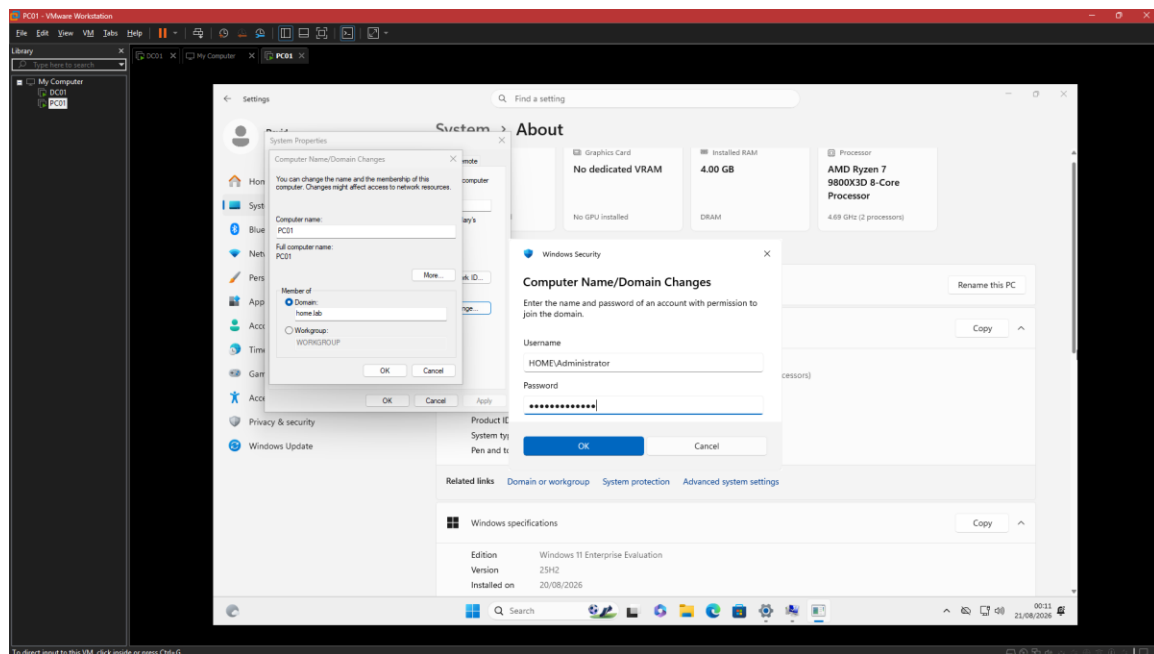
Procedure

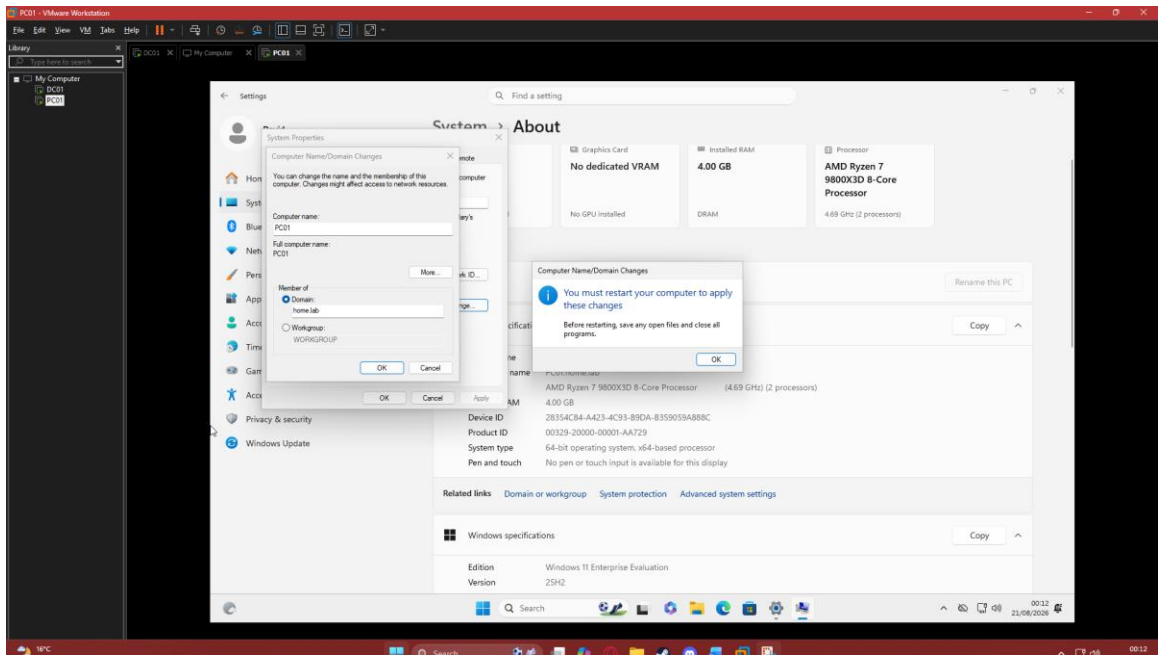
82. Open System Properties.
83. Select Domain and enter home.lab.
84. Authenticate with a domain administrator account.
85. Restart the workstation.

Verification

PC01 is successfully joined to the domain and managed through Active Directory.

Evidence





Step 22 – Validate Domain Authentication

Purpose

Verify that a domain user can successfully authenticate on PC01.

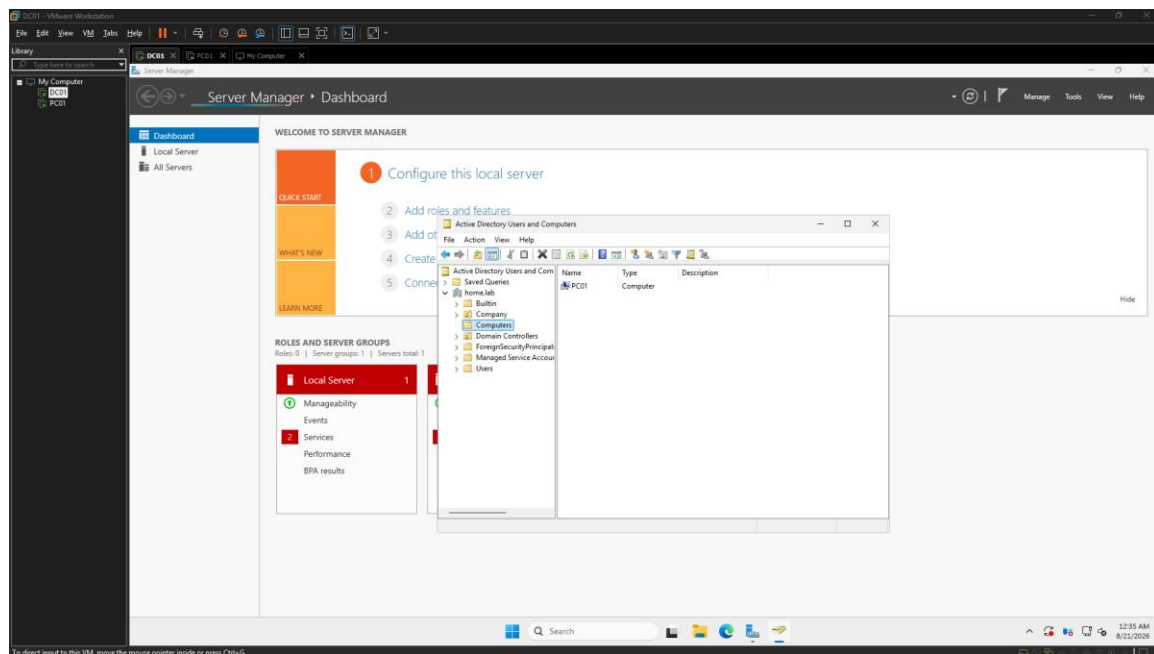
Procedure

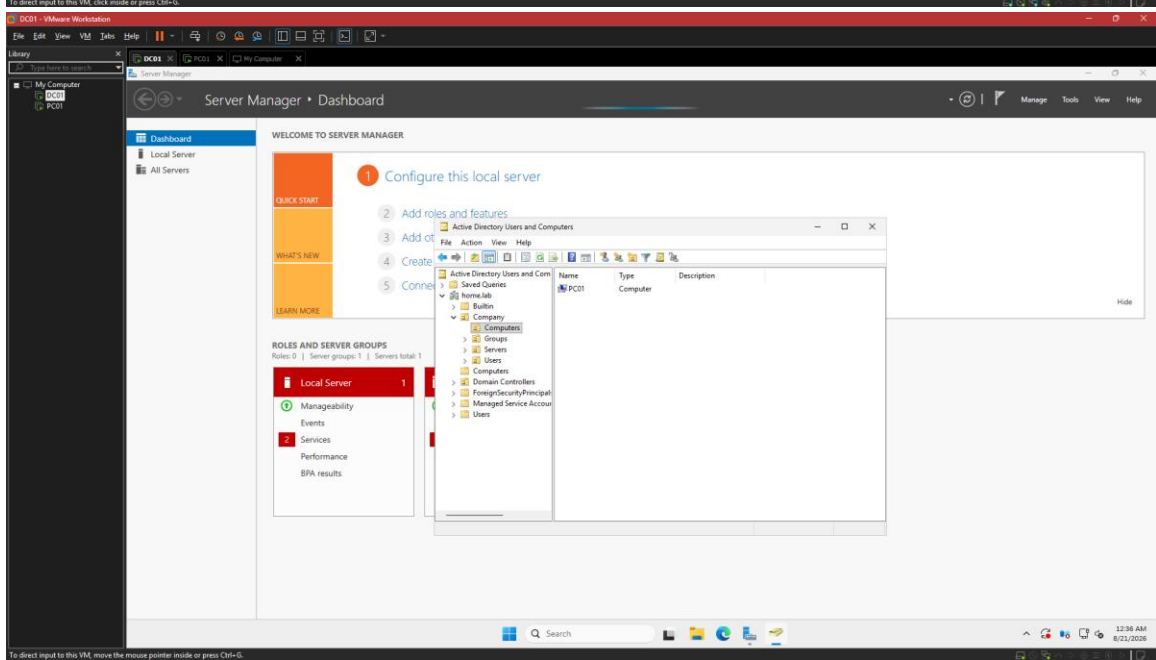
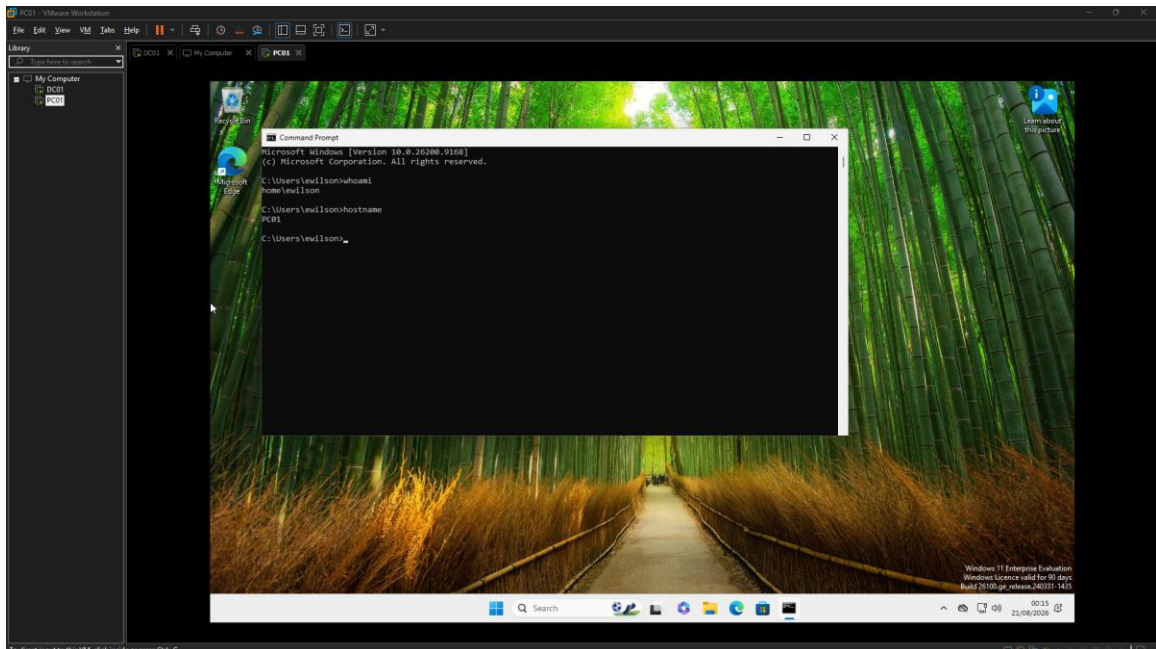
86. Sign in using HOME\ewilson.
87. Allow Windows to create the user profile.
88. Run whoami and hostname.
89. Verify the authenticated domain and workstation.

Verification

Domain authentication is functioning correctly between PC01 and DC01.

Evidence





Step 23 – Create Department File Shares

Purpose

Create departmental folders and publish them as network shares.

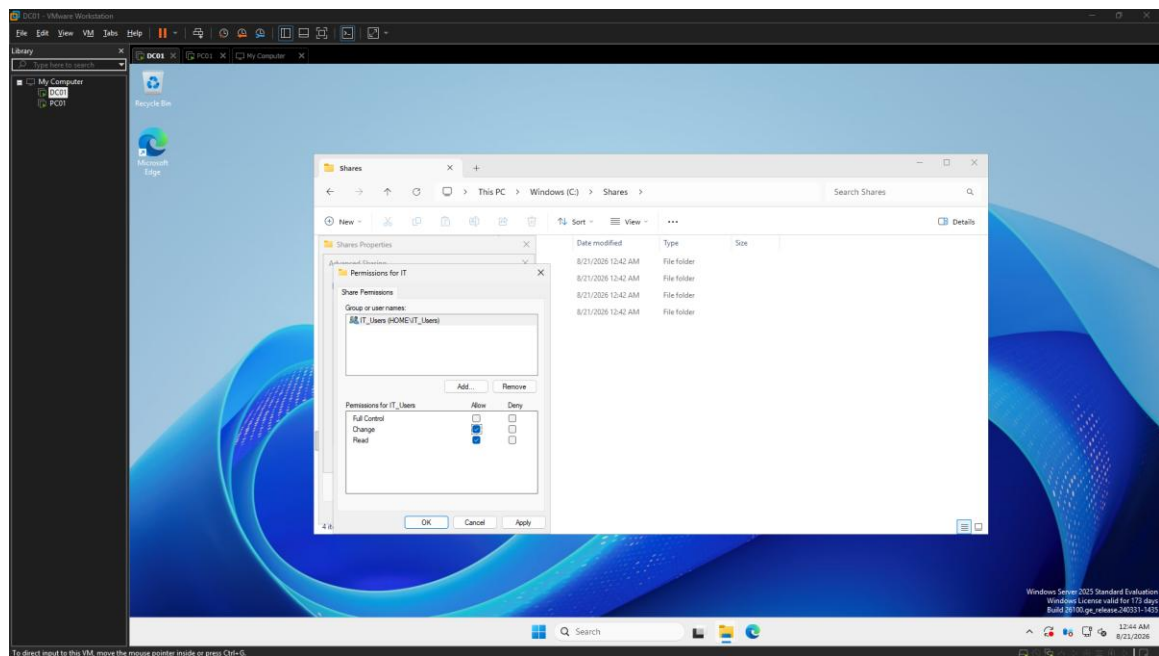
Procedure

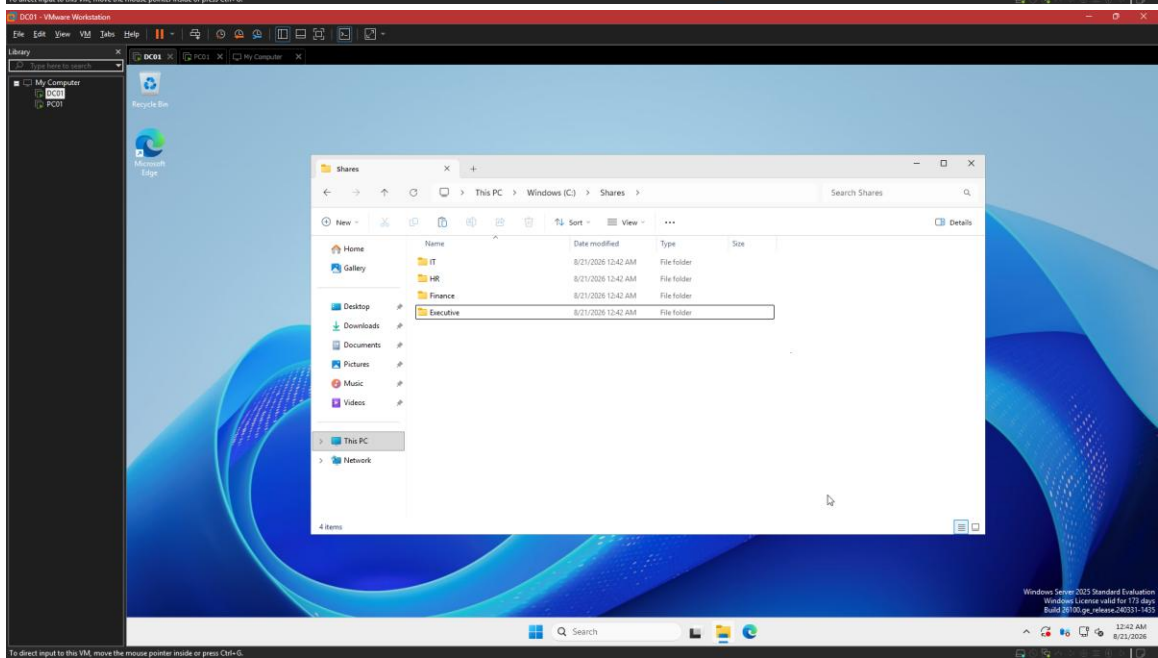
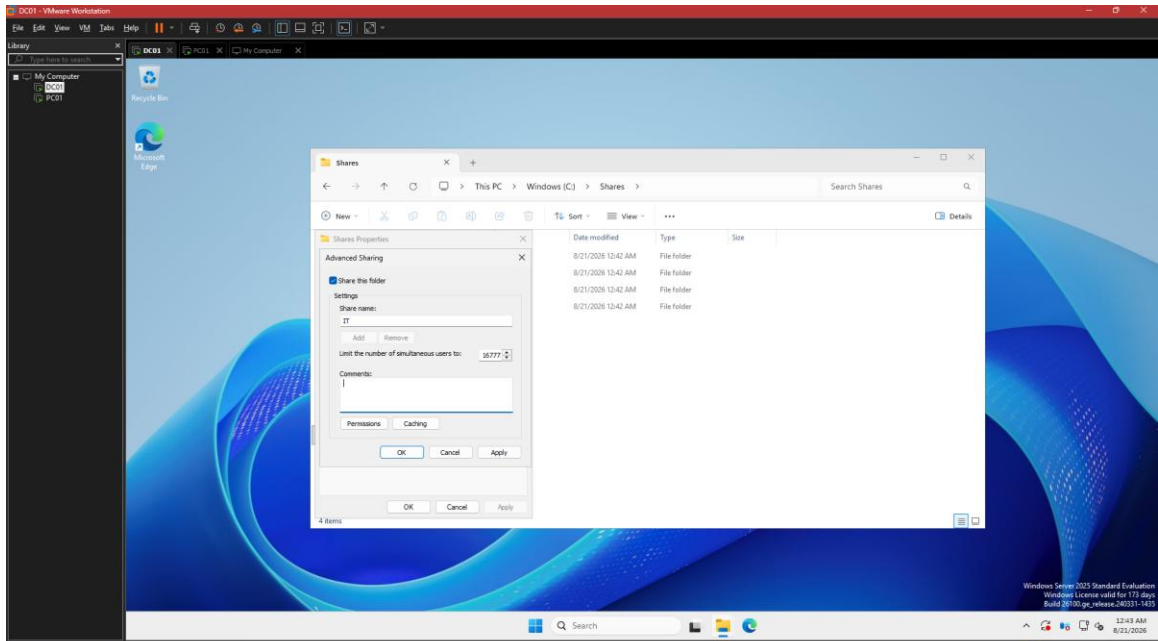
90. Create C:\Shares.
91. Create IT, HR, Finance and Executive folders.
92. Share each folder using Advanced Sharing.
93. Assign appropriate share permissions.

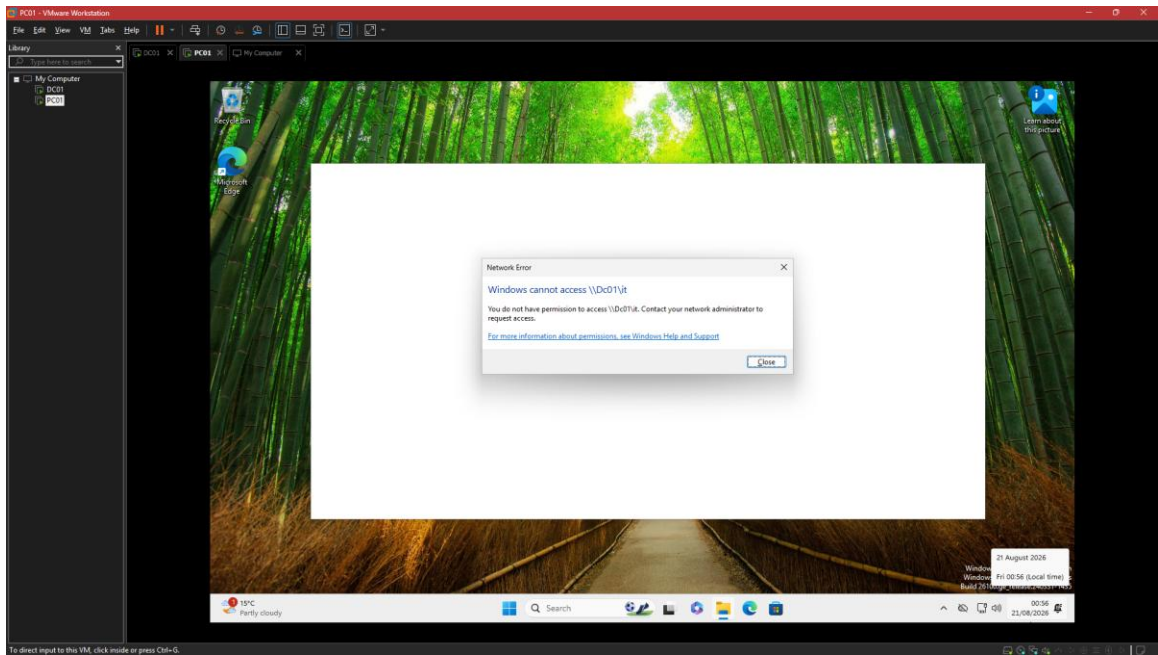
Verification

Departmental shares are accessible over the network according to assigned permissions.

Evidence







Step 24 – Configure NTFS Permissions

Purpose

Secure departmental folders using NTFS permissions and security groups.

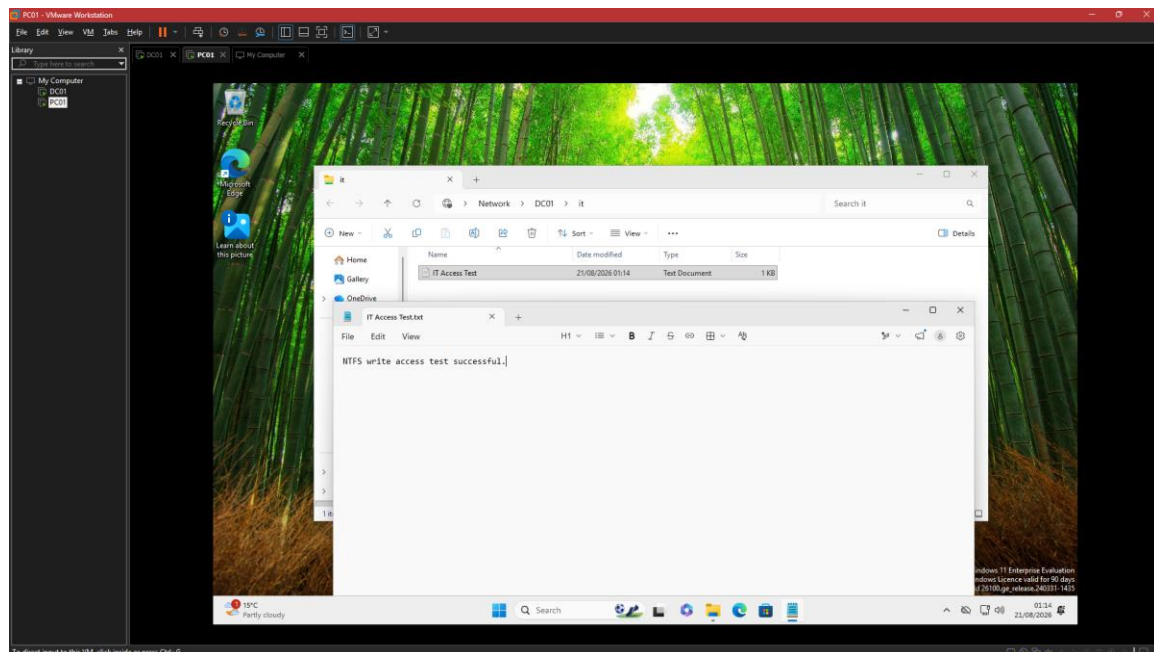
Procedure

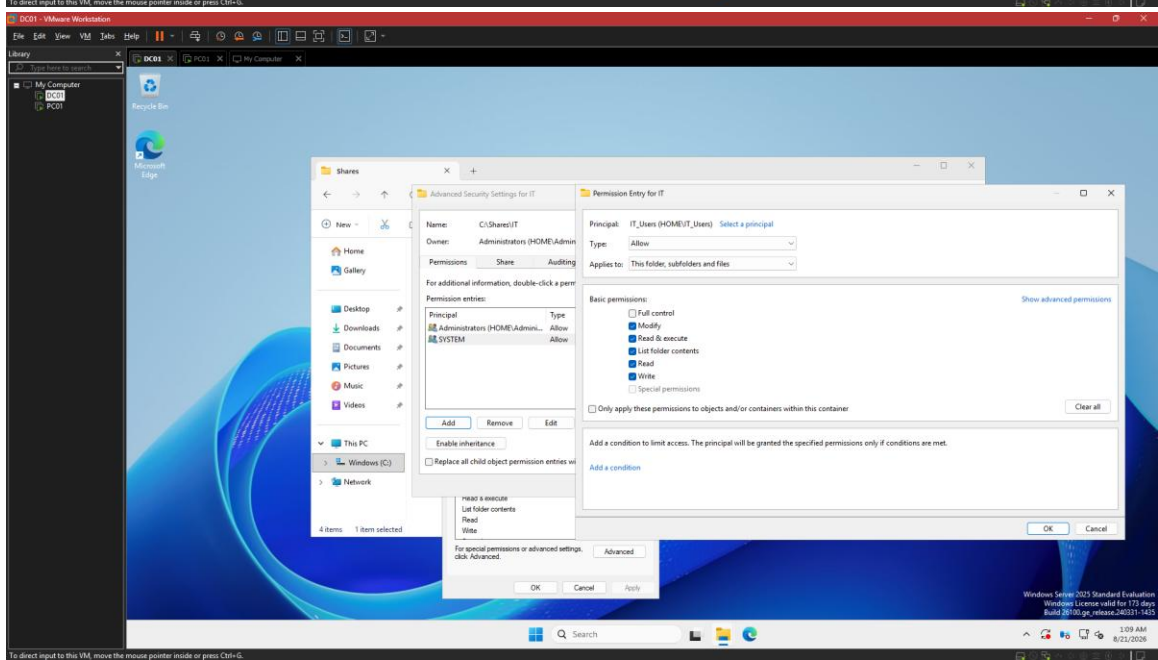
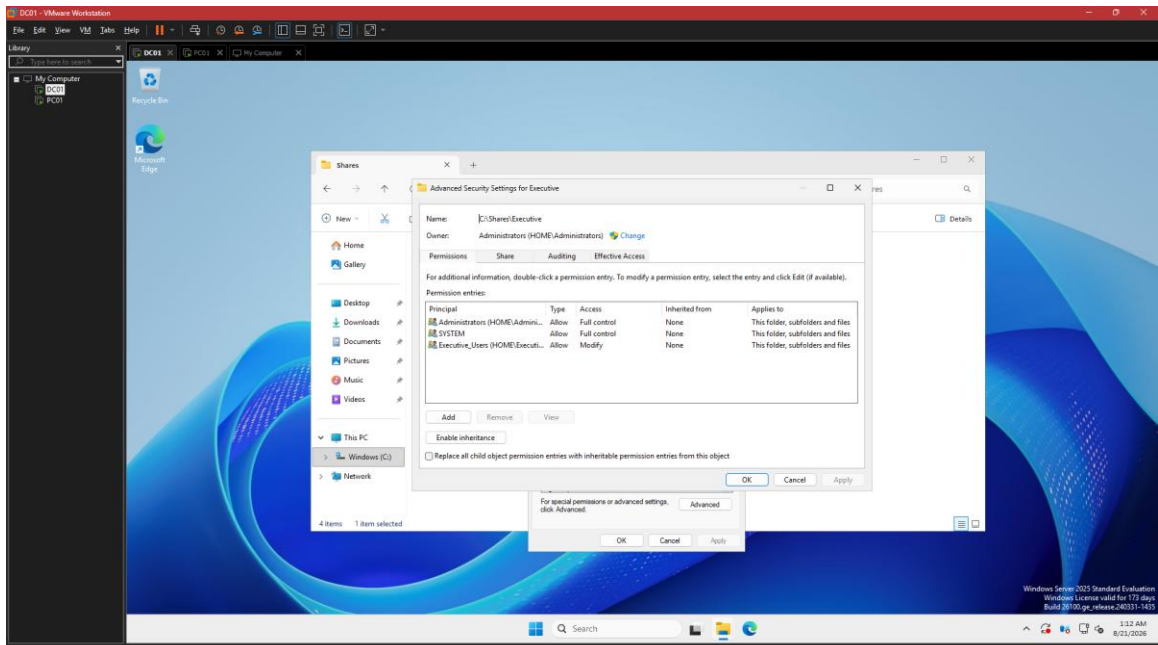
94. Open folder Security settings.
95. Disable inheritance and convert inherited permissions.
96. Grant Modify permissions to the appropriate departmental group.
97. Retain SYSTEM and Administrators access.
98. Repeat for each department.

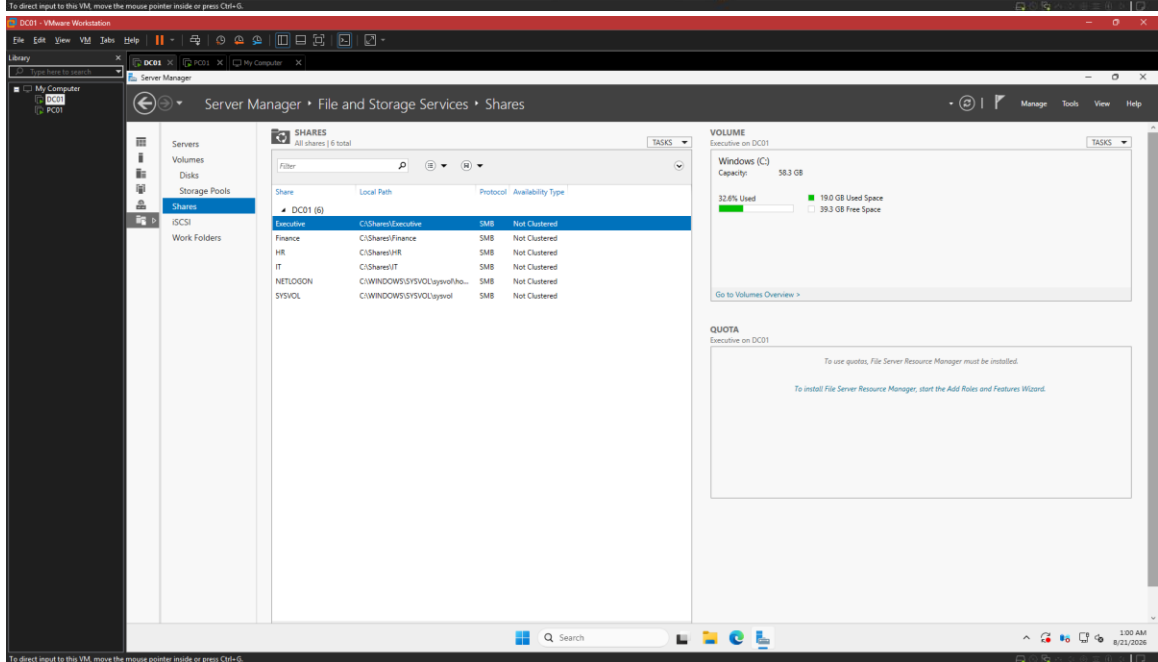
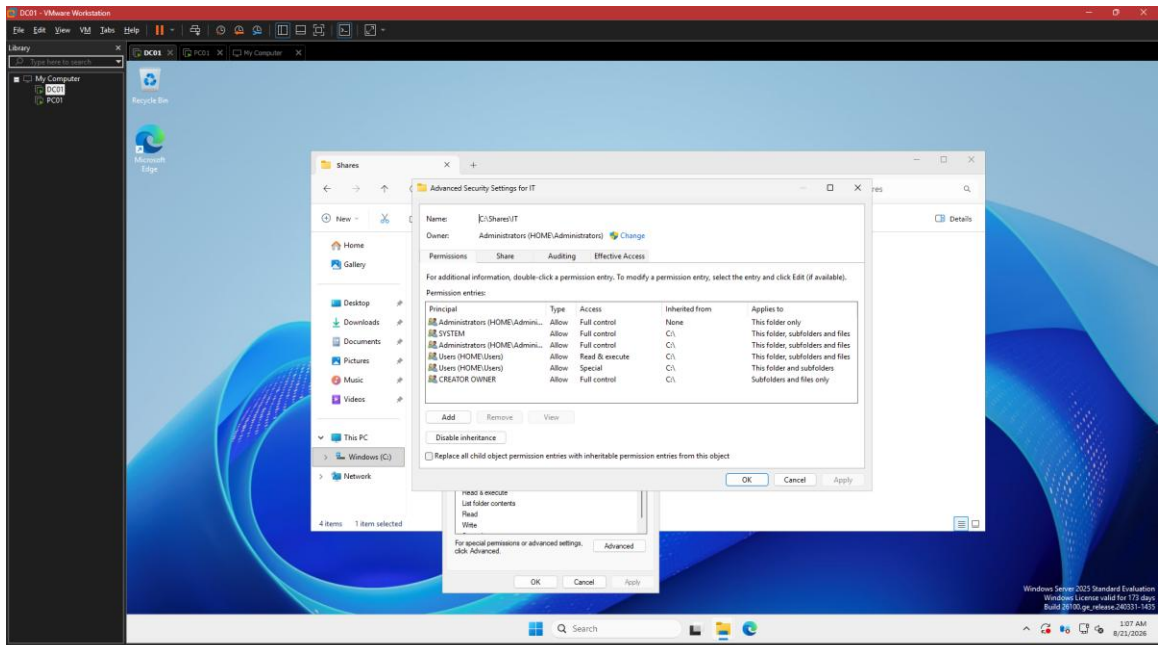
Verification

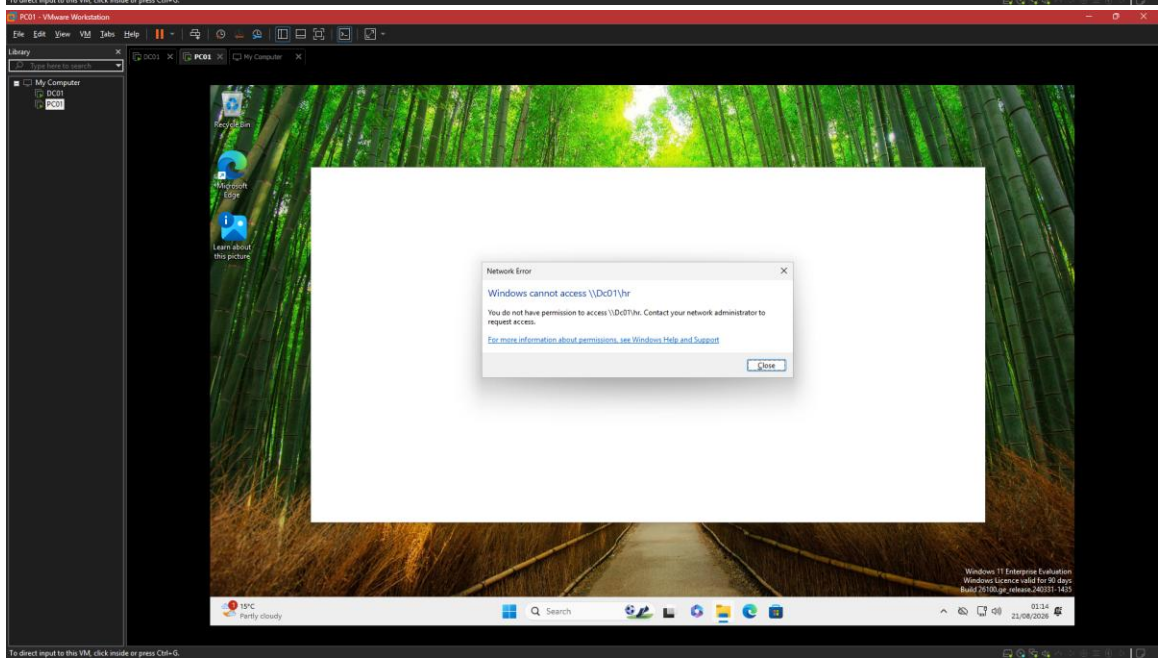
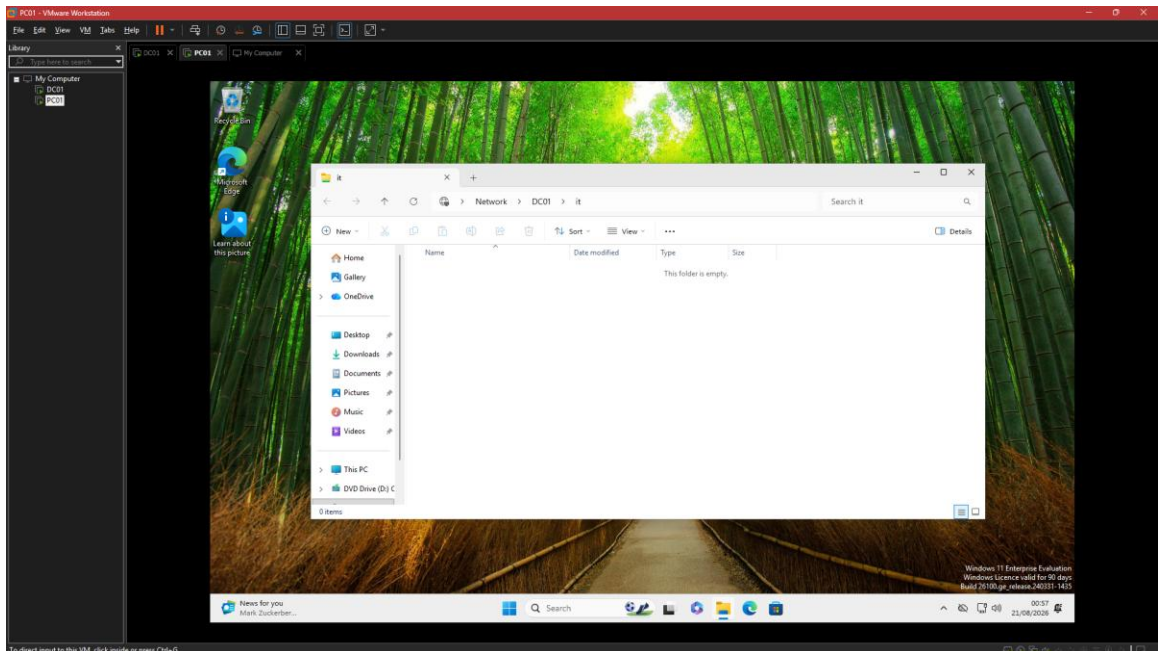
Users can only access resources belonging to their department.

Evidence









Step 25 – Configure Group Policy

Purpose

Create and link a Group Policy Object to manage user settings centrally.

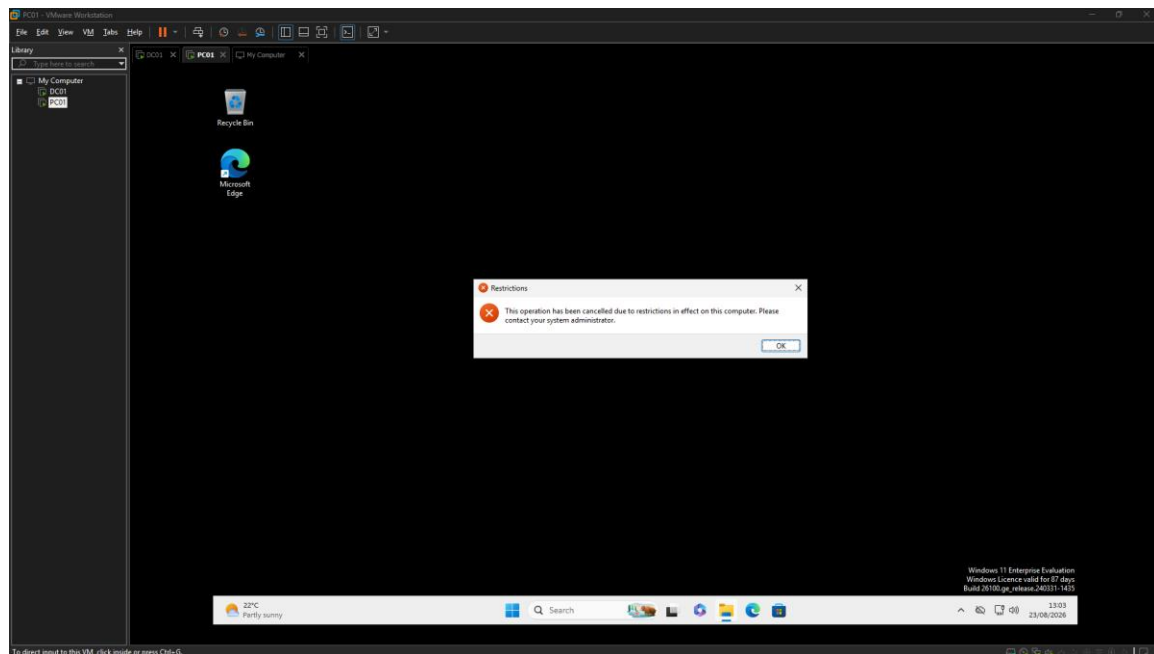
Procedure

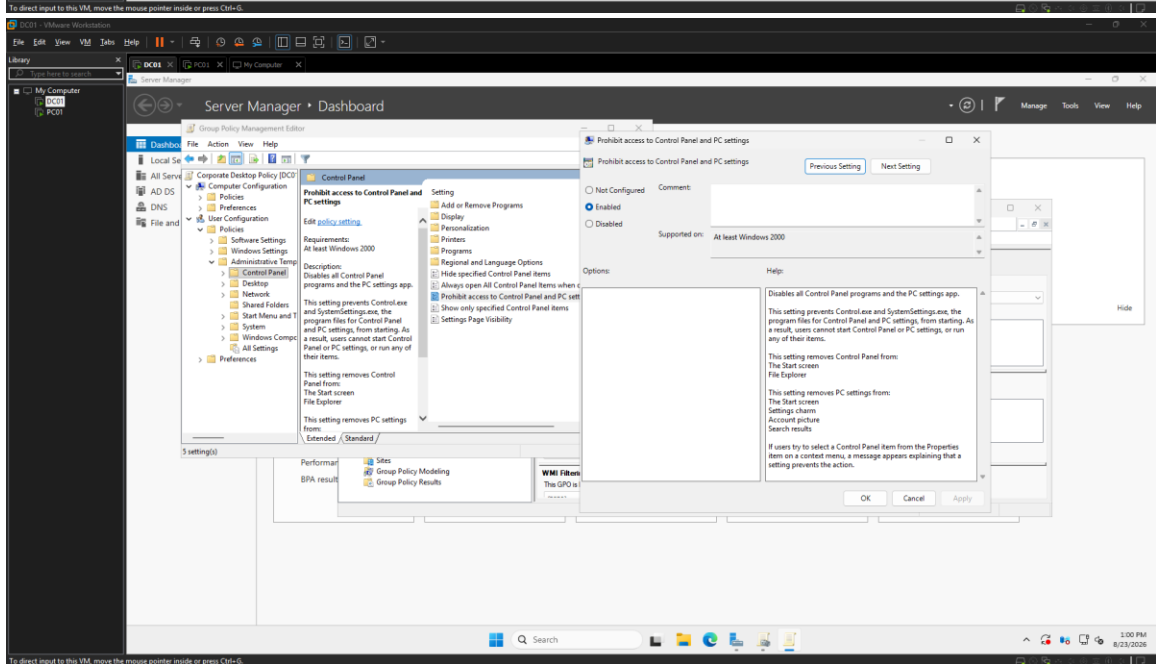
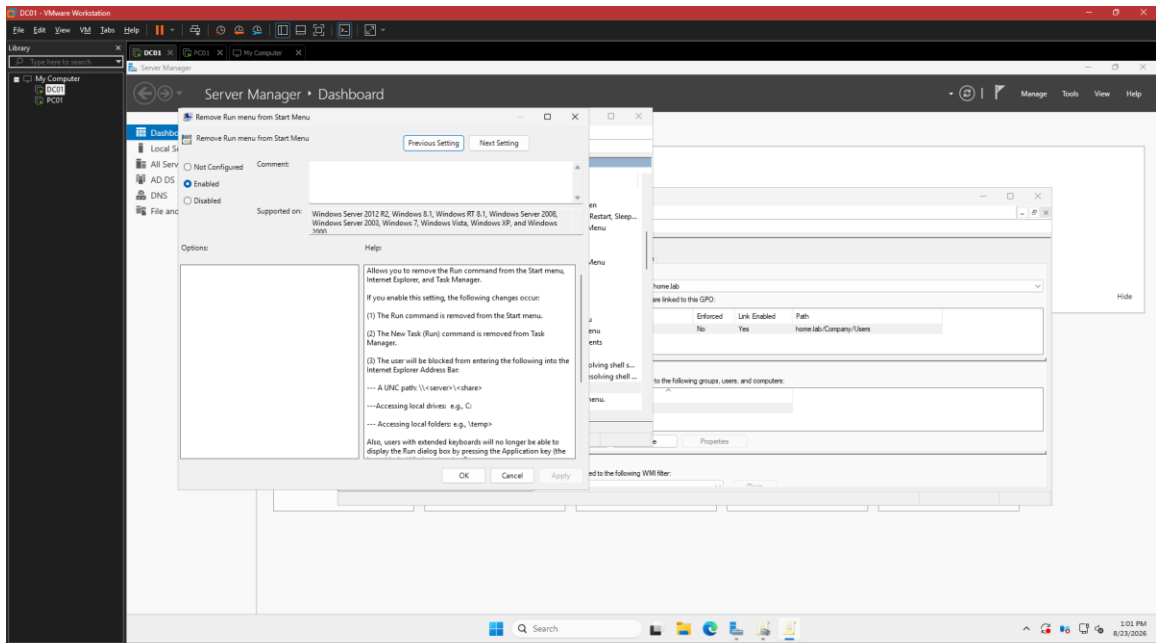
99. Open Group Policy Management.
100. Create 'Corporate Desktop Policy'.
101. Link the GPO to the Users OU.
102. Enable Control Panel and Run menu restrictions.
103. Update Group Policy on PC01.

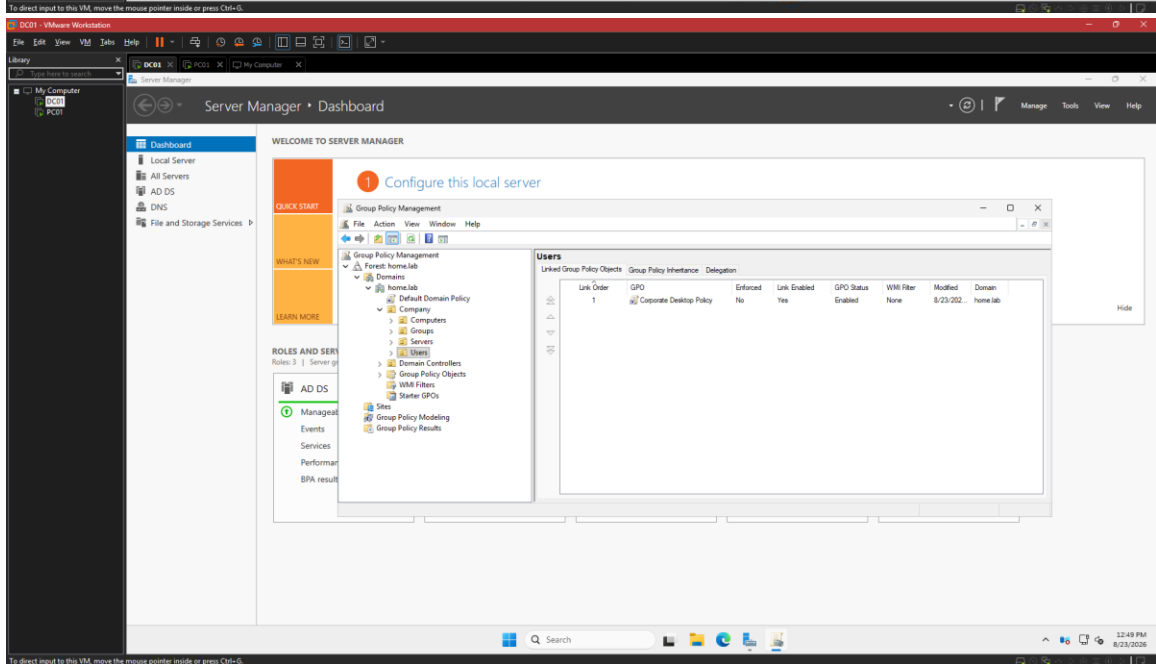
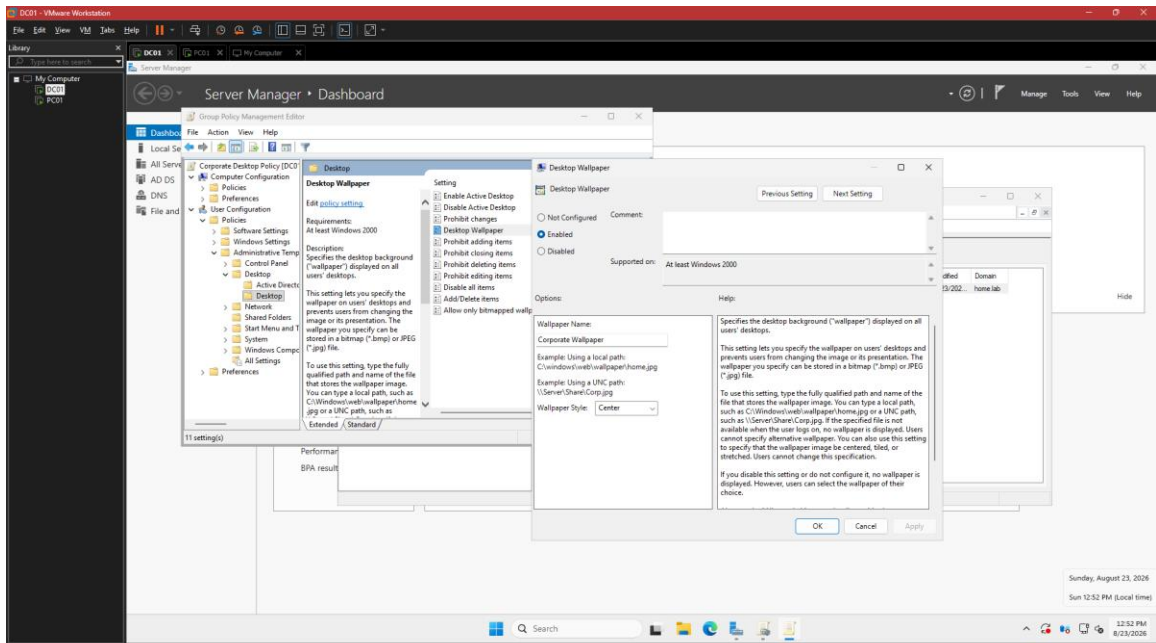
Verification

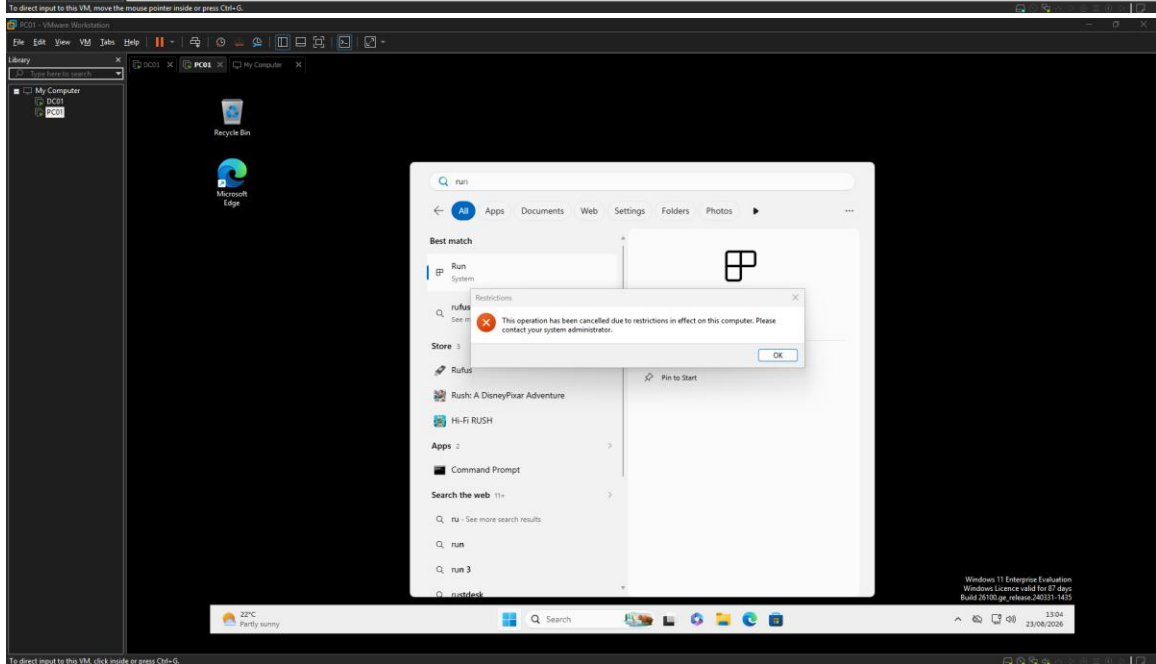
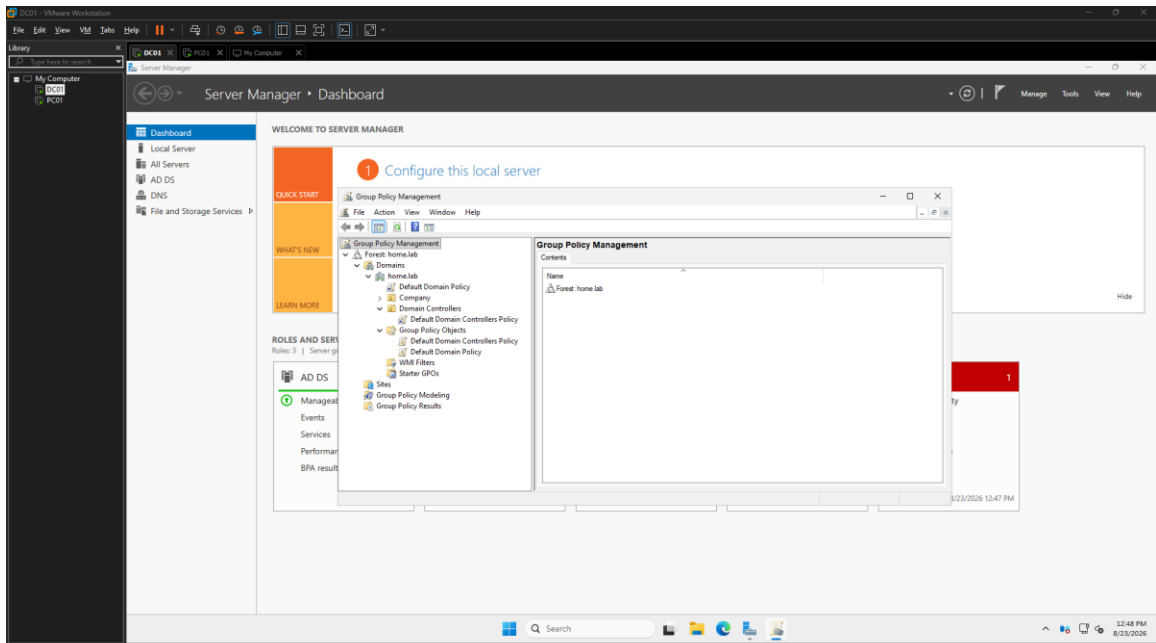
User restrictions are applied successfully after policy refresh.

Evidence









Step 26 – Perform Password Reset

Purpose

Simulate a forgotten password support request.

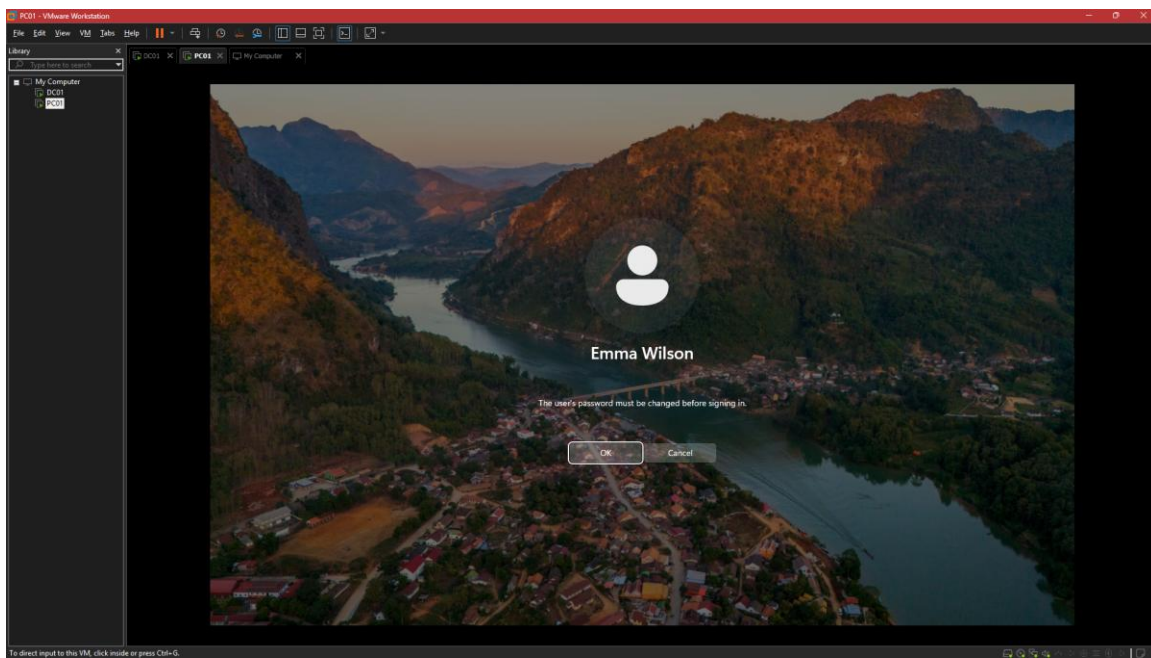
Procedure

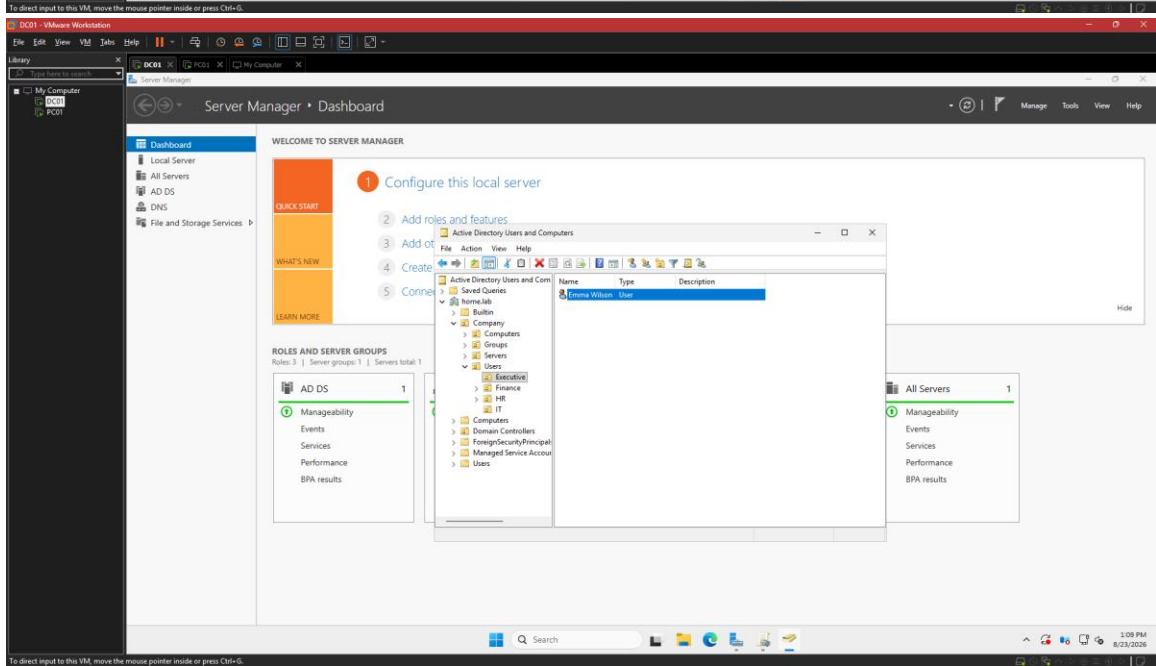
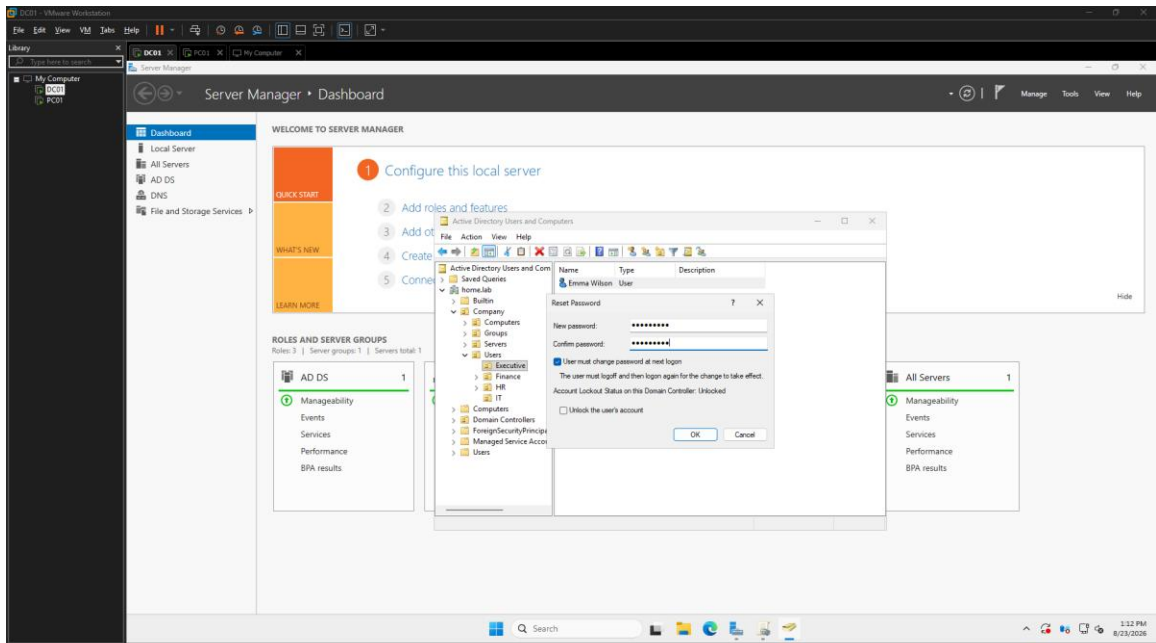
104. Open Active Directory Users and Computers.
105. Locate the user account.
106. Reset the password.
107. Verify successful sign-in on PC01.

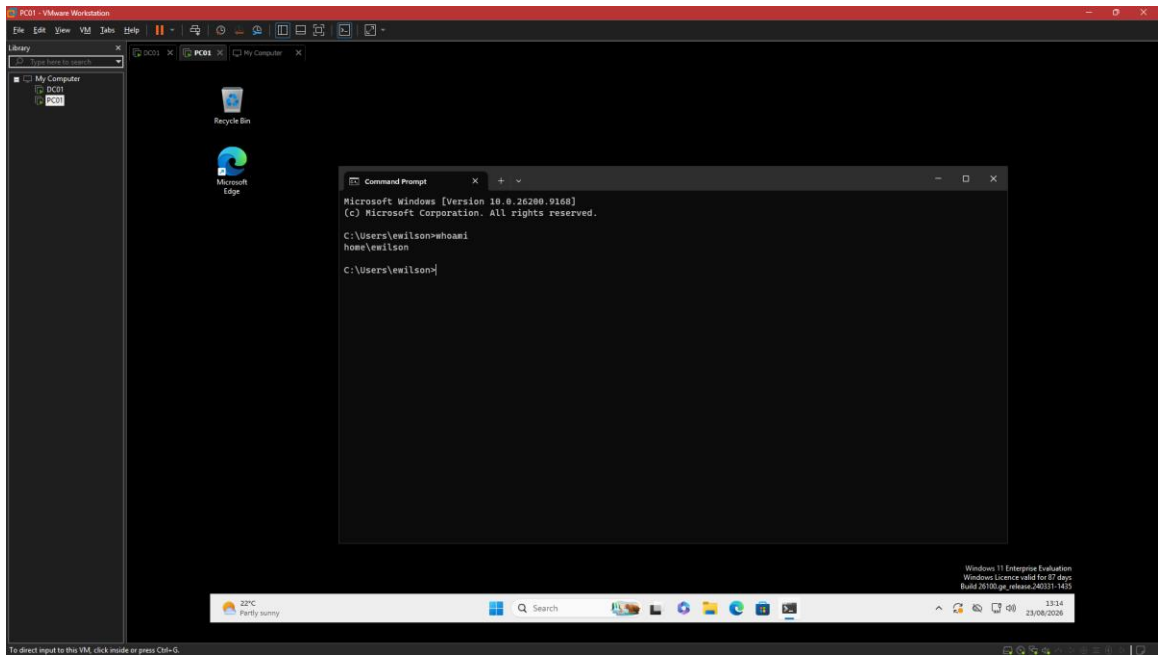
Verification

The user authenticates successfully using the new password.

Evidence







Step 27 – Configure and Test Account Lockout

Purpose

Configure an account lockout policy and demonstrate account recovery.

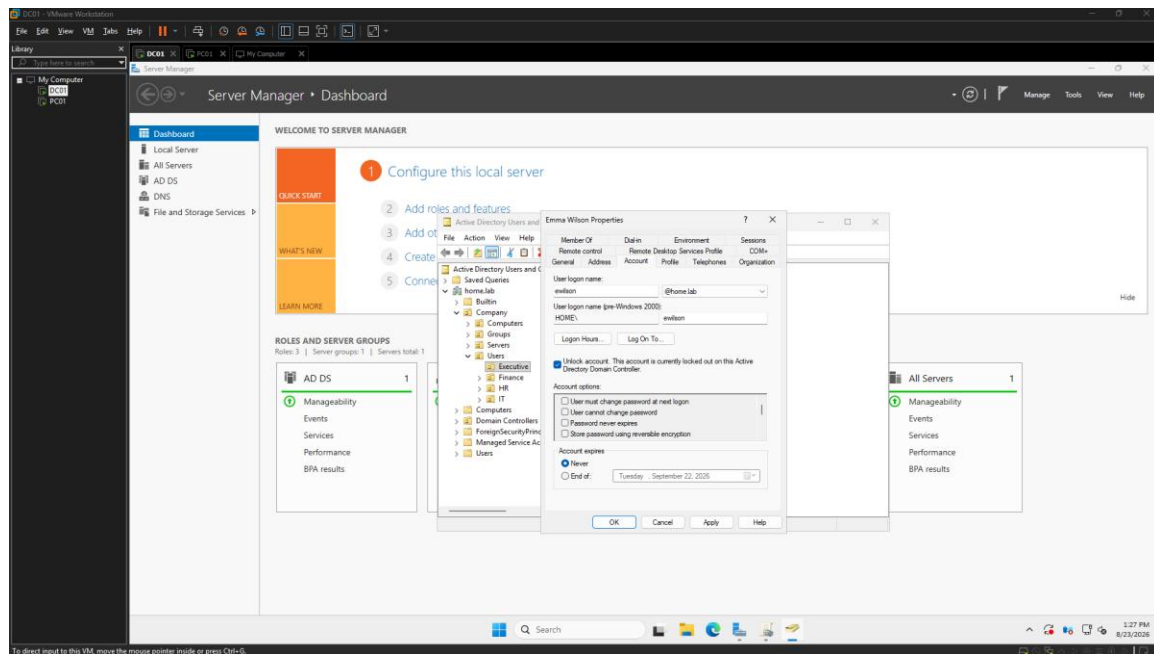
Procedure

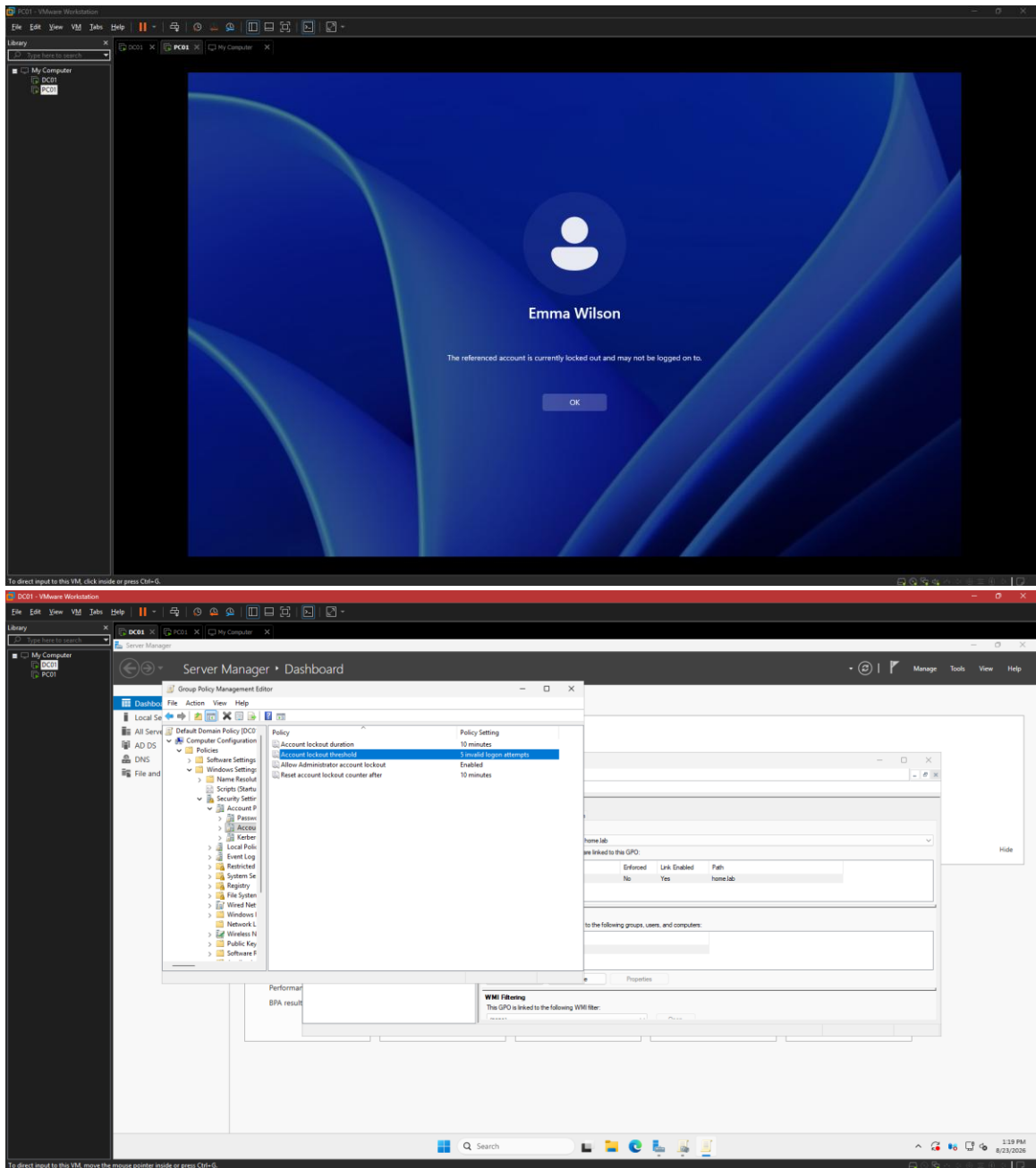
108. Edit the Default Domain Policy.
109. Configure account lockout settings.
110. Force Group Policy update.
111. Trigger a lockout using invalid passwords.
112. Unlock the account in Active Directory.
113. Verify successful sign-in.

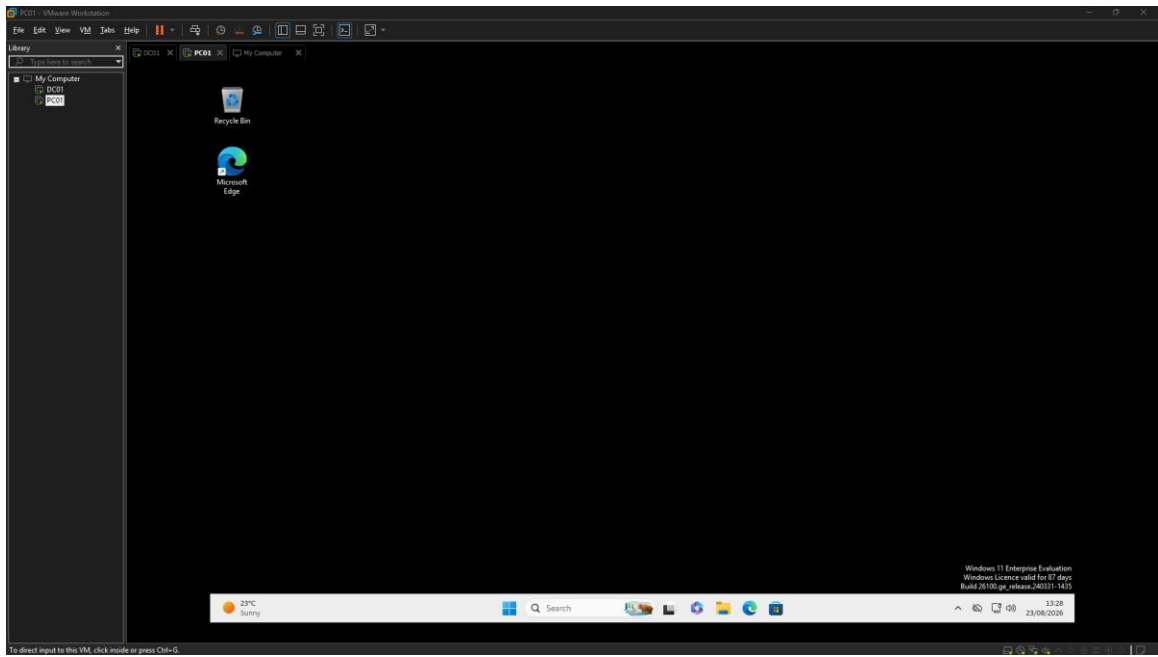
Verification

The account lockout policy functions correctly and the user account is successfully restored.

Evidence







Step 28 – Validate the Environment

Purpose

Perform a final review of all deployed services and configurations.

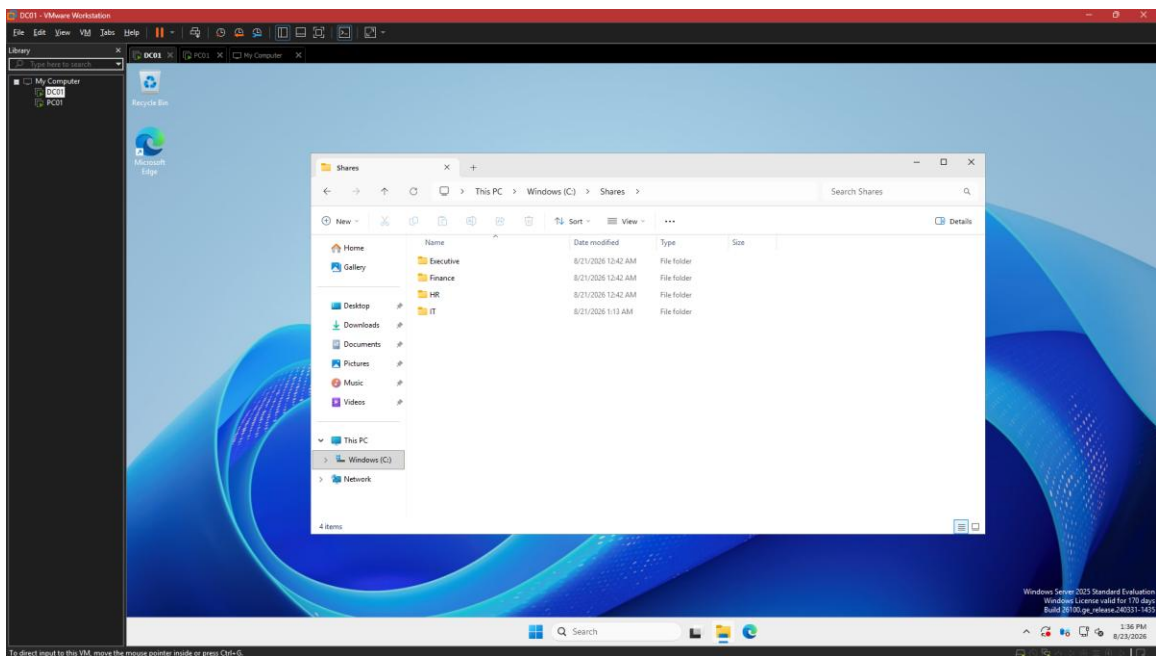
Procedure

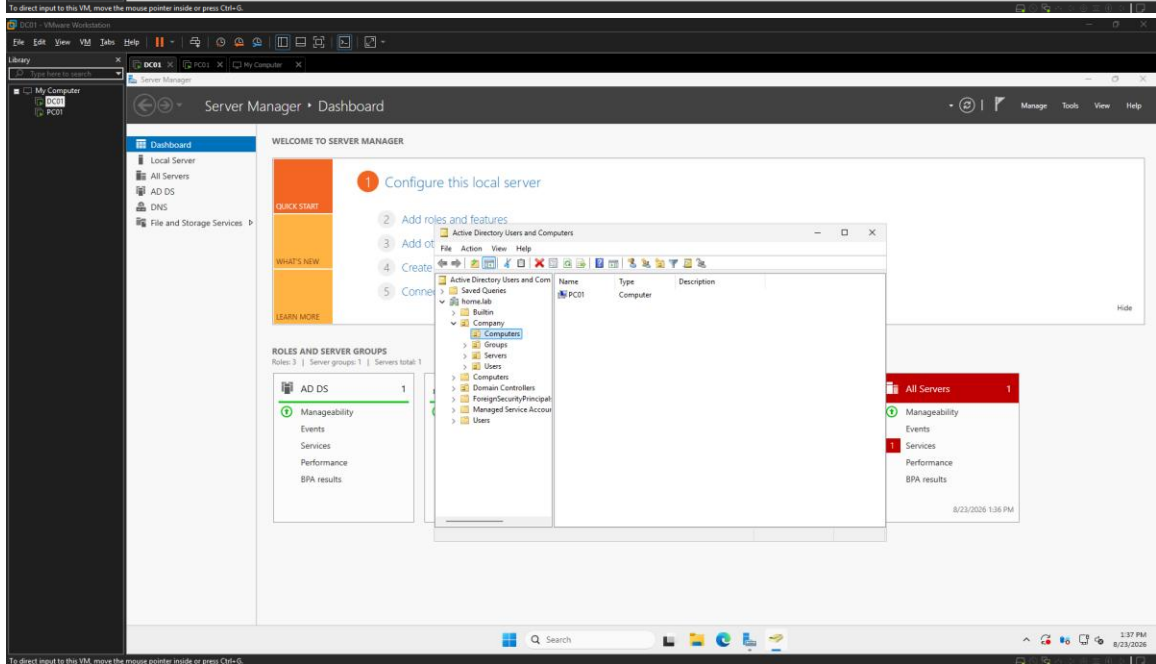
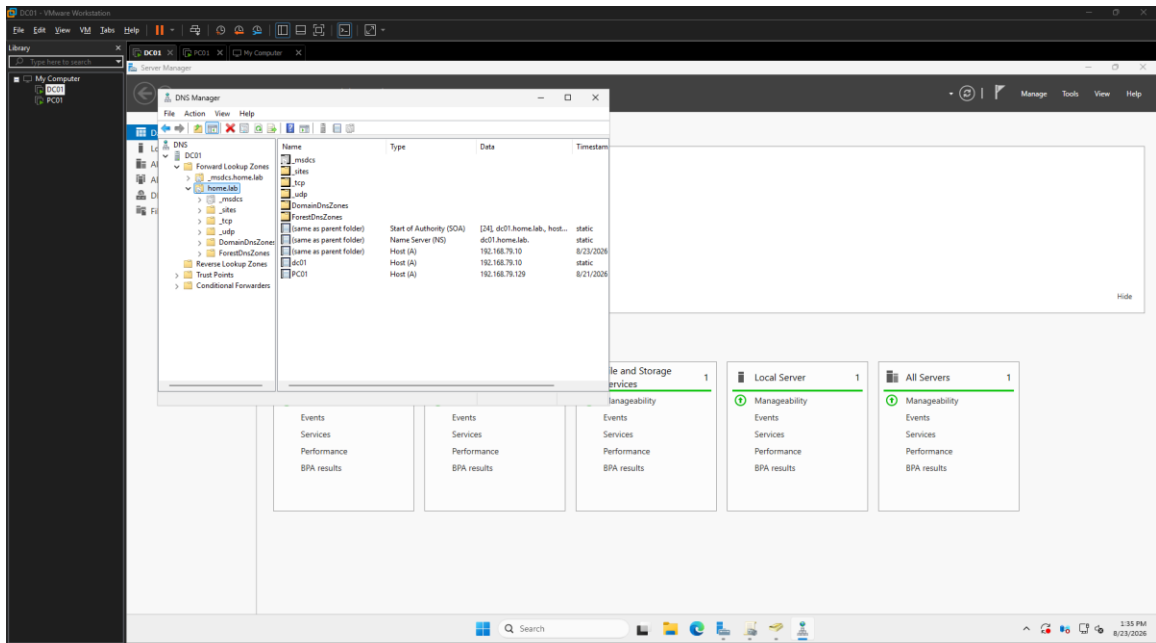
114. Verify Active Directory structure.
115. Verify DNS records.
116. Verify domain membership.
117. Verify shared folders and permissions.
118. Verify Group Policy.
119. Confirm successful authentication.

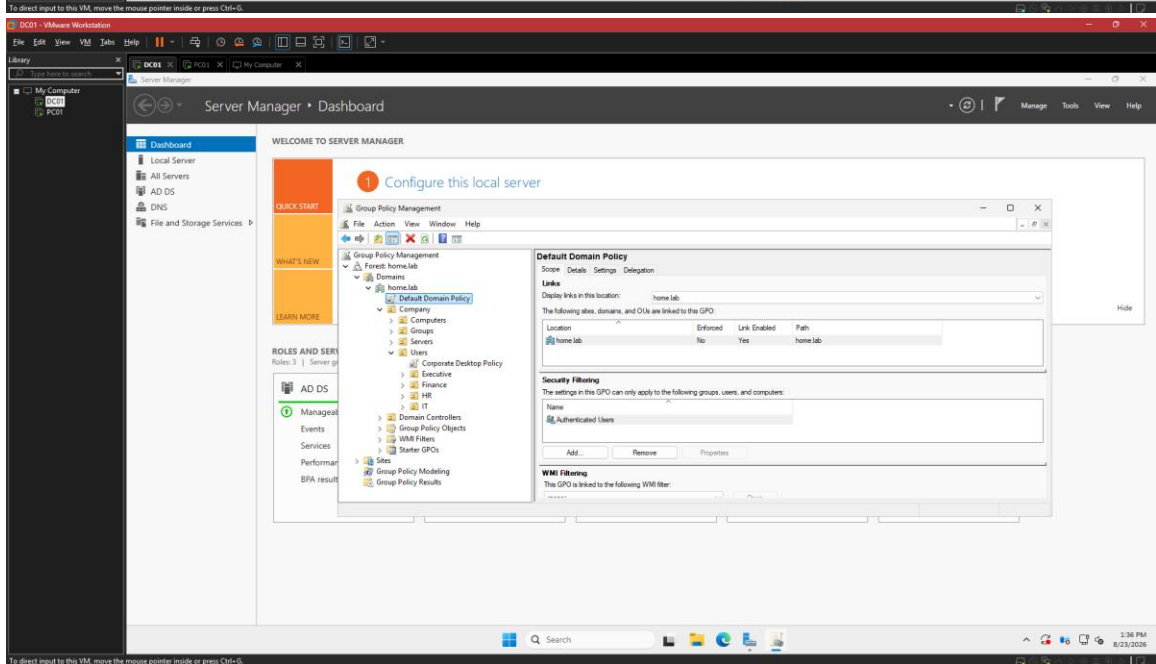
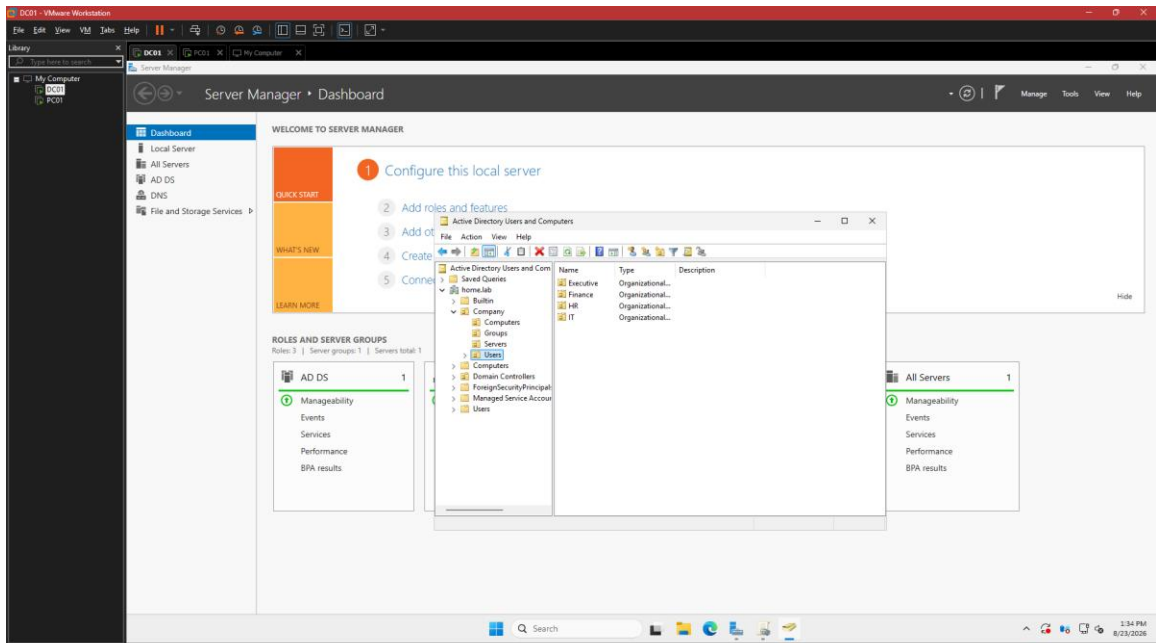
Verification

All infrastructure components are operating as expected.

Evidence







5. Skills Demonstrated

- Windows Server 2025 Administration
- Active Directory Domain Services (AD DS)
- DNS Administration
- Group Policy Management
- VMware Workstation Pro
- Windows 11 Enterprise Administration
- NTFS Permissions
- Network Shares
- Identity and Access Management
- IT Support Troubleshooting
- Technical Documentation

6. Lessons Learned

This project strengthened practical knowledge of enterprise Windows administration by combining infrastructure deployment with real-world IT support scenarios. It highlighted the importance of planning, validation, documentation and troubleshooting throughout a deployment.

7. Future Improvements

- Deploy a secondary Domain Controller.
- Implement DHCP and WSUS.
- Automate user provisioning with PowerShell.
- Integrate Microsoft Entra ID and Microsoft 365.
- Deploy osTicket for help desk simulations.

8. Operational Best Practices

The following practices were followed throughout the deployment to reflect enterprise standards and improve stability, maintainability and recoverability.

- Assign a static IP address before promoting a Domain Controller.
- Use the Domain Controller as the preferred DNS server for all domain members.
- Create VMware snapshots before major configuration changes.
- Manage permissions through security groups instead of individual user accounts.
- Apply the principle of least privilege when configuring share and NTFS permissions.
- Validate each configuration change before progressing to the next deployment stage.
- Document every significant configuration step with supporting evidence.

9. Troubleshooting Summary

Issue	Resolution	Outcome
Domain join failed	Verified DNS pointed to DC01 and confirmed connectivity.	PC01 joined the domain successfully.
Share access denied	Reviewed security group membership and NTFS permissions.	Authorised users received correct access.
GPO not applying	Forced gpupdate /force and verified OU placement.	Policies applied successfully.
Account locked	Unlocked account in Active Directory Users and Computers.	User authenticated successfully.

10. Conclusion

The project successfully delivered a fully functional Windows Server 2025 Active Directory environment capable of supporting centralised authentication, policy management and secure resource access. The completed lab demonstrates practical experience with technologies commonly encountered in entry-level IT Support and Systems Administration roles. The deployment also provides a reusable platform for future projects, including osTicket, Microsoft 365 administration, PowerShell automation and Windows Server services.

Appendix A – Technologies Used

- Windows Server 2025 Standard
- Windows 11 Enterprise
- VMware Workstation Pro
- Active Directory Domain Services (AD DS)
- DNS
- Group Policy
- NTFS
- PowerShell (future expansion)

Appendix B – Acronyms

Acronym	Meaning
AD DS	Active Directory Domain Services
DNS	Domain Name System
GPO	Group Policy Object
OU	Organisational Unit
NTFS	New Technology File System
VM	Virtual Machine